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United States Patent Application Publication

20250281056

Kind Code

A1

Publication Date

September 11, 2025

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WEARABLE DEVICE WITH PHYSIOLOGICAL MONITORING

Abstract

An optical physiological sensor can be integrated into a wearable device and can comprise a frame defining a first emitter chamber housing a first group of emitters; a second emitter chamber housing a second group of emitters; and a plurality of detector chamber housing a plurality of detectors configured to detect optical radiation emitted from the first and second groups of emitters.

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Family ID: 1000008618204

Appl. No.: 19/211874

Filed: May 19, 2025

Related U.S. Application Data

parent US continuation 17812104 20220712 parent-grant-document US 12336796 child US 19211874

us-provisional-application US 63221385 20210713

us-provisional-application US 63230239 20210806

us-provisional-application US 63252893 20211006

Publication Classification

Int. Cl.: A61B5/024 (20060101); A61B5/00 (20060101)

U.S. Cl.:

CPC A61B5/02427 (20130101); A61B5/6824 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/812,104, filed Jul. 12, 2022 which claims the benefit of U.S. Provisional Application No. 63/252,893, filed Oct. 6, 2021, and U.S. Provisional Application No. 63/230,239, filed Aug. 6, 2021, and U.S. Provisional Application No. 63/221,385, filed Jul. 13, 2021. The entire disclosure of each of the above items is hereby made part of this specification as if set forth fully herein and incorporated by reference for all purposes, for all that it contains.

FIELD

[0002] The present disclosure relates to a wearable health monitoring device incorporating a plurality of sensors worn on the wrist.

BACKGROUND

[0003] Spectroscopy is a common technique for measuring the concentration of organic and some inorganic constituents of a solution. The theoretical basis of this technique is the Beer-Lambert law, which states that the concentration $c_{\text{sub } i}$ of an absorbent in solution can be determined by the intensity of light transmitted through the solution, knowing the pathlength $d_{\text{sub } \lambda}$, the intensity of the incident light $I_{\text{sub } 0, \lambda}$, and the extinction coefficient $\epsilon_{\text{sub } i, \lambda}$ at a particular wavelength λ .

[0004] In generalized form, the Beer-Lambert law is expressed as:

$$[00001] \quad I = I_0 \cdot e^{-d \cdot \sum_{i=1}^n \mu_{\text{sub } a, \lambda} \cdot c_i} \quad (1) \quad \mu_{\text{sub } a, \lambda} = \sum_{i=1}^n \epsilon_{\text{sub } i, \lambda} \cdot c_i \quad (2)$$

[0005] where $\mu_{\text{sub } a, \lambda}$ is the bulk absorption coefficient and represents the probability of absorption per unit length. The minimum number of discrete wavelengths that are required to solve equations 1 and 2 is the number of significant absorbers that are present in the solution.

[0006] A practical application of this technique is pulse oximetry or plethysmography, which utilizes a noninvasive sensor to measure oxygen saturation and pulse rate, among other physiological parameters. Pulse oximetry or plethysmography relies on a sensor attached externally to the patient (typically for example, at the fingertip, foot, ear, forehead, or other measurement sites) to output signals indicative of various physiological parameters, such as a patient's blood constituents and/or analytes, including for example a percent value for arterial oxygen saturation, among other physiological parameters. The sensor has at least one emitter that transmits optical radiation of one or more wavelengths into a tissue site and at least one detector that responds to the intensity of the optical radiation (which can be reflected from or transmitted through the tissue site) after absorption by pulsatile arterial blood flowing within the tissue site. Based upon this response, a processor determines the relative concentrations of oxygenated hemoglobin ($\text{HbO}_{\text{sub } 2}$) and deoxygenated hemoglobin (Hb) in the blood so as to derive oxygen saturation, which can provide early detection of potentially hazardous decreases in a patient's oxygen supply, and other physiological parameters.

[0007] A patient monitoring device can include a plethysmograph sensor. The plethysmograph sensor can calculate oxygen saturation ($\text{SpO}_{\text{sub } 2}$), pulse rate, a plethysmograph waveform, perfusion index (PI), pleth variability index (PVI), methemoglobin (MetHb), carboxyhemoglobin (CoHb), total hemoglobin (tHb), respiration rate, glucose, and/or otherwise. The parameters measured by the plethysmograph sensor can display on one or more monitors the foregoing parameters individually, in groups, in trends, as combinations, or as an overall wellness or other index.

[0008] A pulse oximetry sensor is described in U.S. Pat. No. 6,088,607 entitled Low Noise Optical Probe; pulse oximetry signal processing is described in U.S. Pat. Nos. 6,650,917 and 6,699,194 entitled Signal Processing Apparatus and Signal Processing Apparatus and Method, respectively; a pulse oximeter monitor is described in U.S. Pat. No. 6,584,336 entitled Universal/Upgrading Pulse

Oximeter; all of which are assigned to Masimo Corporation, Irvine, CA, and each is incorporated by reference herein in its entirety.

SUMMARY

[0009] A drawback to current pulse oximetry sensors is a need to be located near significant capillary beds on the body, including fingers, ears, toes, nose and forehead. Such locations are often inconvenient for monitoring a user during normal activities, outside of a healthcare facility. Further, although measuring through motion oxygen saturation technology exists, it is directed to the healthcare facility context and is not reliable for normal routines, which include sporting activities or other significant daily movement. Accordingly, the present disclosure provides a sensor which allows for measuring pulse oximetry at sparse capillary bed locations, including the wrist. The present disclosure also provides algorithms for measuring pulse oximetry through higher exertion everyday motion.

[0010] A physiological monitoring sensor or module, also referred to herein as a physiological parameter measurement sensor or module, or a module, can be integrated into a wearable device that is secured to a wrist of a person (the “wearer”), such as a wristwatch or watch. The sensor on the watch can be used to monitor the wearer's physiological parameters. The sensor can detect pulse rate, oxygen saturation, hydration status, respiratory rate, and/or other parameters, such as the parameters disclosed herein, of the wearer. The sensor can include a convex protrusion to improve pressure and contact, and therefore optical coupling, between the wearer's skin and the physiological parameter measurement sensor. The curvature of the sensor can be designed to balance the desired pressure by the watch on the wearer's wrist and the wearer's comfort. The sensor can include a light barrier between emitters and detectors of the module and/or light diffusing materials surrounding the emitters and the detectors, among other features, to improve signal strength and reduce noise. The sensor or the watch can include a connection port to receive another sensor, which can be configured to be coupled to the wearer at a different measurement site of the wearer's body than the wrist. The sensor can be configured to continuously, at certain time intervals, and/or upon the wearer's request, measure one or more of the physiological parameters. For example, the sensor can be configured to continuously measure the wearer's oxygen saturation and/or pulse rate when the watch is worn on the wearer's wrist.

[0011] An example optical physiological sensor of the present disclosure can be integrated into a watch configured to monitor health of a wearer. The optical physiological sensor can be configured to face tissue of the wearer when the watch is worn by the wearer. The optical physiological sensor can be configured to measure physiological parameters of the wearer using information from the optical physiological sensor. The optical physiological sensor can comprise: a printed circuit board (PCB); a first emitter group comprising a first plurality of light emitting diodes (LEDs) at a first location on the PCB; a second emitter group comprising a second plurality of LEDs at a second location on the PCB different from the first location, wherein the second emitter group comprises the same number and type of LEDs as the first emitter group; one or more first light blocks separating the first emitter group from the second emitter group; light diffusing material configured to diffuse light emitted by each of the first and second pluralities of LEDs; a plurality of detectors including six or more photodiodes on the PCB; one or more second light blocks separating each of the detectors of the plurality of detectors from (i) each of the other detectors of the plurality of detectors and from (ii) the first and second emitter groups; and a convex surface configured to be positioned between (i) the first and second emitter groups and the six or more photodiodes and (ii) the tissue of the wearer, the convex surface comprising one or more surface materials.

[0012] In some configurations, the first plurality of LEDs may be configured to emit light of at least five wavelengths and the second plurality of LEDs may be configured to emit light of at least five wavelengths.

[0013] In some configurations, the five wavelengths emitted by the first plurality of LEDs may be the same wavelengths as the five wavelengths emitted by the second plurality of LEDs.

[0014] In some configurations, the first plurality of LEDs may comprise LEDs of a same type as LEDs of the second plurality of LEDs.

[0015] In some configurations, the LEDs of the first plurality of LEDs may be arranged on the PCB to mirror, across a centerline of the sensor, an arrangement of the LEDs of the second plurality of LEDs on the PCB.

[0016] In some configurations, the first plurality of LEDs may comprise: a first LED configured to emit light of a first wavelength, a second LED configured to emit light of a second wavelength, a third LED configured to emit light of a third wavelength, a fourth LED configured to emit light of a fourth wavelength, and a fifth LED configured to emit light of a fifth wavelength, and the second plurality of LEDs may comprise: a sixth LED configured to emit light of the first wavelength, a seventh LED configured to emit light of the second wavelength, an eighth LED configured to emit light of the third wavelength, a ninth LED configured to emit light of the fourth wavelength, and a tenth LED configured to emit light of the fifth wavelength.

[0017] In some configurations, the first LED may be located on the PCB at a location that mirrors, across a centerline of the sensor, a location of the sixth LED on the PCB; the second LED may be located on the PCB at a location that mirrors, across a centerline of the sensor, a location of the seventh LED on the PCB; the third LED may be located on the PCB at a location that mirrors, across a centerline of the sensor, a location of the eighth LED on the PCB; the fourth LED may be located on the PCB at a location that mirrors, across a centerline of the sensor, a location of the ninth LED on the PCB; the fifth LED may be located on the PCB at a location that mirrors, across a centerline of the sensor, a location of the tenth LED on the PCB; and the centerline of the sensor may bisect at least one of the plurality of detectors.

[0018] In some configurations, the centerline of the sensor may bisect two of the plurality of detectors, and each of said two detectors may be configured to generate, responsive to detecting light emitted from the first and second plurality of emitters, one or more signals for normalizing a physiological parameter measurement of the sensor.

[0019] In some configurations, the first and sixth LEDs may each be located a same distance from at least one of the plurality of detectors; the second and seventh LEDs may each be located a same distance from at least one of the plurality of detectors; the third and eighth LEDs may each be located a same distance from at least one of the plurality of detectors; the fourth and ninth LEDs may each be located a same distance from at least one of the plurality of detectors; and the fifth and tenth LEDs may each be located a same distance from at least one of the plurality of detectors.

[0020] In some configurations, the plurality of detectors may comprise: a first detector group comprising at least two detectors; and a second detector group comprising at least two detectors, and the first emitter group may be proximal to the first detector group and distal to the second detector group, and the second emitter group may be proximal to the second detector group and distal to the first detector group.

[0021] In some configurations, the first and second emitter groups may be located at a central location of the PCB of the sensor.

[0022] In some configurations, the plurality of detectors may extend around a circumference of the PCB of the sensor.

[0023] An example optical physiological sensor of the present disclosure can be integrated into a watch configured to monitor health of a wearer. The optical physiological sensor can be configured to face tissue of the wearer when the watch is worn by the wearer. The optical physiological sensor can be configured to measure physiological parameters of the wearer using information from the optical physiological sensor. The optical physiological sensor can comprise: a printed circuit board (PCB); a first emitter group comprising a first plurality of light emitting diodes (LEDs) at a first location on the PCB; a second emitter group comprising a second plurality of LEDs at a second location on the PCB different from the first location, wherein the second emitter group comprises the same number and type of LEDs as the first emitter group; one or more first light blocks

separating the first emitter group from the second emitter group; light diffusing material configured to diffuse light emitted by each of the first and second pluralities of LEDs; a plurality of detectors including six or more photodiodes on the PCB; one or more second light blocks separating each of the detectors of the plurality of detectors from each of the other detectors of the plurality of detectors; and a convex surface configured to be positioned between (i) the first and second emitter groups and the six or more photodiodes and (ii) the tissue of the wearer, the convex surface comprising one or more surface materials.

[0024] In some configurations, the first emitter group can define a first group of emitters, each emitter of the first group of emitters being located on the PCB less than 1.0 mm from each of the other emitters of the first group of emitters, and wherein the second emitter group can define a second group of emitters, each emitter of the second group of emitters being located on the PCB less than 1.0 mm from each of the other emitters of the second group of emitters.

[0025] In some configurations, the plurality of detectors can comprise: a first detector group which can comprise at least two detectors, wherein each detector of the at least two detectors can be proximal to the first emitter group; a second detector group which can comprise at least two detectors, wherein each detector of the at least two detectors can be distal to the first emitter group.

[0026] In some configurations, at least two of the plurality of detectors can each individually comprise an intermediate detector for the first emitter group and the second emitter group.

[0027] In some configurations, the first and second emitter groups can be located at a central location of the PCB of the sensor.

[0028] In some configurations, the plurality of detectors can extend around a circumference of the PCB of the sensor.

[0029] In some configurations, the PCB can comprise a conductive liquid adhesive.

[0030] In some configurations, the PCB can comprise a reflective surface.

[0031] In some configurations, the first plurality of LEDs can be configured to emit light of at least four wavelengths and wherein the second plurality of LEDs can be configured to emit light of at least four wavelengths.

[0032] In some configurations, the first plurality of LEDs can comprise: a first LED which can be configured to emit light of a first wavelength, a second LED which can be configured to emit light of a second wavelength, a third LED which can be configured to emit light of a third wavelength, and a fourth LED which can be configured to emit light of a fourth wavelength, and the second plurality of LEDs can comprise: a fifth LED which can be configured to emit light of the first wavelength, a sixth LED which can be configured to emit light of the second wavelength, a seventh LED which can be configured to emit light of the third wavelength, and an eighth LED which can be configured to emit light of the fourth wavelength.

[0033] In some configurations, the sensor can comprise a sensor processor configured to drive the first plurality of LEDs and the second plurality of LEDs at varying intensities.

[0034] In some configurations, the sensor processor can be configured to drive the intensities of the first plurality of LEDs and the second plurality of LEDs based, at least in part, on one or more of an attenuation of emitted light or a temperature of the first plurality of LEDs and the second plurality of LEDs.

[0035] In some configurations, the sensor may comprise a sensor processor configured to activate or deactivate some of the detectors of the plurality of detectors.

[0036] In some configurations, the sensor may comprise an ECG sensor, and the ECG sensor may comprise a reference electrode, a negative electrode, and a positive electrode and the reference and negative electrodes can be located on the sensor and a portion of a housing of the watch can form the positive electrode.

[0037] In some configurations, the reference and negative electrodes can be releasably coupled to the PCB of the sensor.

[0038] The present disclosure provides a system for monitoring physiological parameters. The

system may comprise a wearable device which may comprise one or more physiological sensors configured to monitor one or more physiological parameters of a user. The system may comprise a computing device external to the wearable device and in wireless communication with the wearable device. The computing device may be configured to: receive, via an interactive user interface, user input; in response to receiving the user input, wirelessly transmit a first signal to the wearable device to instruct the wearable device to perform a physiological monitoring operation; receive, from the wearable device, physiological parameter data measured by the physiological monitoring operation; and cause presentation of the physiological parameter data in the interactive user interface.

[0039] In some configurations, the monitoring operation can comprise continuous measuring of the physiological parameters.

[0040] In some configurations, the monitoring operation can comprise: measuring the physiological parameters for a predetermined length of time; and ceasing to measure the physiological parameters upon expiration of the predetermined length of time.

[0041] In some configurations, the wearable device can be configured to: in response to receiving the first signal from the computing device, determine a power level of a battery of the wearable device; determine whether the power level is sufficient to perform the monitoring operation; and in response to determining that the power level is sufficient, perform the monitoring operation.

[0042] In some configurations, the wearable device can be configured to: in response to receiving the first signal from the computing device, determine a power level of a battery of the wearable device; determine whether the power level is sufficient to perform the monitoring operation; in response to determining that the power level is not sufficient, determine a modified monitoring operation based at least in part on the power level; and perform the modified monitoring operation.

[0043] In some configurations, the computing device can be further configured to: cause presentation, in the interactive user interface, of one or more of a graph, chart, or trend of historical and substantially real-time physiological parameter data.

[0044] In some configurations, the wearable device can be configured to: determine that one or more physiological parameters exceed a respective predetermined threshold; and in response to determining that the one or more physiological parameters exceed the respective predetermined threshold, transmit a second signal to the computing device to cause the computing device to generate an alarm.

[0045] An example optical physiological sensor of the present disclosure can be integrated into a watch configured to monitor health of a wearer. The optical physiological sensor can be configured to face tissue of the wearer when the watch is worn by the wearer and to measure physiological parameters of the wearer using information from the optical physiological sensor. The optical physiological sensor can comprise a first emitter group comprising a first plurality of light emitting diodes (LEDs) at a first location; a second emitter group comprising a second plurality of LEDs at a second location different from the first location, wherein the second emitter group can comprise the same number and type of LEDs as the first emitter groups; one or more light blocks separating the first emitter group from the second emitter group; light diffusing material configured to diffuse light emitted by each of the first and second pluralities of LEDs; a plurality of detectors including four or more photodiodes; and a convex surface configured to be positioned between (i) the first and second emitter groups and the four or more photodiodes and (ii) the tissue of the wearer, the convex surface comprising one or more surface materials.

[0046] In some configurations, the one or more surface materials can comprise at least a portion of the one or more light blocks and a light transmission material.

[0047] In some configurations, the emitters in the first or second emitter groups may not be electrically connected to one another.

[0048] In some configurations, the first or second emitter groups can define a group of emitters located in close proximity.

[0049] In some configurations, the plurality of detectors can be individually both a near detector and far detector for each emitter group.

[0050] In some configurations, the first and second emitter groups can be located at non-central locations of a printed circuit board (PCB) of the sensor.

[0051] In some configurations, the one or more light blocks can extend from a surface of the sensor positioning the first and second pluralities of LEDs towards a tissue of the wearer when the watch is worn.

[0052] In some configurations, each of the first or second emitter group can be surrounded by its own diffusing material.

[0053] In some configurations, the light diffusing material surrounding the first emitter group can be different from the light diffusing material surrounding the second emitter group.

[0054] In some configurations, at least some of the plurality of detectors can extend around a circumference of the sensor.

[0055] In some configurations, the plurality of detectors can be positioned in a grid pattern and/or across from one another.

[0056] In some configurations, locations of the emitter groups can be interleaved with the plurality of detectors.

[0057] In some configurations, at least one of the plurality of detectors can be located between the first plurality of LEDs and the second plurality of LEDs, and at least one of the plurality of detectors can be located on each of at least two sides of each of the first plurality of LEDs and the second plurality of LEDs.

[0058] In some configurations, the sensor can further comprise a processor configured to determine an oxygen saturation measurement based on signals from the optical physiological sensor.

[0059] An example optical physiological sensor of the present disclosure can be integrated into a watch configured to monitor health of a wearer. The optical physiological sensor can be configured to face tissue of the wearer when the watch is worn by the wearer and to measure physiological parameters of the wearer using information from the optical physiological sensor. The optical physiological sensor can comprise a plurality of emitters, the emitters configured to emit light of a plurality of different wavelengths, the plurality of different wavelengths comprising at least three different wavelengths; a plurality of detectors, the detectors configured to detect light emitted by the plurality of emitters and attenuated by tissue of the user when the watch is worn on the wrist of the wearer and output signals to a sensor processor for determining the physiological parameters of the wearer; and a sensor housing, the plurality of emitters and the plurality of detectors enclosed within the housing, wherein the sensor housing can comprise: a convex skin-facing light transmissive cover extending over the plurality of emitters and the plurality of detectors, the cover located at a first side of sensor housing, and a printed circuit board (PCB) located at a second side of the sensor housing opposite the first side, the plurality of emitters and detectors located on a skin-facing side of the PCB; and a plurality of light barriers extending from the PCB to the cover, the plurality of light barriers configured to form walls of chambers to block light or substantially all the light between the chambers, each chamber enclosing one or more emitters without detectors, or one or more detectors without emitters, wherein a skin-facing surface of the cover and at least one of the light barriers can define a skin-facing surface of the sensor, a surface area of the cover extending over the chambers that enclose one or more detectors is at least 50% of a surface area of the skin-facing surface of the sensor.

[0060] In some configurations, the surface area of the cover extending over the chambers that enclose one or more detectors can be at least 100 mm².

[0061] In some configurations, the surface area of the cover extending over the chambers that enclose one or more detectors can be at least 150 mm².

[0062] In some configurations, the surface area of the cover extending over the chambers that enclose one or more detectors can be at least 165 mm².

[0063] In some configurations, a surface area of the light transmissive cover that extends over the chambers that enclose one or more emitters can be at least 25 mm.^{sup.2}

[0064] In some configurations, the surface area of the light transmissive cover that extends over the chambers that enclose one or more detectors can be at least 35 mm.^{sup.2}

[0065] In some configurations, the skin-facing surface of the sensor can have a longer side and a shorter side, the longer side configured to be along a width of the wearer's wrist when the watch is worn.

[0066] In some configurations, more of the plurality of detectors can be located along the longer side than along the shorter side.

[0067] In some configurations, the plurality of emitters can comprise a first group of emitters and a second group of emitters, the chambers comprising a first emitter chamber enclosing the first group and a second emitter chamber enclosing the second group.

[0068] In some configurations, the plurality of detectors can comprise a first ring of detectors and a second ring of detectors, the first ring of detectors surrounding the first group of emitters and the second ring of detectors surrounding the second group of emitters.

[0069] In some configurations, at least one of the plurality of detectors can be located between the first and second group of emitters and can be shared by the first and second rings of detectors.

[0070] In some configurations, some of the plurality of detectors can be closer to the first group of emitters than a remainder of the plurality of detectors and some of the plurality of detectors can be closer to the second group of emitters than a remainder of the plurality of detectors.

[0071] In some configurations, the plurality of light barriers can extend to a skin-facing surface of the cover.

[0072] An example optical physiological sensor of the present disclosure can be integrated into a watch configured to monitor health of a wearer. The optical physiological sensor can be configured to face tissue of the wearer when the watch is worn by the wearer and to measure physiological parameters of the wearer using information from the optical physiological sensor. The optical physiological sensor can comprise a plurality of emitters, the emitters configured to emit light of a plurality of different wavelengths, the plurality of different wavelengths comprising at least three different wavelengths; a plurality of detectors, the detectors configured to detect light emitted by the plurality of emitters and attenuated by tissue of the user when the watch is worn on the wrist of the wearer and output signals to a sensor processor for determining the physiological parameters of the wearer; and a sensor housing, the plurality of emitters and the plurality of detectors enclosed within the housing, wherein the sensor housing can comprise: a convex skin-facing light transmissive cover extending over the plurality of emitters and the plurality of detectors, the cover located at a first side of sensor housing, and a printed circuit board (PCB) located at a second side of the sensor housing opposite the first side, the plurality of emitters and detectors located on a skin-facing side of the PCB; and a plurality of light barriers extending from the PCB to the cover, the plurality of light barriers configured to form walls of chambers to block light or substantially all the light between the chambers, each chamber enclosing one or more emitters without detectors, or one or more detectors without emitters, wherein at least one of the plurality of light barriers can extend to a skin-facing surface of the cover.

[0073] In some configurations, all of the plurality of light barriers can extend to the skin-facing surface of the cover.

[0074] In some configurations, the skin-facing surface of the cover and the at least one of the light barriers can define a skin-facing surface of the sensor.

[0075] In some configurations, the skin-facing surface of the sensor can comprise a continuous curvature.

[0076] In some configurations, the cover can be a single lens or cover.

[0077] In some configurations, the cover can comprise individual lenses, each lens or cover covering a single chamber.

[0078] In some configurations, the cover can comprise a lens or cover covering all the chambers that extend over one or more detectors.

[0079] In some configurations, the lens or cover covering all the chambers that extend over one or more detectors may not cover a chamber that extends over one or more emitters.

[0080] In some configurations, the plurality of light barriers can comprise colored sapphire glass.

[0081] An example optical physiological sensor of the present disclosure can be integrated into a watch configured to monitor health of a wearer. The optical physiological sensor can be configured to face tissue of the wearer when the watch is worn by the wearer and to measure physiological parameters of the wearer using information from the optical physiological sensor. The optical physiological sensor can comprise a first emitter comprising a first plurality of light emitting diodes (LEDs) positioned on a surface of a substrate; a first photodiode positioned on the surface of the substrate; a curved surface extending over all the first plurality of LEDs and the first photodiode; and a first light barrier positioned between the first emitter and the first photodiode, and extending from the surface of the substrate to the curved surface.

[0082] In some configurations, the first light barrier can comprise one or more portions that together extend from the surface of the substrate to the curved surface.

[0083] In some configurations, the sensor can further comprise: a second emitter comprising a second plurality of LEDs positioned on the surface of the substrate; a second photodiode positioned on the surface of the substrate; a second light barrier positioned between (i) both the first and second emitters and (ii) the second photodiode, and extending from the surface of the substrate to the curved surface, wherein the curved surface can extend over all the second plurality of LEDs and the second photodiode.

[0084] In some configurations, the second light barrier can comprise one or more portions that together extend from the surface of the substrate to the curved surface.

[0085] In some configurations, portions of the curved surface positioned above the first and second emitters can comprise at least a first material, portions of the curved surface positioned and the first and second photodiodes can comprise at least a second material, and portions of the first and second barriers extending to the curved surface can comprise at least a third material different from the first and second materials.

[0086] In some configurations, at least the first, second, and third materials together can make up the curved surface.

[0087] In some configurations, the first and second materials can comprise the same material.

[0088] An example optical physiological sensor of the present disclosure can be integrated into a watch configured to monitor health of a wearer. The optical physiological sensor can be configured to face tissue of the wearer when the watch is worn by the wearer and to measure physiological parameters of the wearer using information from the optical physiological sensor. The optical physiological sensor can comprise a plurality of light-emitting diodes (LEDs) configured to emit light to tissue of a wearer; a wall dividing the plurality of LEDs into at least a first group of LEDs and a second group of LEDs, the wall blocking at least some of the light emitted by the first group of LEDs from contacting the second group of LEDs; four or more photodiodes configured to detect the light emitted by the plurality of LEDs after attenuation by the tissue; and one or more covers covering the plurality of LEDs and the four or more photodiodes, the one or more covers together forming part of a convex surface configured to contact the tissue.

[0089] An example optical physiological sensor of the present disclosure can be integrated into a watch configured to monitor health of a wearer. The optical physiological sensor can be configured to face tissue of the wearer when the watch is worn by the wearer and to measure physiological parameters of the wearer using information from the optical physiological sensor. The optical physiological sensor can comprise a plurality of emitters, the emitters configured to emit light of a plurality of different wavelengths, the plurality of different wavelengths comprising at least three different wavelengths; a plurality of detectors, the detectors configured to detect light emitted by

the plurality of emitters and attenuated by tissue of the user when the watch is worn on the wrist of the wearer and output signals to a sensor processor for determining the physiological parameters of the wearer; and a sensor housing, the plurality of emitters and the plurality of detectors enclosed within the housing, wherein the sensor housing can comprise: a convex skin-facing light transmissive cover extending over the plurality of emitters and the plurality of detectors, the cover located at a first side of sensor housing, and a printed circuit board (PCB) located at a second side of the sensor housing opposite the first side, the plurality of emitters and detectors located on a skin-facing side of the PCB; and a plurality of light barriers extending from the PCB to the cover, the plurality of light barriers configured to form walls of chambers to block light or substantially all the light between the chambers, each chamber enclosing one or more emitters without detectors, or one or more detectors without emitters, wherein the plurality of detectors can comprise a plurality of far detectors that are further from at least some of the plurality of emitters than a remainder of the plurality of detectors.

[0090] In some configurations, the plurality of emitters can comprise a first group of emitters and a second group of emitters, the chambers comprising a first emitter chamber enclosing the first group and a second emitter chamber enclosing the second group.

[0091] In some configurations, the plurality of detectors can comprise a first ring of detectors and a second ring of detectors, the first ring of detectors surrounding the first group of emitters and the second ring of detectors surrounding the second group of emitters.

[0092] In some configurations, at least one of the plurality of detectors can be located between the first and second group of emitters and is shared by the first and second rings of detectors.

[0093] In some configurations, some of the plurality of detectors can be closer to the first group of emitters than a remainder of the plurality of detectors and some of the plurality of detectors are closer to the second group of emitters than a remainder of the plurality of detectors.

[0094] In some configurations, the sensor can further comprise the sensor processor, wherein the sensor processor is configured to determine a hydration status of a user based on signals from the plurality of far detectors.

[0095] In some configurations, at least one of the emitters can be configured to emit light of a wavelength more sensitive to water than a remainder of the different wavelengths.

[0096] In some configurations, the wavelength more sensitive to water can be about 970 nm.

[0097] In some configurations, the sensor processor can be configured to compare signals of the reflected light of the wavelength more sensitive to water and another wavelength less sensitive to water from the plurality of far detectors.

[0098] In some configurations, the sensor processor can be configured to selectively drive some of the plurality of emitters and/or activate or deactivate some of the plurality of detectors.

[0099] An example optical physiological sensor of the present disclosure can be integrated into a watch configured to monitor health of a wearer. The optical physiological sensor can be configured to face tissue of the wearer when the watch is worn by the wearer and to measure physiological parameters of the wearer using information from the optical physiological sensor. The optical physiological sensor can comprise a plurality of emitters, the emitters configured to emit light of a plurality of different wavelengths, wherein at least one of the emitters can be configured to emit light of a reference wavelength; a plurality of detectors, the detectors configured to detect light emitted by the plurality of emitters and attenuated by tissue of the user when the watch is worn on the wrist of the wearer; a sensor processor, wherein the plurality of detectors can be configured to output signals to the sensor processor for determining at least some of the physiological parameters of the wearer based in part on a signal of the reflected light of the reference wavelength; and a sensor housing, the plurality of emitters and the plurality of detectors enclosed within the housing, wherein the sensor housing can comprise: a convex skin-facing light transmissive cover extending over the plurality of emitters and the plurality of detectors, the cover located at a first side of sensor housing, and a printed circuit board (PCB) located at a second side of the sensor housing opposite

the first side, the plurality of emitters and detectors located on a skin-facing side of the PCB; and a plurality of light barriers extending from the printed circuit board to the cover, the plurality of light barriers configured to form walls of chambers to block light or substantially all the light between the chambers, each chamber enclosing one or more emitters without detectors, or one or more detectors without emitters.

[0100] In some configurations, the reference wavelength can be about 525 nm.

[0101] In some configurations, the light of the reference wavelength can be green or yellow.

[0102] In some configurations, the sensor processor can be configured to extract features from signals of other wavelengths based on the signal of the reflected light of the reference wavelength and calculate the at least some of the physiological parameters based on the extracted features.

[0103] In some configurations, at least one of the emitters can be configured to emit light of a wavelength more sensitive to oxygen saturation.

[0104] In some configurations, at least one of the emitters can be configured to emit light of a wavelength more sensitive to water.

[0105] In some configurations, at least one of the emitters can be configured to emit light of a normalizing wavelength.

[0106] In some configurations, the sensor processor can be configured to determine a hydration status of a user based on signals of the reflected light of the wavelength more sensitive to water and of the normalizing wavelength.

[0107] In some configurations, one or more physiological parameters can comprise a pulse rate, respiration rate, SpO₂, PVI, PI, RRP, hydration, or a combination thereof.

[0108] In some configurations, the sensor can further comprise a thermistor located near the plurality of emitters.

[0109] In some configurations, the sensor can further comprise an accelerometer and/or gyroscope.

[0110] In some configurations, the sensor processor can be configured to selectively drive some of the plurality of emitters and/or activate or deactivate some of the plurality of detectors.

[0111] An example optical physiological sensor of the present disclosure can be integrated into a watch configured to monitor health of a wearer. The optical physiological sensor can be configured to face tissue of the wearer when the watch is worn by the wearer and to measure physiological parameters of the wearer using information from the optical physiological sensor. The optical physiological sensor can comprise a plurality of emitters, the emitters configured to emit light of a plurality of different wavelengths, the plurality of different wavelengths comprising at least three different wavelengths; a plurality of detectors, the detectors configured to detect light emitted by the plurality of emitters and attenuated by tissue of the user when the watch is worn on the wrist of the wearer and output signals to a sensor processor for determining the physiological parameters of the wearer; and a sensor housing, the plurality of emitters and the plurality of detectors enclosed within the housing, wherein the sensor housing can comprise: a convex skin-facing light transmissive cover extending over the plurality of emitters and the plurality of detectors, the cover located at a first side of sensor housing, and a printed circuit board (PCB) located at a second side of the sensor housing opposite the first side, the plurality of emitters and detectors located on a skin-facing side of the PCB; a plurality of light barriers extending from the PCB to the cover, the plurality of light barriers configured to form walls of chambers to block light or substantially all the light between the chambers, each chamber enclosing one or more emitters without detectors, or one or more detectors without emitters, wherein each chamber that encloses one or more emitters can be filled with a diffusing material such that there is no air gap between the plurality of emitters and the cover.

[0112] In some configurations, the light diffusing material can comprise glass microspheres.

[0113] In some configurations, the cover can comprise glass microspheres.

[0114] In some configurations, the sensor housing can comprise one or more openings configured to receive a flow of light diffusing solution.

[0115] In some configurations, the light diffusion solution can be UV-cured after being injected into each chamber that encloses one or more emitters.

[0116] In some configurations, the sensor housing can comprise one or more air vent openings configured to receive air displaced from the chamber(s) by the flow of light diffusing solution.

[0117] In some configurations, each chamber that encloses one or more detectors can be filled with the diffusing material such that there is no air gap between the plurality of detectors and the cover.

[0118] In some configurations, the diffusing material in each chamber that encloses one or more emitters can be configured to improve mixing of light such that light emitted by one of the emitter in the same chamber appears to be emitted from the entire same chamber.

[0119] An example watch of the present disclosure can be configured to monitor physiological parameters of a wearer. The watch can comprise any of the optical sensor or physiological parameter measurement sensor configurations disclosed above; a watch processor separate from and in electrical communication with the sensor processor; a power source configured to power the watch and the sensor, and a display in communication with the processor, the display configured to display the plurality of physiological parameters monitored by the sensor.

[0120] In some configurations, the display can be configured to display the wearer's SpO2 and pulse rate that are monitored by the sensor.

[0121] In some configurations, the sensor can be configured to continuously monitor the wearer's SpO2 and pulse rate.

[0122] In some configurations, the display can be configured to continuously display the wearer's SpO2 and pulse rate.

[0123] In some configurations, the watch can further comprise an ECG sensor.

[0124] In some configurations, the ECG sensor can comprise a reference electrode, a negative electrode, and a positive electrode.

[0125] In some configurations, the reference and negative electrodes can be located on the sensor.

[0126] In some configurations, a portion of a housing of the watch can form the positive electrode.

[0127] In some configurations, the ECG sensor can be in electrical communication with the sensor processor.

[0128] In some configurations, the watch can further comprise a wireless transmitter such that the watch is configured to wireless connect to external devices and/or external sensors.

[0129] In some configurations, the wireless transmitter can be a Bluetooth chip.

[0130] In some configurations, the external devices and/or external sensors can comprise a bedside monitor, a mobile communication device, a tablet, a nurses' station system, or a different medical device.

[0131] A health monitoring watch of the present disclosure can comprise a strap and a housing. The housing can comprise: a first chamber comprising a first well comprising a first depth below a first surface configured to be in contact with a skin of a user; a first plurality of light emitting diodes positioned at the first depth inside the first well, said first plurality of light emitting diodes comprising a first light emitting diode configured to emit light at a first wavelength, a second light emitting diode configured to emit light at a second wavelength different than the first wavelength, and a third light emitting diode configured to emit light at a third wavelength different than the first wavelength and the second wavelength, and a first wall surrounding the first well; a second chamber comprising a second well comprising a second depth below a second surface configured to be in contact with the skin of the user, a second plurality of light emitting diodes positioned at the second depth inside the second well, said second plurality of light emitting diodes comprising a fourth light emitting diode configured to emit light at the first wavelength, a fifth light emitting diode configured to emit light at the second wavelength different than the first wavelength, and a sixth light emitting diode configured to emit light at the third wavelength different than the first wavelength and the second wavelength, and a second wall surrounding the second well; and four or more light sensors.

[0132] A wearable health monitoring device can be configured to be worn on a wrist of a user and monitor one or more physiological parameters indicative of the user's health. The wearable health monitoring device can comprise: a first emitter group, the first emitter group comprising a first plurality of light-emitting diodes (LEDs) configured to emit light of one or more wavelengths, wherein the first emitter group can be arranged at a first location, the first location being spaced from an axis extending through a center of the wearable health monitoring device; a second emitter group, the second emitter group comprising a second plurality of LEDs configured to emit light of one or more wavelengths, wherein the second emitter group can be arranged at a second location, the second location being spaced from the first location and spaced from the axis extending through the center of the wearable health monitoring device; one or more light blocks separating the first emitter group from the second emitter group; a first light diffusing material, the first light diffusing material configured to be positioned between the first emitter group and tissue of the user when the wearable health monitoring device is in use, wherein the first light diffusing material can be configured to spread light emitted from one or more of the first plurality of LEDs before the emitted light reaches the tissue; a second light diffusing material, the second light diffusing material configured to be positioned between the second emitter group and the tissue of the user when the wearable health monitoring device is in use, wherein the second light diffusing material can be configured to spread light emitted from one or more of the second plurality of LEDs before the emitted light reaches the tissue; a plurality of photodiodes configured to detect at least a portion of the light emitted from one or more of the first plurality of LEDs or one or more of the second plurality of LEDs after attenuation through the user's tissue, the plurality of photodiodes configured to output one or more signals responsive to the detected light; and a processor configured to receive and process one or more signals responsive to the one or more signals outputted by the plurality of photodiodes and further configured to determine a physiological parameter of the user based on the received and processed one or more signals.

[0133] The present disclosure provides an optical physiological sensor integrated into a wearable device. The optical physiological sensor can comprise: a substrate having an optical center; a first emitter group of light emitting diodes (LEDs) positioned adjacent to the optical center of the substrate and spaced at an offset from the optical center; a second emitter group of LEDs positioned adjacent to the optical center of the substrate at an offset to the optical center and spaced at an offset from the optical center opposite the first emitter group of LEDs relative to the optical center; and a plurality of detectors arranged in a spatial configuration that surrounds the first and the second emitter group. Each of the plurality of detectors can be positioned on the substrate a same distance away from the optical center of the substrate.

[0134] In some configurations, the spatial configuration includes an annulus.

[0135] In some configurations, the annulus has a same optical center as the substrate.

[0136] In some configurations, the annulus is concentric with a perimeter of the sensor.

[0137] In some configurations, the annulus includes a radius of between about 6.2 mm and about 6.6 mm.

[0138] In some configurations, the annulus includes a radius of between about 6.15 mm and about 6.45 mm.

[0139] In some configurations, the sensor includes a radius of between about 15.0 mm and about 15.5 mm.

[0140] In some configurations, a ratio of a radius of the annulus to a radius of the sensor is between about 40% and about 45%.

[0141] In some configurations, the second emitter group is spaced less than 1.5 mm away from the first emitter group.

[0142] In some configurations, the second emitter group is spaced about 1.28 mm away from the first emitter group.

[0143] In some configurations, the second emitter group is spaced about 1.2 mm away from the

first emitter group.

[0144] In some configurations, the sensor can further comprise a light barrier construct mounted on the substrate and configured to isolate the first emitter group, the second emitter group, and the plurality of detectors.

[0145] In some configurations, the light barrier construct is a single integrated unit comprising a plurality of light barriers defining one or more chambers.

[0146] In some configurations, the light barrier construct comprises a plurality of light barriers configured to isolate the first and second emitter groups from each of the plurality of detectors.

[0147] In some configurations, the light barrier construct comprises a light barrier configured to isolate the first emitter group from the second emitter group.

[0148] In some configurations, the light barrier includes a width of between about 1.25 mm and about 1.35 mm separating the first and second emitter groups.

[0149] In some configurations, the light barrier includes a width of between about 1.1 mm and about 1.3 mm separating the first and second emitter groups.

[0150] In some configurations, the light barrier includes a width of less than a distance from an emitter chamber defined by the light barrier construct and housing the first emitter group to a detector chamber defined by the light barrier construct and housing a detector bisected by a centerline of the sensor.

[0151] In some configurations, the light barrier construct includes a height of between about 2.80 mm and about 2.90 mm extending away from the optical center of the substrate.

[0152] In some configurations, the light barrier construct includes a height of between about 2.55 mm and about 2.65 mm extending away from the optical center of the substrate.

[0153] In some configurations, the light barrier construct includes a height of between about 1.25 mm and about 1.35 mm extending away from a perimeter of the substrate.

[0154] In some configurations, the light barrier construct includes a height of between about 1.75 mm and about 1.85 mm extending away from a perimeter of the substrate.

[0155] In some configurations, a ratio of a maximum height of the light barrier construct extending away from the substrate to a minimum height of the light barrier construct extending away from the substrate is between about 215% and about 225%.

[0156] In some configurations, a ratio of a maximum height of the light barrier construct extending away from the substrate to a minimum height of the light barrier construct extending away from the substrate is between about 145% and about 155%.

[0157] In some configurations, the light barrier construct includes a maximum height extending away from the substrate at the optical center of the substrate.

[0158] In some configurations, the light barrier defines a plurality of chambers, including, at least, a detector chamber housing a detector bisected by a centerline of the sensor; a first emitter chamber housing the first emitter group; and a second emitter chamber housing the second emitter group.

[0159] In some configurations, a distance from the first emitter chamber to the detector chamber extending along a length parallel to the centerline of the sensor is the same as a distance from the second emitter chamber to the detector chamber extending along a length parallel to the centerline of the sensor.

[0160] In some configurations, a distance from the first emitter chamber to the detector chamber extending along a length parallel to the centerline of the sensor is less than half a width of the first emitter chamber along a length parallel to the centerline of the sensor.

[0161] In some configurations, a distance from the first emitter chamber to the detector chamber extending along a length parallel to the centerline of the sensor is greater than half a width of the first emitter chamber along a length parallel to the centerline of the sensor.

[0162] In some configurations, the first emitter group includes two or more LEDs and wherein the second emitter group includes two or more LEDs.

[0163] In some configurations, the first emitter group includes three or more LEDs and wherein the

second emitter group includes three or more LEDs.

[0164] In some configurations, the first emitter group includes four or more LEDs and wherein the second emitter group includes four or more LEDs.

[0165] In some configurations, the first emitter group includes five LEDs and wherein the second emitter group includes five LEDs.

[0166] In some configurations, each LED in the first emitter group emits light at a different wavelength than each of the other LEDs in the first emitter group.

[0167] In some configurations, each LED in the second emitter group emits light at a different wavelength than each of the other LEDs in the second emitter group.

[0168] In some configurations, the first emitter group is configured to emit a same plurality of light wavelengths as the second emitter group.

[0169] In some configurations, an arrangement of the LEDs of the first emitter group on the substrate mirrors an arrangement of the LEDs of the second emitter group on the substrate across a centerline of the sensor that bisects the sensor.

[0170] In some configurations, the optical center of the substrate is located on the centerline of the sensor.

[0171] In some configurations, the centerline of the sensor bisects at least one of the plurality of detectors.

[0172] In some configurations, a distance from the first emitter group to the at least one of the plurality of detectors is substantially similar to a distance from the second emitter group to the at least one of the plurality of detectors.

[0173] In some configurations, the sensor further comprises an ECG sensor, and the ECG sensor comprises a reference electrode, a negative electrode, and a positive electrode and the reference and negative electrodes are located on the sensor and the positive electrode is located on a housing of the wearable device.

[0174] In some configurations, the reference and negative electrodes are releasably coupled to the substrate of the sensor via one or more springs configured to bias the reference and negative electrodes away from the substrate.

[0175] In some configurations, the reference electrode extends along a circumference of the sensor on a first side of the sensor, and wherein the negative electrode extends along a circumference of the sensor on a second side of the sensor.

[0176] In some configurations, the reference electrode is substantially semi-annular and wherein the negative electrode is substantially semi-annular.

[0177] In some configurations, the reference and negative electrodes surround the plurality of detectors.

[0178] In some configurations, the substrate comprises a conductive liquid adhesive configured to facilitate a conductive electrical connection between the electrodes of the ECG sensor and the substrate.

[0179] In some configurations, the substrate comprises a printed circuit board (PCB).

[0180] In some configurations, the substrate comprises a reflective surface.

[0181] In some configurations, the sensor further comprises a first temperature sensor positioned on the substrate adjacent to the first emitter group of LEDs, and a second temperature sensor adjacent to the second emitter group of LEDs.

[0182] In some configurations, the first emitter group of LEDs is configured to emit light of at least four wavelengths and wherein the second emitter group of LEDs is configured to emit light of at least four wavelengths.

[0183] In some configurations, the four wavelengths emitted by the first emitter group of LEDs are the same wavelengths as the four wavelengths emitted by the second emitter group of LEDs.

[0184] In some configurations, the first emitter group of LEDs comprise LEDs of a same type as the second emitter group of LEDs.

[0185] In some configurations, the first emitter group of LEDs comprises: a first LED configured to emit light of a first wavelength, a second LED configured to emit light of a second wavelength, a third LED configured to emit light of a third wavelength, and a fourth LED configured to emit light of a fourth wavelength, and the second emitter group of LEDs comprises: a fifth LED configured to emit light of the first wavelength, a sixth LED configured to emit light of the second wavelength, a seventh LED configured to emit light of the third wavelength, and an eighth LED configured to emit light of the fourth wavelength.

[0186] In some configurations, the first LED is located on the PCB at a location that mirrors, across a centerline of the sensor, a location of the fifth LED on the PCB; the second LED is located on the PCB at a location that mirrors, across the centerline of the sensor, a location of the sixth LED on the PCB; the third LED is located on the PCB at a location that mirrors, across the centerline of the sensor, a location of the seventh LED on the PCB; and the fourth LED is located on the PCB at a location that mirrors, across the centerline of the sensor, a location of the eighth LED on the PCB.

[0187] In some configurations, the first and fifth LEDs are each located a same distance from at least one of the plurality of detectors; the second and sixth LEDs are each located a same distance from at least one of the plurality of detectors; the third and seventh LEDs are each located a same distance from at least one of the plurality of detectors; and the fourth and eighth LEDs are each located a same distance from at least one of the plurality of detectors.

[0188] In some configurations, light emitted from the first emitter group of LEDs travels along a light path to a first detector of the plurality of detectors that is substantially similar in length to a light path along which light emitted from the second emitter group of LEDs travels to the first detector.

[0189] In some configurations, the plurality of detectors includes a first detector group and a second detector group.

[0190] In some configurations, the first detector group includes at least two detectors housed in respective detector chambers, and wherein the second detector group includes at least two detectors housed in respective detector chambers.

[0191] In some configurations, a distance between the first emitter group and the first detector group is greater than a distance between the second emitter group and the first detector group.

[0192] In some configurations, a distance between the first emitter group and a first detector of the first detector group is substantially similar as a distance between the first emitter group and a second detector of the first detector group.

[0193] In some configurations, a distance between the first emitter group and the first detector group is substantially similar to a distance between the second emitter group and the second detector group.

[0194] In some configurations, light emitted from the first emitter group travels to the first detector group along a light path that is shorter than a light path along which light emitted from the second emitter group travels to the first detector group.

[0195] In some configurations, light emitted from the second emitter group and detected by first detector group penetrates deeper into a tissue of the wearer than light emitted from the first emitter group and detected by first detector group.

[0196] In some configurations, the sensor further comprises a light barrier construct mounted on the substrate and configured to isolate the first emitter group, the second emitter group, and the plurality of detectors, wherein the light barrier construct comprises a plurality of light barriers defining one or more chambers.

[0197] In some configurations, the one or more chambers includes a first emitter chamber configured to house the first emitter group of light emitting diodes (LEDs).

[0198] In some configurations, the one or more chambers includes a second emitter chamber configured to house the second emitter group of light emitting diodes (LEDs).

[0199] In some configurations, the one or more chambers includes a plurality of detector chambers

configured to house the plurality of detectors.

[0200] In some configurations, the light barrier construct is further configured to prevent the transmission of light therethrough.

[0201] In some configurations, the sensor further comprises a convex surface configured to be positioned between the tissue of the wearer and the emitter and detector chambers, the convex surface comprising at least a portion of the light barrier construct.

[0202] In some configurations, the convex surface further includes one or more lenses defining a surface of the emitter and detector chambers, wherein the one more lenses are further configured to transmit light.

[0203] In some configurations, the one or more lenses include polycarbonate and are configured to diffuse light.

[0204] In some configurations, the emitter and detector chambers enclose light diffusing material or encapsulant.

[0205] In some configurations, the light diffusing material or encapsulant includes glass beads or microspheres.

[0206] In some configurations, a height of the convex surface from the optical center of the substrate is between about 2.80 mm and about 2.90 mm.

[0207] In some configurations, a height of the convex surface from the optical center of the substrate is between about 2.55 mm and about 2.65 mm.

[0208] In some configurations, a centerline of the sensor bisects a first detector of the plurality of detectors, and wherein said first detector is configured to generate, responsive to detecting light emitted from the first and second emitter groups, one or more signals for calibrating or normalizing a physiological parameter measurement of the sensor.

[0209] In some configurations, the sensor processor is further configured to calibrate or normalize the physiological parameter measurement of the sensor based at least in part on comparing a first signal generated by the first detector responsive to light emitted from the first emitter group with a second signal generated by the first detector responsive to light emitted from the second emitter group.

[0210] In some configurations, the sensor further comprises a sensor processor configured to drive the LEDs of the first and second emitter groups at varying intensities.

[0211] In some configurations, the sensor processor is configured to drive the intensities of the LEDs of the first and second emitter groups based, at least in part, on one or more of an attenuation of emitted light or a temperature of the LEDs of the first and second emitter groups.

[0212] In some configurations, the sensor further comprises a sensor processor configured to activate or deactivate some of the detectors of the plurality of detectors.

[0213] The present disclosure provides an optical physiological sensor integrated into a wearable device. The optical physiological sensor can comprise: a first emitter group of light emitting diodes (LEDs) mounted on a substrate; a second emitter group of LEDs mounted on the substrate and spaced away from the first emitter group of LEDs, wherein the first emitter group is isolated from the second emitter group; and at least six detectors mounted on the substrate and arranged in a spatial configuration that surrounds the first and the second emitter group and isolated from each other, the first emitter group, and the second emitter group.

[0214] In some configurations, the spatial configuration includes an annulus.

[0215] In some configurations, the annulus has a same geometric center as the substrate or as the sensor.

[0216] In some configurations, the first and second emitter groups are located at a central location of the substrate.

[0217] In some configurations, the first and second emitter groups are located adjacent to a geometric center of the substrate or a geometric center of the sensor.

[0218] In some configurations, the sensor further comprises a light barrier construct mounted on

the substrate and configured to isolate the first emitter group, the second emitter group, and the at least six detectors.

[0219] In some configurations, the sensor further comprises an ECG sensor, the ECG sensor comprises a reference electrode, a negative electrode, and a positive electrode and the reference and negative electrodes are located on the sensor and the positive electrode is located on a housing of the wearable device.

[0220] In some configurations, the reference and negative electrodes are releasably coupled to the substrate of the sensor via one or more springs configured to bias the reference and negative electrodes away from the substrate.

[0221] In some configurations, the reference electrode extends along a circumference of the sensor on a first side of the sensor, and wherein the negative electrode extends along a circumference of the sensor on a second side of the sensor.

[0222] In some configurations, the reference electrode is substantially semi-annular and the negative electrode is substantially semi-annular.

[0223] In some configurations, the reference and negative electrodes surround the plurality of detectors.

[0224] In some configurations, the substrate comprises a conductive liquid adhesive configured to facilitate a conductive electrical connection between the electrodes of the ECG sensor and the substrate.

[0225] In some configurations, the substrate comprises a printed circuit board (PCB).

[0226] In some configurations, the substrate comprises a reflective surface.

[0227] In some configurations, the first emitter group includes five LEDs and wherein the second emitter group includes five LEDs.

[0228] In some configurations, the sensor further comprises a first temperature sensor positioned on the substrate adjacent to the first emitter group of LEDs, and a second temperature sensor adjacent to the second emitter group of LEDs.

[0229] In some configurations, a centerline of the sensor bisects a first detector of the plurality of detectors, and said first detector is configured to generate, responsive to detecting light emitted from the first and second emitter groups, one or more signals for calibrating or normalizing a physiological parameter measurement of the sensor.

[0230] In some configurations, the sensor processor is further configured to calibrate or normalize the physiological parameter measurement of the sensor based at least in part on comparing a first signal generated by the first detector responsive to light emitted from the first emitter group with a second signal generated by the first detector responsive to light emitted from the second emitter group.

[0231] In some configurations, the sensor further comprises a sensor processor configured to drive the LEDs of the first and second emitter groups at varying intensities.

[0232] In some configurations, the sensor processor is configured to drive the intensities of the LEDs of the first and second emitter groups based, at least in part, on one or more of an attenuation of emitted light or a temperature of the LEDs of the first and second emitter groups.

[0233] In some configurations, the sensor further comprises a sensor processor configured to activate or deactivate some of the detectors of the plurality of detectors.

[0234] The present disclosure provides an optical physiological sensor integrated into a wearable device. The optical physiological sensor can comprise: a first emitter group of light emitting diodes (LEDs) mounted on a substrate; a second emitter group of LEDs mounted on the substrate and spaced away from the first emitter group of LEDs; and a plurality of detectors mounted on the substrate and arranged in an annular configuration that surrounds the first and the second emitter group, the detectors can be substantially rectangular having a first side of a length and a second side of a width, wherein the detectors are oriented on the substrate such that the first side of each detector having the length is substantially orthogonal to a radius of the annular configuration.

[0235] The present disclosure provides an optical physiological sensor integrated into a wearable device. The optical physiological sensor can comprise: a first emitter group of light emitting diodes (LEDs) mounted on a substrate; a second emitter group of LEDs mounted on the substrate and spaced away from the first emitter group of LEDs, wherein the LEDs of the first emitter group are arranged on the substrate to mirror, across a centerline of the sensor, an arrangement of the LEDs of the second emitter group on the substrate; and a plurality of detectors mounted on the substrate and arranged in a spatial configuration that surrounds the first and the second emitter group.

[0236] In some configurations, the centerline of the sensor bisects at least one of the plurality of detectors.

[0237] In some configurations, the first emitter group of LEDs is configured to emit light of at least four wavelengths and wherein the second emitter group of LEDs is configured to emit light of at least four wavelengths.

[0238] In some configurations, the four wavelengths emitted by the first emitter group of LEDs are the same wavelengths as the four wavelengths emitted by the second emitter group of LEDs.

[0239] In some configurations, the first emitter group of LEDs comprise LEDs of a same type as the second emitter group of LEDs.

[0240] In some configurations, the first emitter group of LEDs comprise: a first LED configured to emit light of a first wavelength, a second LED configured to emit light of a second wavelength, a third LED configured to emit light of a third wavelength, and a fourth LED configured to emit light of a fourth wavelength, and the second emitter group of LEDs comprises: a fifth LED configured to emit light of the first wavelength, a sixth LED configured to emit light of the second wavelength, a seventh LED configured to emit light of the third wavelength, and an eighth LED configured to emit light of the fourth wavelength.

[0241] In some configurations, the first LED is located on the PCB at a location that mirrors, across a centerline of the sensor, a location of the fifth LED on the PCB; and the second LED is located on the PCB at a location that mirrors, across the centerline of the sensor, a location of the sixth LED on the PCB; the third LED is located on the PCB at a location that mirrors, across the centerline of the sensor, a location of the seventh LED on the PCB; and the fourth LED is located on the PCB at a location that mirrors, across the centerline of the sensor, a location of the eighth LED on the PCB.

[0242] In some configurations, the first and fifth LEDs are each located a same distance from at least one of the plurality of detectors; the second and sixth LEDs are each located a same distance from at least one of the plurality of detectors; the third and seventh LEDs are each located a same distance from at least one of the plurality of detectors; and the fourth and eighth LEDs are each located a same distance from at least one of the plurality of detectors.

[0243] In some configurations, light emitted from the first emitter group of LEDs travels along a light path to a first detector of the plurality of detectors that is substantially similar in length to a light path along which light emitted from the second emitter group of LEDs travels to the first detector.

[0244] In some configurations, a centerline of the sensor bisects a first detector of the plurality of detectors, and wherein said first detector is configured to generate, responsive to detecting light emitted from the first and second emitter groups, one or more signals for calibrating or normalizing a physiological parameter measurement of the sensor.

[0245] In some configurations, the sensor processor is further configured to calibrate or normalize the physiological parameter measurement of the sensor based at least in part on comparing a first signal generated by the first detector responsive to light emitted from the first emitter group with a second signal generated by the first detector responsive to light emitted from the second emitter group.

[0246] In some configurations, the sensor further comprises a sensor processor configured to drive the LEDs of the first and second emitter groups at varying intensities.

[0247] In some configurations, the sensor processor is configured to drive the intensities of the

LEDs of the first and second emitter groups based, at least in part, on one or more of an attenuation of emitted light or a temperature of the LEDs of the first and second emitter groups.

[0248] In some configurations, the sensor further comprises a sensor processor configured to activate or deactivate some of the detectors of the plurality of detectors.

[0249] The present disclosure provides an optical physiological sensor integrated into a wearable device. The optical physiological sensor can comprise: a first emitter group of light emitting diodes (LEDs) mounted on a substrate; a second emitter group of LEDs mounted on the substrate and spaced away from the first emitter group of LEDs, wherein the second emitter group is isolated from the first emitter group; and a plurality of detectors mounted on the substrate, wherein the plurality of detectors is arranged in a spatial configuration that surrounds the first and the second emitter group and isolated from each other, wherein the plurality of detectors includes a first detector group and a second detector group, and the first emitter group can be proximal to the first detector group and distal to the second detector group, and the second emitter group can be proximal to the second detector group and distal to the first detector group.

[0250] In some configurations, the first detector group includes at least two detectors housed in respective detector chambers, and wherein the second detector group includes at least two detectors housed in respective detector chambers.

[0251] In some configurations, a distance between the first emitter group and the first detector group is greater than a distance between the second emitter group and the first detector group.

[0252] In some configurations, a distance between the first emitter group and a first detector of the first detector group is substantially similar as a distance between the first emitter group and a second detector of the first detector group.

[0253] In some configurations, a distance between the first emitter group and the first detector group is substantially similar to a distance between the second emitter group and the second detector group.

[0254] In some configurations, light emitted from the first emitter group travels to the first detector group along a light path that is shorter than a light path along which light emitted from the second emitter group travels to the first detector group.

[0255] In some configurations, light emitted from the second emitter group and detected by first detector group penetrates deeper into a tissue of the wearer than light emitted from the first emitter group and detected by first detector group.

[0256] In some configurations, a centerline of the sensor bisects a first detector of the plurality of detectors, and wherein said first detector is configured to generate, responsive to detecting light emitted from the first and second emitter groups, one or more signals for calibrating or normalizing a physiological parameter measurement of the sensor.

[0257] In some configurations, the sensor processor is further configured to calibrate or normalize the physiological parameter measurement of the sensor based at least in part on comparing a first signal generated by the first detector responsive to light emitted from the first emitter group with a second signal generated by the first detector responsive to light emitted from the second emitter group.

[0258] In some configurations, the sensor further comprises a sensor processor configured to drive the LEDs of the first and second emitter groups at varying intensities.

[0259] In some configurations, the sensor processor is configured to drive the intensities of the LEDs of the first and second emitter groups based, at least in part, on one or more of an attenuation of emitted light or a temperature of the LEDs of the first and second emitter groups.

[0260] In some configurations, the sensor further comprises a sensor processor configured to activate or deactivate some of the detectors of the plurality of detectors.

[0261] The present disclosure provides an optical physiological sensor integrated into a wearable device. The optical physiological sensor can comprise: first and second emitter groups of light emitting diodes mounted on a substrate; a first emitter chamber housing the first emitter group of

light emitting diodes (LEDs); a second emitter chamber housing the second emitter group of LEDs wherein the first emitter group is isolated from the second emitter group; a plurality of detector chambers housing detectors mounted on the substrate and arranged in a spatial configuration that surrounds the first and the second emitter chambers; a light barrier construct mounted on the substrate and including one or more light barriers configured to prevent the transmission of light therethrough, wherein the light barriers of the light barrier construct isolate the emitter chambers and the detector chambers; and a convex surface configured to be positioned between the tissue of the wearer and the emitter and detector chambers, the convex surface comprising at least a portion of the light barrier construct.

[0262] In some configurations, the convex surface further includes one or more lenses defining a surface of the emitter and detector chambers, wherein the one more lenses are further configured to transmit light.

[0263] In some configurations, the one or more lenses include polycarbonate and are configured to diffuse light.

[0264] In some configurations, the emitter and detector chambers enclose light diffusing material or encapsulant.

[0265] In some configurations, the light diffusing material or encapsulant includes glass beads or microspheres.

[0266] In some configurations, a height of the convex surface from a center of the substrate is between about 2.80 mm and about 2.90 mm.

[0267] In some configurations, a height of the convex surface from a center of the substrate is between about 2.55 mm and about 2.65 mm.

[0268] In some configurations, a centerline of the sensor bisects a first detector of the detectors, and wherein said first detector is configured to generate, responsive to detecting light emitted from the first and second emitter groups, one or more signals for calibrating or normalizing a physiological parameter measurement of the sensor.

[0269] In some configurations, the sensor processor is further configured to calibrate or normalize the physiological parameter measurement of the sensor based at least in part on comparing a first signal generated by the first detector responsive to light emitted from the first emitter group with a second signal generated by the first detector responsive to light emitted from the second emitter group.

[0270] In some configurations, the sensor further comprises a sensor processor configured to drive the LEDs of the first and second emitter groups at varying intensities.

[0271] In some configurations, the sensor processor is configured to drive the intensities of the LEDs of the first and second emitter groups based, at least in part, on one or more of an attenuation of emitted light or a temperature of the LEDs of the first and second emitter groups.

[0272] In some configurations, the sensor further comprises a sensor processor configured to activate or deactivate some of the detectors of the plurality of detectors.

[0273] The present disclosure provides an optical physiological sensor integrated into a wearable device. The optical physiological sensor can comprise: a circuit layer; a light barrier construct positioned proximate said circuit layer, the light barrier construct comprising a plurality of openings defining at least a first emitter chamber, a second emitter chamber, and a plurality of detector chambers. The first and second emitter chambers can be spaced from one another and are spaced inward from and surrounded by the plurality of detector chambers; and the first emitter chamber, the second emitter chamber, and the plurality of detector chambers can be isolated from one another by portions of the light barrier construct. In some configurations, the sensor comprises: a first plurality of emitters mounted on said circuit layer and positioned within the first emitter chamber of the light barrier construct; a second plurality of emitters mounted on said circuit layer and positioned within the second emitter chamber of the light barrier construct; and a plurality of detectors, each of the plurality of detectors mounted on said circuit layer and positioned within a

different one of the plurality of detector chambers of the light barrier construct. In some configurations, each of the plurality of detector chambers includes only one detector.

[0274] In some configurations, the sensor further comprises a first electrocardiogram (ECG) electrode and a second ECG electrode, and wherein the first and second ECG electrodes partially surround the first plurality of emitters, second plurality of emitters, and plurality of detectors.

[0275] In some configurations, each of the first and second ECG electrodes comprises a semi-annular shape having a first end and a second end; and the first ends of the first and second ECG electrodes are separated from one another by a first gap and the second ends of the first and second ECG electrodes are separated from one another by a second gap.

[0276] In some configurations, the light barrier construct further comprises: a first recess sized and shaped to receive the first ECG electrode; and a second recess sized and shaped to receive the second ECG electrode; each of the first and second recesses comprises a semi-annular shape having a first end and a second end; the first ends of the first and second recesses are separated from one another by said first gap and the second ends of the first and second recesses are separated from one another by said second gap; and the first and second recesses partially surround the first emitter chamber, second emitter chamber, and plurality of detector chambers.

[0277] In some configurations, the plurality of detector chambers are spaced inward from the first and second recesses.

[0278] In some configurations, the sensor does not include any emitters positioned outward from the first and second ECG electrodes.

[0279] In some configurations, the sensor does not include any detectors positioned outward from the first and second ECG electrodes.

[0280] In some configurations, the first and second gaps are substantially equal.

[0281] In some configurations, first and second ends of the first and second ECG electrodes comprise a rounded shape.

[0282] In some configurations, each of the first and second gaps are between approximately 1.5 mm and approximately 1.75 mm.

[0283] In some configurations, each of the first and second gaps are between approximately 0.5 mm and approximately 0.75 mm.

[0284] In some configurations, the sensor does not include any emitters positioned within any of the plurality of detector chambers.

[0285] In some configurations, the sensor does not include any detectors positioned within any of the first and second emitter chambers.

[0286] In some configurations, the plurality of detector chambers are arranged in a circular configuration around said first and second emitter chambers.

[0287] In some configurations, said sensor does not include any emitters other than the first and second plurality of emitters.

[0288] In some configurations, said sensor does not include any detectors other than the plurality of detectors.

[0289] In some configurations, said circuit layer comprises a printed circuit board.

[0290] In some configurations, each of the first and second plurality of emitters comprises a light emitting diode (LED).

[0291] In some configurations, each of the plurality of detectors comprises a photodiode.

[0292] In some configurations, each of the first and second plurality of emitters are configured to emit light into tissue of a user; each of the plurality of detectors are configured to: (i) detect at least a portion of light emitted by at least one of the first and second plurality of emitters after attenuation by said tissue; and (ii) output at least one signal responsive to detected light; and the sensor further comprises one or more processors configured to receive and process the outputted at least one signal and determine at least one physiological parameter of the user.

[0293] The present disclosure provides a system for monitoring physiological parameters. The

system can comprise: a wearable device comprising one or more physiological sensors configured to monitor one or more physiological parameters of a user; and a computing device external to the wearable device and in wireless communication with the wearable device, the computing device configured to: receive, via an interactive user interface, user input; in response to receiving the user input, wirelessly transmit a first signal to the wearable device to instruct the wearable device to perform a physiological monitoring operation; receive, from the wearable device, physiological parameter data measured by the physiological monitoring operation; and cause presentation of the physiological parameter data in the interactive user interface.

[0294] In some configurations, the monitoring operation comprises continuous measuring of the physiological parameters.

[0295] In some configurations, the monitoring operation comprises: measuring the physiological parameters for a predetermined length of time; and ceasing to measure the physiological parameters upon expiration of the predetermined length of time.

[0296] In some configurations, the wearable device is configured to: in response to receiving the first signal from the computing device, determine a power level of a battery of the wearable device; determine whether the power level is sufficient to perform the monitoring operation; and in response to determining that the power level is sufficient, perform the monitoring operation.

[0297] In some configurations, the wearable device is configured to: in response to receiving the first signal from the computing device, determine a power level of a battery of the wearable device; determine whether the power level is sufficient to perform the monitoring operation; in response to determining that the power level is not sufficient, determine a modified monitoring operation based at least in part on the power level; and perform the modified monitoring operation.

[0298] In some configurations, the wearable device is configured to: determine that one or more physiological parameters exceed a respective predetermined threshold; and in response to determining that the one or more physiological parameters exceed the respective predetermined threshold, transmit a second signal to the computing device to cause the computing device to generate an alarm.

[0299] In some configurations, the computing device is further configured to: cause presentation, in the interactive user interface, of one or more of a graph, chart, or trend of historical and substantially real-time physiological parameter data.

[0300] It is noted that “plethysmograph” as used herein (commonly referred to as “photoplethysmograph”), encompasses its broad ordinary meaning known to one of skill in the art, which includes at least data representative of a change in the absorption of particular wavelengths of light as a function of the changes in body tissue resulting from pulsing blood. Moreover, “oximetry” as used herein encompasses its broad ordinary meaning known to one of skill in the art, which includes at least those noninvasive procedures for measuring parameters of circulating blood through spectroscopy.

[0301] For purposes of summarization, certain aspects, advantages and novel features are described herein. Of course, it is to be understood that not necessarily all such aspects, advantages or features need to be present in any particular aspect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0302] The drawings and the associated descriptions are provided to illustrate aspects of the disclosure and not to limit the scope of the claims. In the present disclosure, “bottom” refers to the side facing a wearer's wrist when an example wearable device disclosed herein is worn on the wearer's wrist and “top” refers to the side facing away from the wearer's wrist.

[0303] FIG. 1A illustrates a first view of an example wearable device including a physiological

parameter measurement sensor or module worn on a wrist using straps.

[0304] FIG. 1B illustrates a second view of the example wearable device of FIG. 1A worn on the wrist.

[0305] FIG. 1C illustrates an example fingertip sensor that can be coupled to the wearable device of the present disclosure.

[0306] FIG. 1D illustrates a top perspective view of the example wearable device of FIGS. 1A-1C with a partial view of the straps.

[0307] FIG. 1E illustrates a bottom perspective view of the example wearable device of FIG. 1D.

[0308] FIG. 1F illustrates a side view of an example wearable device without the straps when the device is interfacing with a wearer's skin.

[0309] FIG. 1G illustrates a top perspective view of the example wearable device of FIG. 1F.

[0310] FIG. 1H illustrates a bottom perspective view of an example wearable device.

[0311] FIG. 1I illustrates a perspective view of an example strap configured to secure the wearable device disclosed herein to a wearer's wrist.

[0312] FIGS. 1J-1K illustrate perspective views of an example wearable device and physiological sensor or module removably secured to a wearable apparatus.

[0313] FIG. 2 is a diagram illustrating schematically a network of non-limiting examples of devices that can communicate with the wearable device disclosed herein.

[0314] FIG. 3 illustrates an example display of physiological parameter measurements on the wearable device disclosed herein.

[0315] FIG. 4 illustrates an example physiological parameter measurement module of the wearable device.

[0316] FIG. 5A illustrates a side view of an example wearable device incorporating an example physiological parameter measurement module.

[0317] FIG. 5B illustrates a cross-sectional view of the example wearable device of FIG. 5A.

[0318] FIG. 5C illustrates a perspective view of the wearable device of FIG. 5A.

[0319] FIG. 5D illustrates a bottom view of the wearable device of FIG. 5A.

[0320] FIG. 6 illustrates schematically arteries and capillaries of a human hand and a proximal portion of a human forearm.

[0321] FIG. 7A illustrates a schematic system diagram of a wearable device including a physiological parameter measurement module.

[0322] FIG. 7B illustrates a partially exploded view of an example wearable device.

[0323] FIG. 7C illustrates an example light transmissive cover of the physiological parameter measurement module of FIG. 7B.

[0324] FIG. 7D illustrates an exploded view of ECG electrodes, light transmissive cover(s), and an opaque frame of the physiological parameter measurement module of FIG. 7B.

[0325] FIG. 7E illustrates a bottom perspective view of a physiological parameter measurement module incorporating the ECG electrodes, light transmissive cover(s), and an opaque frame of FIG. 7C or 7D.

[0326] FIG. 7F illustrates a top perspective view of the example physiological parameter measurement module of FIG. 7E.

[0327] FIGS. 7G and 7H illustrate schematically top and bottom views of an example device processor board of the wearable device disclosed herein.

[0328] FIGS. 8A and 8B illustrate schematically top and bottom views of an example sensor or module processor board of an example physiological parameter measurement module.

[0329] FIGS. 8C-8E illustrate various views of bonding of detectors to a PCB substrate of a physiological parameter measurement module.

[0330] FIG. 8F illustrates a perspective view of a PCB substrate of a physiological parameter measurement module with different wire bonding arrangements than shown in FIGS. 8C-8E.

[0331] FIGS. 9A and 9B illustrate light diffusing material fill channels and air venting channels in

a opaque frame of an example physiological parameter measurement module.

[0332] FIG. **10** illustrates a longitudinal cross-sectional view of an example physiological parameter measurement module and example light paths between emitters and detectors of the module.

[0333] FIG. **11A** illustrates a schematic system diagram of an example wearable device including a physiological parameter measurement module.

[0334] FIG. **11B** illustrate a schematic diagram of an example device processor shown in FIG. **11A**.

[0335] FIG. **11C** illustrates a schematic system diagram of an example sensor or module processor shown in FIG. **11A**.

[0336] FIG. **11D** illustrates a block diagram of an example front end circuitry of the sensor or module processor of FIG. **11C**.

[0337] FIG. **12A** illustrates a bottom view of an example physiological parameter measurement module with first and second ECG electrodes.

[0338] FIG. **12B** illustrates a top perspective view of the example wearable device including a third ECG electrode.

[0339] FIG. **12C** illustrates a partial top perspective view of the example wearable device of FIG. **12B** with the third ECG electrode shown as transparent to illustrate contact springs underneath the third ECG electrode.

[0340] FIG. **13A** illustrates an example block diagram of LED drive circuitry of the physiological parameter measurement module disclosed herein.

[0341] FIG. **13B** illustrates an example block diagram of emitters circuitry of the physiological parameter measurement module disclosed herein.

[0342] FIG. **13C** illustrates an example block diagram of detectors circuitry of the physiological parameter measurement module disclosed herein.

[0343] FIG. **13D** illustrates an example block diagram of temperature sensors circuitry of the physiological parameter measurement module disclosed herein.

[0344] FIGS. **14A** and **14B** are example block diagrams illustrating signal processing of a conventional plethysmograph sensor.

[0345] FIGS. **15A** and **15B** illustrate example schematic input and output flow diagrams of a physiological parameter measurement module disclosed herein.

[0346] FIG. **15C** illustrates an example schematic input and output flow diagram of the gyroscope and accelerometer of a physiological parameter measurement module disclosed herein.

[0347] FIG. **15D** illustrates an example schematic block diagram for determining pulse rate using a physiological parameter measurement module disclosed herein.

[0348] FIG. **15E** illustrates an example decision logic for determining pulse rate using a physiological parameter measurement module disclosed herein.

[0349] FIG. **15F** illustrates an example schematic input and output flow diagram for determining oxygen saturation using a physiological parameter measurement module disclosed herein.

[0350] FIG. **15G** illustrates an example decision logic for determining oxygen saturation using a physiological parameter measurement module disclosed herein.

[0351] FIG. **16A** illustrates schematically an example plethysmograph sensor arrangement on a sensor or module processor board of a physiological parameter measurement module of a wearable device.

[0352] FIG. **16B** illustrates a bottom view of an example physiological parameter measurement module incorporating the plethysmograph sensor arrangement of FIG. **16A**.

[0353] FIG. **16C** illustrates a side view of the example physiological parameter measurement module of FIG. **16B**.

[0354] FIG. **16D** illustrates a bottom perspective view of the example physiological parameter measurement module of FIG. **16B**.

[0355] FIG. **16E** illustrates a bottom view of a variation of the example physiological parameter

measurement module of FIG. 16B including ECG electrodes.

[0356] FIG. 16F illustrates a side view of the example physiological parameter measurement module of FIG. 16E.

[0357] FIG. 16G illustrates a bottom perspective view of the example physiological parameter measurement module of FIG. 16E with the opaque frame and light transmissive cover hidden to show ECG electrodes assembled with the sensor or module processor board.

[0358] FIG. 17A illustrates a bottom perspective view of an example physiological parameter measurement module incorporating the plethysmograph sensor arrangement of FIG. 16A.

[0359] FIG. 17B illustrates a bottom view of the example physiological parameter measurement module of FIG. 17A.

[0360] FIG. 17C illustrates a side view of the example physiological parameter measurement module of FIG. 17A.

[0361] FIG. 18A illustrates schematically an example plethysmograph sensor arrangement on a sensor or module processor board of a physiological parameter measurement module of a wearable device.

[0362] FIG. 18B illustrate schematically an example plethysmograph sensor arrangement on a sensor or module processor board of a physiological parameter measurement module of a wearable device.

[0363] FIG. 19A illustrate schematically an example plethysmograph sensor arrangement on a sensor or module processor board of a physiological parameter measurement module of a wearable device.

[0364] FIG. 19B illustrates a bottom view of an example physiological parameter measurement module incorporating the plethysmograph sensor arrangement of FIG. 19A.

[0365] FIG. 19C illustrates a side view of the physiological parameter measurement module of FIG. 19B.

[0366] FIG. 20A illustrates a bottom view of an example physiological parameter measurement module of a wearable device as worn on a schematic representation of a wearer's wrist.

[0367] FIG. 20B illustrates a side view of the physiological parameter measurement module of FIG. 20A.

[0368] FIGS. 20C and 20D illustrate exploded views of the physiological parameter measurement module of FIG. 20A.

[0369] FIG. 20E illustrates a first side view of an example wearable device incorporating the physiological parameter measurement module of FIGS. 20A-20D.

[0370] FIG. 20F illustrates a bottom view of the wearable device of FIG. 20E.

[0371] FIG. 20G illustrates a second side view of the wearable device of FIG. 20E.

[0372] FIG. 20H illustrates a third side view of the wearable device of FIG. 20E.

[0373] FIG. 20I illustrates a bottom perspective view of the wearable device of FIG. 20E.

[0374] FIG. 20J illustrates a top perspective view of the wearable device of FIG. 20E.

[0375] FIGS. 21A and 21B illustrate perspective views of an example physiological parameter measurement module with alternative light transmissive cover curvatures from the module in FIG. 20A.

[0376] FIG. 21C illustrates a longitudinal cross-sectional view of the physiological parameter measurement module of FIGS. 21A and 21B.

[0377] FIGS. 22A and 22B illustrate perspective views of an example physiological parameter measurement module with another alternative light transmissive cover curvatures from the module in FIG. 20A.

[0378] FIG. 22C illustrates a longitudinal cross-sectional view of the physiological parameter measurement module of FIGS. 22A and 22B.

[0379] FIG. 23A illustrates a bottom perspective view of an example wearable device incorporating the physiological parameter measurement module of FIGS. 20A-20D.

[0380] FIG. 23B illustrates a side view of the wearable device of FIG. 23A.

[0381] FIG. 23C illustrates a top perspective view of the wearable device of FIG. 23A.

[0382] FIG. 23D illustrates a top view of the wearable device of FIG. 23A.

[0383] FIG. 23E illustrates a bottom view of the wearable device of FIG. 23A.

[0384] FIG. 24A illustrates a bottom view of another example physiological parameter measurement module of a wearable device.

[0385] FIG. 24B illustrates a side view of the physiological parameter measurement module of FIG. 24A.

[0386] FIG. 25A illustrates a bottom view of another example physiological parameter measurement module of a wearable device.

[0387] FIG. 25B illustrates a side view of the physiological parameter measurement module of FIG. 25A.

[0388] FIG. 25C illustrates a first side view of another example wearable device incorporating the physiological parameter measurement module of FIGS. 25A-25B.

[0389] FIG. 25D illustrates a bottom view of the wearable device of FIG. 25C.

[0390] FIG. 25E illustrates a second side view of the wearable device of FIG. 25C.

[0391] FIG. 25F illustrates a top perspective view of the wearable device of FIG. 25C.

[0392] FIG. 25G illustrates a third side view of the wearable device of FIG. 25C.

[0393] FIG. 25H illustrates a bottom perspective view of the wearable device of FIG. 25C.

[0394] FIG. 26A illustrates schematically a microneedle inserted into skin of a wearer.

[0395] FIG. 26B illustrates schematically a microneedle patch coupled to a body of the wearable device disclosed herein.

[0396] FIG. 26C illustrates schematically a microneedle patch coupled to a strap of the wearable device disclosed herein.

[0397] FIG. 26D illustrates schematically a simplified system diagram of the microneedle patch and the wearable device.

[0398] FIG. 27A illustrates a front view of an example aspect of a physiological parameter measurement sensor or module.

[0399] FIG. 27B illustrates an exploded view of an example aspect of a physiological parameter measurement sensor or module.

[0400] FIGS. 27C and 27D illustrate light diffusing material fill channels and air venting channels in an opaque frame of an example physiological parameter measurement sensor or module.

[0401] FIG. 27E illustrates an example physiological parameter measurement sensor or module and example light paths between emitters and detectors of the module.

[0402] FIG. 27F illustrates a perspective view of PCB substrate of a physiological parameter measurement sensor or module with example plethysmograph sensor arrangement.

[0403] FIG. 27G illustrates a longitudinal cross-sectional view of an example physiological parameter measurement sensor or module.

[0404] FIG. 27H illustrates a longitudinal cross-sectional, that is orthogonal to the view of FIG. 27G, of an example physiological parameter measurement sensor or module.

[0405] FIG. 27I illustrates a perspective view of PCB substrate of a physiological parameter measurement sensor or module with example plethysmograph sensor arrangement.

[0406] FIG. 27J illustrates a perspective cross-sectional view of an example physiological parameter measurement sensor or module.

[0407] FIGS. 27K-27L illustrate an example physiological parameter measurement sensor or module and example light paths between emitters and detectors of the module.

[0408] FIGS. 27M-27P illustrate an example physiological parameter measurement sensor or module and example light barriers or light blocks between emitter and detector chambers of the module.

[0409] FIG. 27Q illustrates an example physiological parameter measurement sensor or module

and example light diffusing material and light transmissive lens(es) or cover(s).

[0410] FIG. **28** shows a block diagram illustrating an example aspect of the wearable device in communication with an external device via a network.

[0411] FIG. **29** illustrates an example dashboard user interface of the health application.

[0412] FIGS. **30A-30D** illustrate example spot check monitoring user interfaces of the health application.

[0413] FIGS. **30E-30H** illustrate example continuous monitoring user interfaces of the health application.

[0414] FIGS. **31A-31D** illustrate example measurement settings user interfaces of the health application.

[0415] FIG. **32** illustrates an example activity user interface of the health application.

[0416] FIGS. **33A-33C** illustrate example workout user interfaces of the health application.

[0417] FIGS. **34A-34C** illustrate example sleep user interfaces of the health application.

[0418] FIGS. **35A-35B** illustrate example history user interfaces of the health application.

[0419] FIG. **36A** illustrates an example aspect of a wearable device with a display screen.

[0420] FIGS. **36B-36E** illustrate example display screens of a wearable device.

[0421] FIG. **37A** illustrates an example aspect of a wearable device with a display screen.

[0422] FIGS. **37B-37E** illustrate example display screens of a wearable device.

[0423] FIG. **38A-38B** illustrate example display screens of a wearable device.

[0424] FIG. **39** is a flowchart illustrating an example process for performing a monitoring operation of the wearable device.

[0425] FIG. **40** illustrates a block diagram of an example computing device that may implement one or more aspects of the present disclosure, according to various aspects of the present disclosure.

DETAILED DESCRIPTION

[0426] Although certain aspects and examples are described below, those of skill in the art will appreciate that the disclosure extends beyond the specifically disclosed aspects and/or uses and obvious modifications and equivalents thereof based on the disclosure herein. Thus, it is intended that the scope of the disclosure herein disclosed should not be limited by any particular aspects described below.

Overview of Wearable Device Including a Physiological Parameter Measurement Sensor or Module

[0427] Daily use of a wearable healthcare monitoring device, which can include oximetry- or plethysmograph-based and/or ECG physiological parameters, can be beneficial to the wearer. The device, such as a device **10** as shown in FIGS. **1A-1H**, can be a wristwatch incorporating a physiological parameter measurement sensor **100** or a wrist-worn physiological parameter measurement sensor with built-in watch or time-indicating functions. The device **10** can include an adjustable strap **30**. Accordingly, the wearer needs not wear an additional sensor when going about daily activities and the appearance of the device attracts less attention from the general public so that the wearer may feel less self-conscious about wearing a pulse oximeter sensor on the wearer's body. The wearer can also connect additional sensors (for example, a fingertip plethysmograph sensor shown in FIG. **1C**) and/or other physiological monitoring devices to the wearable device to expand the functionality of the wearable device.

[0428] The wearer can be informed of physiological parameters, such as vital signs including but not limited to heart rate (or pulse rate), and oxygen saturation by the wearable device **10**. The device **10** can display one or more of the measured physiological parameters on its display **12**. The information can be helpful in providing feedback to the wearer and/or a third party user, for example, a healthcare professional or the wearer's family member, when the wearer is exercising, or otherwise for warning the wearer of possible health-related conditions, including but not limited to changes in the wearer's physiological parameters in response to medication that is being

administered to the wearer.

[0429] As shown in FIGS. 1A-1H, the wearable device **10** can be a watch, which can include a physiological parameter measurement sensor or module **100** configured to measure an indication of the wearer's physiological parameters, which can include, for example, pulse rate, respiration rate, oxygen saturation (SpO₂), Pleth Variability Index (PVI), Perfusion Index (PI), Respiration from the pleth (RRp), hydration, glucose, blood pressure, and/or other parameters. The physiological parameter measurement sensor or module **100** can be an optical sensor. Additionally, the sensor or module **100** can optionally calculate a wellness index based on more than one individual physiological parameter measured by the module and/or received by the sensor or module **100** based on externally connected sensors and/or patient monitoring devices. The sensor or module **100** can perform intermittent and/or continuous monitoring of the measured parameters. The sensor or module **100** can additionally and/or alternatively perform a spot check of the measured parameters, for example, upon request by the wearer.

[0430] As shown in FIGS. 1E and 1H, a bottom side of a device (or watch) housing **101** can include an opening sized to retain the physiological parameter measurement module **100** while still allowing the tissue-facing surface of the sensor or module **100** to be exposed. The retaining of the sensor or module **100** in the device housing **101** can be aided by any suitable retaining mechanisms. As shown in FIGS. 1F and 1H, the physiological parameter measurement module **100** can include a skin-interfacing light transmissive cover **102** that encloses a plurality of light emitters **104** (such as LEDs) and one or more photodetectors (also referred to as “detectors”) **106**. Additionally, the sensor or module **100** can optionally include an electrocardiogram (ECG) sensor, which can include a plurality of ECG electrodes **124**, **125**. As shown in FIGS. 1G and 1H, some of the ECG electrodes **125** can be located away from the sensor or module **100** and some of the ECG electrodes **124** can be located on the sensor or module **100**. The cover **102** can include a plurality of lenses or covers or a single construct of lens or cover. The physiological parameter measurement module **100** is designed to reduce noise in the signals detected by the detectors **106**, for example, by reducing mixing of the emitted light and the reflected light using light barriers that are substantially opaque. As shown in FIG. 1F, the light barrier **120** can include a first light barrier which can be placed between the emitters and the detectors of the sensor or module **100**. The first light barrier can extend (for example, entirely extend) along an inner portion of the cover **102**. The first light barrier can also suppress light emitted by the emitters at an angle. The sensor or module **100** can include additional light barriers, including for example, a side perimeter wall and additional light barriers to separate the detectors from the emitters, and/or separate different detector groups from one another.

[0431] FIG. 1F illustrates the device **10** being worn on the wrist **2** of the wearer, with the physiological parameter measurement module **100** facing the wrist **2**. The physiological parameter measurement module **100** on the device **10** is designed so as to reduce and/or eliminate a gap between a surface of the physiological parameter measurement module **100** and the wearer's skin at the measurement site where the device **10** is worn. At the wrist, if the device **10** is worn too loosely (which can be the case when the device **10** is able to slide over the skin when the device **10** is moved), the gap between the tissue-facing surface of the physiological parameter measurement module **100** and the wearer's skin can cause inaccurate measurements. This is because the gap can result in both light-piping and in the emitted light not penetrating deep enough into the wearer's tissue, for example, by going no deeper than within a top skin layer (for example, the epidermis) of the wearer's tissue, which typically does not have any blood vessels present. Therefore, light cannot reach and or interact with tissues, such as the arterial blood in the dermis, located below the top skin layer. The gap can also result in loss of the attenuated and reflected light through the gap so that less of the attenuated and reflected light can arrive at the detectors **106**.

[0432] The tightness of the device **10** on the wearer's body (for example, the wrist) can be adjusted by adjusting any suitable strap(s) **30** used to secure the device to the wearer's body. The strap(s) can

be connected to the device **10** using any suitable strap connections **22**. For example, the strap connections **22** can be compatible with third party watch bands, wearable blood pressure monitors, and/or the like. As shown in FIG. **1I**, an example strap **30** can be stretchable and evenly distribute the pressure of the device **10** around the wrist so as to provide better contact between the sensor or module **100** and the wrist **2** while not compromising the comfort of the wearer and/or reducing the blood flow across the wrist **2** in a way that reduces the accuracy of the measurement by the sensor or module **100**. As shown in FIG. **1L**, a rubber base **302** can be molded through a plurality of metal loops **304** arranged along a length of a strap **30** to form the strap **30**. The metal loops **304** can include a thin (for example, less than about 1 mm) wall of metal forming a closed loop with a through-hole in a direction generally transverse to the length (that is, along a width) of the strap **30** and perpendicular to a thickness of the strap **30**. During the overmolding process, the rubber material can fill up or substantially fill up the space in the through-hole. The metal loops **304** can be arranged in two rows along the length of the strap **30**. Alternatively, the metal loops can include a partial loop with an opening, or the strap may include more than one partial metal loop snapped onto each other around the rubber base. Additional details of the strap **30** are described in U.S. Provisional Application No. 63/068,256, filed Aug. 20, 2020 and titled "WEARABLE PHYSIOLOGICAL MONITORING DEVICE WITH ADJUSTABLE STRAPS", the entirety of which is incorporated herein by reference.

[0433] Additionally, the gap between a surface of the physiological parameter measurement module **100** and the wearer's skin at the measurement site can be reduced by the design of the light transmissive cover **102**. As shown in FIG. **1F**, a cover **102** of the physiological parameter measurement module **100** can include a convex curvature or convex protrusion on its skin-interfacing cover **102**. As will be described in greater detail below, the curvature of the cover **102** of the sensor or module **100**, which can include a plurality of lenses or covers or a single lens or cover, can be discontinuous or continuous.

[0434] As shown in FIG. **1F**, when the device **10** is worn by the wearer, the convex cover **102** can be pressed onto the skin and the tissue **2** of the wearer can conform around the convex curvature. The contact between the convex cover **102** and the tissue **2** of the wearer can leave no air gaps between the tissue **2** and the convex cover **102**. And as the emitters and/or detectors can be surrounded by a light-diffusing material (as will be described below), the physiological parameter measurement module **100** may leave no air gap between the tissue **2** and any of the emitters and/or detectors. Optionally, certain portion(s) of the cover **102** can protrude more into the skin than the remainder of the cover. The pressure exerted by the curvature of the cover **102** on the skin and/or the absence of air gap can increase a light illuminated and/or detection area, improve the optical coupling of the emitted light and the blood vessels and/or of the reflected light and the detectors, reduce light piping, and/or reduce stagnation of the blood. The cover curvature can be configured so as to balance the pressure needed to improve contact between the cover **102** and the skin, and the comfort of the wearer.

[0435] FIGS. **1J-1K** illustrate perspective views of an example wearable device **10** removably secured to a wearable apparatus **135**. The wearable apparatus **135** includes one or more straps **30** and a frame **177**. The wearable device **10** may include a physiological sensor or module **100** and an electrode **58**. The electrode may be a positive electrode. In some aspects, the wearable device **10** may include a screen or display. In some aspects, the wearable device **10** may not include a screen or display.

[0436] The device **10** may be removably secured to the frame **177**. The frame **177** may hold the device **10** within the frame **177** via force-fit, friction-fit, snap-fit, or the like. The straps **30** may secure the apparatus **135** to a body part of a wearer. The wearable apparatus **135** may be configured to hold the wearable device **10** (e.g., the sensor or module **100**) in contact with the skin of a wearer when the device **10** is secured within the frame **177** and the wearable apparatus **135** is secured to the body of a wearer. The wearable apparatus **135** (e.g., straps **30**) may be sized and/or shaped to

secure to a wrist of a wearer. In some aspects, the wearable apparatus **135** (e.g., straps **30**) may be sized and/or shaped to secure to other portions of a wearer's body, including an ankle, a leg, an arm, a finger, a chest, a torso, a stomach, a back, a neck, a head, a forehead, as non-limiting examples. [0437] The wearable device **10** can be used in a standalone manner and/or in combination with other devices and/or sensors. As shown in FIG. 2, the device **10** can connect (for example, wirelessly) with a plurality of devices, including but not limited to a patient monitor **202** (for example, a bedside monitor such as Masimo's Radical-7®, Rad-97® (optionally with noninvasive blood pressure or NomoLine capnography), and Rad-8® bedside monitors, a patient monitoring and connectivity hub such as Masimo's Root® Platform, any handheld patient monitoring devices, and any other wearable patient monitoring devices), a mobile communication device **204** (for example, a smartphone), a computer **206** (which can be a laptop or a desktop), a tablet **208**, a nurses' station system **201**, glasses such as smart glasses configured to display images on a surface of the glasses and/or the like. The wireless connection can be based on Bluetooth technology, near-field communication (NFC) technology, and/or the like. Additionally, the wearable device **10** can connect to a computing network **212** (for example, via any of the connected devices disclosed herein, or directly). The network **212** may comprise a local area network (LAN), a personal area network (PAN) a metropolitan area network (MAN), a wide area network (WAN) or the like, and may allow geographically dispersed devices, systems, databases, servers and the like to connect (e.g., wirelessly) and to communicate (e.g., transfer data) with each other. The wearable device **10** can establish connection via the network **212** to one or more electronic medical record system **214**, a remote server with a database **216**, and/or the like.

[0438] Optionally, the device **10** can be integrated with more sensors and/or configured to connect to a plurality of external sensors, wirelessly or with a connecting cable. The connecting cable can be a universal connector configured to connect to any of the medical devices and/or sensors disclosed herein to provide communication between the wearable device **10** and the connected medical devices and/or sensors. The cable can optionally include a board-in-cable device that includes its own processor, but may not include its own display.

[0439] The device **10** can act as hub for the external sensors, for example, the sensors described in U.S. Patent Publication No. 2020/0138288, published on May 7, 2020 (the entirety of which is hereby incorporated herein by reference). The sensors described in U.S. Patent Publication No. 2020/0138288 can collect patient physiological data and provide power for a reusable pairing device. The reusable pairing device can establish wireless communication with a patient monitoring device. The wearable device **10** can replace the patient monitoring device in U.S. Patent Publication No. 2020/0138288. As another example, the device **10** can replace a patient monitor device described in U.S. Patent Publication No. 2020/0329993, published on Oct. 22, 2020, the entirety of which is hereby incorporated herein by reference. By replacing the patient monitor device in U.S. Patent Publication No. 2020/0329993, the wearable device **10** performs all the computations based on the sensor data so that the connected external sensors, for example, the ECG sensors disclosed in U.S. Patent Publication No. 2020/0329993, do not require heavy computing power.

[0440] The device **10** can include open architecture to allow connection of third party wireless sensor, and/or allow third party access to a plurality of sensors on the wearable device **10** or connected to the wearable device **10**. The plurality of sensors can include, for example, a temperature sensor, an altimeter, a gyroscope, an accelerometer, emitters, LEDs, etc. Third party applications can be installed on the wearable device **10** and can use data from one or more of the sensors on the wearable device **10** and/or in electrical communication with the wearable device.

[0441] Optionally, the wearable device **10** can communicate with any other suitable noninvasive sensor, such as an acoustic sensor, a blood pressure sensor, temperature sensor, movement sensor, ECG sensor, etc. Examples of some of these devices include Masimo's Radius PPG™ sensor, Radius T™ sensor, and Centroid™ sensor, or otherwise. One or more of those sensors, for

example, the Centroid™ sensor, can be used for stroke detection. The wearable device **10** can output an alert of stroke detection of the wearer and/or automatically initiate communication with a first respondent and/or the wearer's guardian or next-of-kin upon stroke detection.

[0442] The wearable device **10** can optionally communicate with chemical sensors, which can detect, for example, chemicals on the wearer's skin, and/or sweat, and/or the odor of certain chemicals in the air. The chemical sensors can include electrochemical sensors or any other suitable types of chemical sensors. A chemical sensor configured to analyze compositions of sweat can output measurements aiding the wearable device **10** in detecting stress and/or the wearer's hydration status. The wearable device **10** can optionally communicate with a skin impedance sensor, which can be used for monitoring the hydration status of the wearer.

[0443] Another example sensor that can be integrated into or connected to the device **10** and/or the sensor or module **100** can include a toxin and/or radiation detector configured to detect toxins in air (for example, pollution or contaminant particulates, carbon monoxide, smoke, and the like in the air). The toxin detection can aid care providers and/or firefighters who wear the device **10**.

Alternatively, the device **10** can be connected wirelessly to an external toxin and/or radiation detector. The toxin and/or radiation detector can be used with a smart mask. For example, the external toxin and/or radiation detector can be located on the mask, which can allow the mask to output a warning to the wearer of the mask when the mask filter or cartridge needs replacement.

[0444] Optionally, the wearable device **10** can communicate with glucose monitors, which can be invasive or minimally invasive such as finger prick type of glucose monitors, or a continuous noninvasive glucose monitor. The wearable device **10** can receive and display the wearer's glucose level from the glucose monitor. The wearable device **10** can also optionally be in communication with an insulin pump. The wearable device **10** can send a control signal to dispense insulin from the insulin pump to the wearer based on the monitored glucose level of the wearer.

[0445] As shown in FIG. 3, the device **10** can include a display screen **12** positioned at a top side of the device housing **101**. In addition to time and date indicators, one display layout (for example, the default display layout) of the display screen **12** can display the wearer's SpO2 measurement, the pulse rate (PR) measurement, the respiration rate (RR) measurement, and/or hydration status (H2O). The format of the measurement displayed is not limiting. For example, some measurements, such as the SpO2 measurement and the PR measurements, can be displayed as numerical values. As another example, some measurements, such as the RR measurements and hydration status, can be displayed as a sliding scale. In the illustrated example, the hydration status can be displayed as having three levels from low (L) to high (H). In the illustrated example, the respiration rate can be displayed as ranging from 5 bpm to 25 bpm. The wearer can optionally view individual display layouts for each measurements or a group of measurements by tapping on the display screen **12**, which can be a touch screen, and/or pressing a button on the device **10**. Each of the measurements can be displayed constantly, at certain intervals, and/or upon receiving instructions for display (for example, by the wearer tapping on the display screen **12** and/or pressing a button on the device **10**). Each of the measurements can be configured to be displayed with different or the same frequencies. Time and certain physiological parameters (for example, SpO2 and pulse rate) can be immediately and/or intermittently available, and/or continuously measured (for example, at least every 5 to 10 measurements per minute or more) and the displayed values constantly updated. Optionally, the display can further show a trend line for some parameters, such as SpO2 and pulse rate. In one example, the display of the wearable device can be configured to display only time, SpO2, and pulse rate.

[0446] As shown in FIG. 4, the physiological parameter measurement module **100** can be preassembled before being integrated into the device **10**. The physiological parameter measurement module **100** can be characterized before being assembled with the rest of the device **10**. The preassembled physiological parameter measurement module **100** can be secured within the device housing **101** using various mechanical assembly mechanisms, for example, one or more screws or

other fasteners. The sensor or module **100** of a wearable device **10** can be interchangeable and be replaced without replacing the memory in the device **10**. For example, the sensor or module **100** can include a quick-connect (and/or quick-release) feature for attaching the sensor or module **100** to the remainder of the device **10**, such as being attachable to the device **10** by magnets. An electrical connection can be established between the physiological parameter measurement sensor or module processor board and the circuit of the rest of the device **10**, including for example, a device processor and the display **12**. Optionally, the electrical connection can include a connector **32** on the sensor or module **100**. The connector **32** is configured to be electrically connected to a flex circuit. The wearable device **10** and the sensor or module **100** are portable and can be moved from place to place. As described above, the functionality of the wearable device **10** can be integrated and/or interchangeable with various other patient monitoring devices, displays, etc. [0447] The sensor or module **100** can be applied to locations on the body other than the wrist. Alternatively or additionally, multiple modules **100** can be applied to different locations of the body of the wearer. Other types of straps or fastening mechanism may be used to attach the multiple modules **100** onto other parts of the body. The other types of straps or fastening mechanism can optionally include a power source (for example, battery) to power a module **100** that is not integrated into the wearable device **10**, but may not have its own display. For example, an optical sensor can be placed on the wearer's neck to measure arterial and venous oxygen saturation, which can be transmitted to and displayed on the wearable device **10**. The wearer can view his or her oxygen consumption information on the wearable device **10** based on the signals from the optical sensor on the neck and/or the signals from the sensor or module **100** that is located on the wearable device **10**.

[0448] As shown in FIGS. 5A-5D, an example wearable device **500** can include a watch housing **501**. Features of the device **500** can be incorporated into features of the device **10** and features of the device **10** can be incorporated into features of the device **500**. The watch housing **501** can have a length, for example, between about 40 mm and 50 mm, or between about 42 mm and 46 mm. The watch housing can have a width, for example, between about 32 mm to about 40 mm, or between about 35 mm to about 38 mm. When fully assembled, the watch **500** can have a thickness or height, for example, between 10 mm to about 15 mm, or between 12 mm to about 14 mm.

[0449] As described above, the physiological parameter measurement module can include a plurality of emitters and a plurality of detectors. The emitters can transmit optical radiation of a plurality of wavelengths into a tissue site (near the wrist of the wearer) and the detectors can respond to the intensity of the optical radiation (which can be reflected from the tissue site) after absorption by pulsatile arterial blood flowing within the tissue site. In addition to the light being attenuated by blood in the arteries, light interaction also happens at the capillary level. Arteries are located deeper below the skin surface than the capillaries, requiring LED emitters of greater light intensity and thus greater power consumption in order for the emitted light to reach the arteries. Moreover, measuring the light intensities signal of the light after attenuation by blood in the artery requires more selective placement of the emitters and detectors directly above the arteries to capture the pulsation of the blood. The physiological parameter measurement module disclosed herein is designed to utilize attenuation by blood in the capillaries and is not reliant on the blood flow in arteries. The patient parameter measurements made by the module disclosed herein can be accurate enough for clinical use. The module disclosed herein can provide plethysmograph-based patient parameter measurements with an accuracy of within about 4% error, or about 2% error. As shown in FIG. 6, the wrist **62** has fewer capillaries per volume than the fingertip **64**. Accordingly, the module is designed to have a width to provide greater coverage area of the wearer's wrist, which can boost the signal from the sensors located on the module (which will be described in greater detail below).

[0450] When measuring oxygen saturation based on attenuation by blood in the capillaries, it is desirable to avoid veins. Because venous blood contains less oxygen, intensity signals of light

attenuated by venous blood can cause errant readings oxygen saturation measurement. Optionally, the sensor or module processor of the physiological parameter measurement modules disclosed herein can reduce the effect of pulsing vein on the signal by comparing the signals from the plurality of detectors to determine which detectors receive better and/or clearer signals and deactivating the detectors that are more likely to cover and/or be around the pulsing veins. The sensor or module processor can dynamically adjust which detectors to deactivate. Deactivating the detectors can include deactivating operation of that detector and/or ignoring signals from that detector.

[0451] Optionally, the sensor or module processor of the physiological parameter measurement module can map the physiological parameter measurements calculated from signals received at the detectors and/or clusters of detectors located at different regions of the module. Variations (for example, if outside a certain range) in the mapped measurements can be an indication that the pressure distribution of the wearable device on the body of the wearer is unbalanced, and therefore the pressure of the device on the wearer is either too high or too low and/or the wearable device is tilted on the wrist. The wearable device can output an instruction to the wearer to readjust the tightness of the straps and/or to re-center of the wearable device on the wrist. Variations (for example, if outside a certain range) in the mapped measurements can additionally or alternatively provide an indication that a certain detector or cluster of detectors is/are placed over a large pulsing vein as described above. Readings from that certain detector or cluster of detectors can be ignored or the detector(s) suspected to be cover a pulsing vein may be deactivated. When two or more physiological parameter measurements, such as oxygen saturation measurements, do not agree among two or more detectors (for example, having a variation exceeding a certain range), the sensor or module processor can use the higher or highest measurement value, or alternatively use a combination of the measurement values from the two or more detectors (for example, using one of the two measurement values at different times or otherwise).

[0452] Alternatively or additionally, the mapped measurements can be compared with experimentally determined data at the same detector location or detector cluster location. The experimentally determined data can be obtained using, for example, a conventional reflectance type pulse oximeter taped over the corresponding detector location, or any other suitable known methods for making the same measurements, including the same wrist-based sensor arrangements described herein. The comparison between the mapped measurements and the experimentally determined data can provide indication of whether the device has achieved a desired pressure on the body of the wearer, whether certain detectors and/or clusters of detectors are placed over or near a pulsing vein, which may interfere with the physiological parameter measurements, or otherwise. For example, if the difference between the mapped measurements and the experimental data at a certain location falls outside a predetermined range, the sensor or module processor can determine that pressure is too high or too low at that location, and/or that the pressure distribution over the body is not sufficiently balanced to make accurate measurements, and/or a detector or cluster of detectors is/are placed over the wearer's pulsing vein. The experimental data can be stored in a memory device of the sensor or module processor.

[0453] The comparison among the mapped measurements and/or between the mapped measurements and the experimental data can be performed when the wearer first puts on the device and/or at certain time intervals in the duration when the device is worn on the wearer. Additionally, running the comparison-based diagnostics can allow the sensor or module processor to determine, at the start of the measurement and/or dynamically during use of the device, which detector(s) provide the most accurate and/or reliable measurements.

Various Example Components of the Wearable Device

[0454] Components of the wearable device will now be described. As shown in FIGS. 7A and 7B, the device **10** can include its own device processor **14**, which can be a digital/analog chip or other processor(s), such as a digital watch processor or a smartwatch processor. As shown in FIGS. 7B,

7G, and 7H, the device processor **14** can be located on a PCB. FIGS. 7G and 7H illustrate example layouts of the PCB for the device processor **14**. As shown in FIGS. 7A and 7B, the device **10** can include a power source **16**, which can be a battery, for powering the device processor **14**, the display screen **12**, and/or the physiological parameter measurement module **100**. The battery **16** can last at least 10 hours, or at least 12 hours, or at least 14 hours, or at least about 16 hours after each charge, with continuous measurements and/or displaying of certain physiological parameters, such as SpO2 and pulse rate.

[0455] The device **10** can be configured to display time after the battery **16** has been depleted, even if other features (for example, measuring physiological parameters using the module) may not be available when the battery **16** has been depleted. Additionally, when the device **10** is used clinically, the display **12** can also continue displaying critical patient information (for example, the patient's name, date of admission, etc.) after the battery **16** has been depleted. The device **10** may include nonvolatile memory to store the critical patient information. The device **10** can include a dual-battery configuration with a main battery and a backup battery. Power management of the device **10** may switch automatically for the device **10** to be powered by the backup battery when the main battery has been depleted. The device can additionally or alternatively be configured to be solar-powered, for example, by including a solar panel on the dial or elsewhere of the wearable device **10**. The display **12** of the device **10** can use e-ink or ULP (ultra low power screen) technology, which draws little amount of current for displaying information. The display **12** may automatically adjust the brightness, being brighter when outdoors and dimmer when indoors to further prolong battery life.

[0456] As shown in FIGS. 7A and 7B, the sensor or module **100** of the wearable device **10** can include a sensor or module processor **108** (which can include a memory and/or other electronics, such as shown in FIG. 11C). The sensor or module processor **108** can process signals from one or more of the sensors in the sensor or module **100** (or optionally other sensors in communication with the device **10**) to determine a plurality of physiological parameters. All the processing of the raw sensor data of the sensors in communication (via a wired and/or wireless connection) with the sensor or module processor **108** is performed by the sensor or module processor **108**. The sensor or module processor **108** can be configured to drive the emitters **104** to emit light of different wavelengths and/or to process signals of attenuated light after absorption by the body tissue of the wearer from the detectors **106**. The sensor or module processor **108** can determine and output for display on the device display screen **12** the physiological parameters based on the detected signals. Optionally, the sensor or module **100** can send the signals from the detectors **106** (for example, preprocessed signals) to the device processor **14**, which can determine and output for display the physiological parameters based on the detected signals. The absorption of light can be via transreflectance by the wearer's body tissue, for example, by the pulsatile arterial blood flowing through the capillaries (and optionally also the arteries) within a tissue site where the device **10** is worn (for example, the wrist). The sensor or module processor **108** can be located on a PCB **116**, such as shown in FIG. 7B.

[0457] The sensor or module **100** can include more than one group or cluster of light emitters (such as LEDs) **104** and more than one group of photodetectors (also referred to as “detectors”) **106**. Each group of emitters **104** can be configured to emit four (or three) different wavelengths described herein. The sensor or module **100** can include one or more thermistors **110** or other types of temperature sensors. The thermistor(s) **110** can be placed near one or more groups of emitters **104**. There can be at least one thermistor **110** near each group of emitters **104**. The thermistor(s) **110** can provide for wavelength correction of the light emitted by the emitters **104**. Optionally, the thermistor(s) **110** can additionally measure a temperature of the wearer of the device **10**. Optionally there can be one or more thermistors **110** located at other places of the sensor or module **100**. The emitters **104**, the thermistor(s) **110**, and/or the detectors **106** can be positioned on the PCB **116**.

[0458] As shown in FIG. 7A, the device **100** can include a gyroscope **112**, an accelerometer **114**,

and/or other position and/or posture detection sensor(s). The gyroscope **112** and/or the accelerometer **114** can be in electrical communication with the sensor or module processor **108**. The sensor or module processor **108** can determine motion information from signals from the gyroscope **112** and/or the accelerometer **114**. The motion information can provide noise reference for analysis of the pleth information and other signal processing (for example, processing of ECG signals) performed by the sensor or module processor **108**. The gyroscope **112** and/or the accelerometer **114** can be located on the PCB **116**.

[0459] FIG. **8A** illustrates example layouts of a top side of the PCB **116**. FIG. **8B** illustrates example layouts of a bottom side of the PCB **116**. The first or bottom side of the PCB **116** can include the emitters **104**, the detectors **106**, the temperature sensor **110**, and any other sensors, for example, the gyroscope, the accelerometer, and/or the like. FIGS. **8C-8E** illustrate the detectors **106** being connected electrically to the PCB **116** via wire bonds **107**. The module can include wires **105** extending over the detector **106** for shielding purposes. The number of wires **105** extending over the detector **106** may vary. The manner in which the wires **105** extend over the detector **106** may vary. The wires **105** may not extend all the way over the detectors **106** across the detector's width or length. For example, as shown in FIG. **8F**, the detectors of detector groups **106a**, **106b**, **106a/b** can each be connected electrically to the first side of the PCB **816** via wire bonds **107**. A wire **105** can extend along each side of the detector for noise shielding. In the illustrated example, the wire **105** can extend along each long side of the detector. The wire **105** may extend parallel with the length of the detector. The wire **105** may not extend over the body of the detector **106a**, **106b**, **106a/b**. In some aspects, a single wire **105** may extend along all four sides of the emitter for shielding. In some aspects, four wires **15** may each extend along one of the four sides of the emitter for shielding. In some aspects, a metal cover, such as a can shaped metal cover, may cover each of the detectors and/or emitters for shielding. The emitters in the emitter groups **104a**, **104b** can each be electrically connected to the first side of the PCB **816** via wire bonds **107**. The thermistors **110** at each of the emitter groups **104a**, **104b** can be electrically connected to the first side of the PCB **816** via wire bonds **107**. The detectors, emitters, and/or thermistor can alternatively be electrically connected to the PCB **116** via other suitable types of electrical connectors.

[0460] The second or top side of the PCB **116** can include the sensor or module processor **108** and other circuit hardware. The second side of the PCB **116** can be electrically noisy and is isolated from the sensors on the first side of the PCB **116** by the board. Electronics on the same side of the PCB **116** can be substantially entirely overmoulded to reduce or avoid components shifting in place or being damaged during use. On the second side of the PCB **116**, which faces away from the light transmissive cover **102**, the PCB **116** can be covered by melt plastic or other suitable electronics protective material **130**, such as shown in FIGS. **7B** and **7F**. As shown in FIG. **7F**, the electronic components on the second side of the PCB **116** can be generally sealed by the protective material **130** except that a connector **132** can extend from the second side of the PCB **116** and be exposed. The connector **132** can electronically connect the sensor or module **100** to circuitry of the wearable device **10**.

[0461] Optionally, as shown in FIGS. **7A**, **7B**, and **7D**, the device **10** can include an electrocardiogram (ECG) sensor including a plurality of electrodes **124**, **125** configured to make contact with the wearer's skin. One or more ECG electrodes **124** may be located on the sensor or module **100** (such as shown in FIGS. **7B** and **7E**). One or more ECG electrodes **125** may be located elsewhere on the device (for example, an ECG electrode **125** can form a part of the housing of the wearable device **10** as shown in FIG. **7B**). The ECG sensor can be in electrical communication with the sensor or module processor **108** via an ECG connector.

[0462] As shown in FIGS. **7B-7E**, the physiological parameter measurement module **100** can include a skin-interfacing light transmissive cover **102** that encloses the first side of the PCB **116**, which positions the plurality of light emitters **104** and detectors **106**. The sensor or module **100** can include a light barrier construct **120** that is configured to divide the emitters **104** and the detectors

106 into different chambers such that light cannot travel or substantially cannot travel between the chambers. The light transmissive cover **102** can extend over the various emitter and detector chambers formed by the light barrier construct **120** and the PCB **116**. The light transmissive cover **102** can include individual lenses or covers such as shown in FIG. 7D, a single lens or cover such as shown in FIGS. 17A-17C, or a combination of individual emitter chamber covering lenses or covers and a single lens or cover covering a plurality of detector chambers, such as shown in FIG. 7C. In the example lens or cover **102b** shown in FIG. 7C, the individual lenses or covers that are configured to cover the detector chambers such as shown in FIG. 7D can be interconnected with bridging portions **103** between the detector chambers, forming a single piece of lens or cover. The lens or cover **102b** can be combined with the lenses or covers **102a** covering the emitter chambers to cover all the openings in the light barrier construct **120** for forming sealed emitter and detector chambers. The light barrier construct **120** can be overmoulded to the lens or cover **102b** and the lenses or covers **102a**. The lens or cover **102b** may not be configured to cover the emitter chambers, which can be covered by individual lenses, so as to avoid any light traveling between an emitter chamber and a detector chamber.

[0463] As shown in FIG. 7B, the physiological parameter measurement module **100** can include an opaque frame **126**. The opaque frame **126** can accommodate the light barrier construct **120**. Alternatively, the opaque frame **126** and the light barrier construct **120** can form an integral piece, such as shown in FIG. 7D. The opaque frame **126** can include indentations having the shape and size to accommodate the ECG electrodes **124** or other components with a suitable shape and size. A front side of the electrodes **124** can have one or more posts **137** extending past openings in the opaque frame **126** into corresponding openings on the PCB **116**. The posts **137** of the electrodes **124** can establish an electrical connection with the corresponding openings of the PCB **116**. A plurality of screws (or other types of fasteners) can extend into the corresponding openings of the PCB **116** from the front side of the PCB **116** to secure the electrodes **124** to the sensor or module **100** by threadedly mating or otherwise with the posts **137**. When a wearer puts the wearable device incorporating the sensor or module **100** onto the wearer's wrist, the electrodes **124** can make contact with the wearer's skin.

[0464] FIGS. 27A, 27B, and 27E illustrate an additional example aspect of an optional electrocardiogram (ECG) sensor. The electrocardiogram (ECG) sensor may include a plurality of electrodes **2724** configured to make contact with the wearer's skin. The plurality of ECG electrodes **2724** may be located on the sensor or module **2700** (such as shown in FIGS. 27A, 27B and 7E). As disclosed herein, the wearable device incorporating the module can include another ECG electrode **125** located on the housing of the wearable device configured to make contact with the wearer's skin.

[0465] FIG. 27B is an exploded perspective view of an example aspects of a sensor or module **2700**. As shown in FIG. 27B, the opaque frame **2726** can include recesses (which may also be referred to as "indentations") having the shape and size to accommodate the ECG electrodes **2724** or other components with a suitable shape and size. For example, in some implementations, frame **2726** includes recesses **2824**. Recesses **2824** can be sized and/or shaped to receive ECG electrodes **2724**. In some implementations, recesses **2824** have a depth (for example, measured from a plane of the frame **2726**) that is substantially equal to a thickness of the ECG electrodes **2724**. In some implementations, recesses **2824** have a size and/or shape that matches a size and/or shape of the ECG electrodes **2724**. For example, in some implementations in which the ECG electrodes have a semi-annular shape (such as that illustrated in at least FIGS. 27A-27B), the recesses **2824** can have a semi-annular shape.

[0466] A front side of the electrodes **2724** can have one or more posts **2737** extending past openings in the opaque frame **2726** into corresponding openings on the substrate **2716**. The posts **2737** of the electrodes **2724** can establish an electrical connection with the corresponding openings of the substrate **2716**. A plurality of screws (or other types of fasteners) can extend into the

corresponding openings of the substrate **2716** from the front side of the substrate **2716** to secure the electrodes **2724** to the sensor or module **2700** by threadedly mating or otherwise with the posts **2737**. When a wearer puts the wearable device incorporating the sensor or module **2700** onto the wearer's wrist, the electrodes **2724** can make contact with the wearer's skin.

[0467] With continued reference to FIG. **27B**, the substrate **2716** can include a printed circuit board (PCB). The substrate **2716** can include a conductive liquid adhesive **2739**. The conductive liquid adhesive **2739** may be provided on the copper of the substrate **2716**. The conductive liquid adhesive **2739** may facilitate conductive electrical connection between the electrodes **2724** and the substrate **2716**.

[0468] With continued reference to FIG. **27B**, one or more spring contacts (such as spring contacts **2755'** shown in FIG. **27I**) may be located between the electrodes **2724** and the substrate **2716**. The shape, size, and/or number of the spring contacts can vary. The spring contacts can establish an electrical connection between the electrodes **2724** and the substrate **2716**. The spring contacts can be biased toward the electrodes **2724** to ensure a firm electrical connection between the spring contacts and the electrodes **2724** and the substrate **2716**.

[0469] The physiological parameter measurement module **100** can include diffusing materials or encapsulant, which can include, for example, microspheres or glass microspheres. As described above, the encapsulant can eliminate air gaps between the surface of the light transmissive cover **102** and the emitters **104** and/or the detectors **106**. The encapsulant can be included around the emitters **104** to more evenly spread the emitted light, which appears to be emitted from an entire emitter chamber rather than from a point source (that is, a single LED emitter) if the encapsulant is absent. The encapsulant can allow the emitted light to travel through a greater volume of the tissue at the tissue site. The diffusing material can act as a beam shaper that can homogenize the input light beam from the emitter, shape the output intensity profile of the received light, and define the way (for example, the shape or pattern) the emitted light is distributed to a tissue measurement site. Such diffuser materials can, for example, deliver substantially uniform illumination over a specified target area in an energy-efficient manner. According to the Beer-Lambert law, the amount of light absorbed by a substance is proportional to the concentration of the light-absorbing substance in the irradiated solution (for example, the arterial blood). Therefore, by irradiating a larger volume of tissue and/or by increasing the amount of detected light, a larger sample size of light attenuated by the wearer's tissue can be measured. The larger sample size provides a data set that can be more representative of the complete interaction of the emitted light as it passes through the patient's blood as compared to a smaller sample size.

[0470] The diffusing materials can be any suitable materials, for example, glass, ground glass, glass beads, opal glass, greyed glass, polytetrafluoroethylene, or a microlens-based, band-limited, engineered diffuser that can deliver efficient and uniform illumination UV-cured flow glass microspheres injected into one or more openings on the sensor or module **100** (for example, after the sensor or module **100** has been assembled). Examples of engineered diffusers can include molded plastics with specific shapes, patterns, and/or textures designed to diffuse the emitter light across the entirety of a tissue surface. The diffusing material can be made of ground glass, which spreads the emitted light with a Gaussian intensity profile. The diffusing material can include glass beads. The diffusing material can be constructed so as to diffuse the emitted light in a Lambertian pattern. A Lambertian pattern is one in which the radiation intensity is substantially constant throughout the area of dispersion. One such diffusing material can be made from opal glass. Opal glass is similar to ground glass, but has one surface coated with a milky white coating to diffuse light evenly. The diffusing material can be capable of distributing the emitted light on the surface of a plane (for example, the surface of the tissue measurement site) in a predefined geometry (for example, a rectangle, square, circle, or otherwise), and with a substantially uniform intensity profile and energy distribution. The efficiency, or the amount of light transmitted by the diffusing material, can be greater than 70% of the light emitted by the emitter. The efficiency can be greater than 90%

of the emitted light. Additional examples of the diffusing material are described in U.S. Pat. No. 10,448,871, the entirety of which is hereby incorporated herein by reference and should be considered part of the disclosure.

[0471] Additionally or alternatively, the physiological parameter measurement module **100** can include encapsulant or light diffusing materials in the detector chambers to more evenly spread the reflected light to so as to increase the amount of the reflected light reaching the detectors. The module can include light diffusing materials positioned around the detectors to scatter and/or deflect the reflected light so that more reflected light can be detected by the detectors. For example, the reflected light can keep bouncing off the diffusing materials until the reflected light reaches the detector. Accordingly, the light detecting surface area in the module can be greater than the surface area of the detectors. Having the light diffusing materials can reduce the power needed to drive the LEDs of the emitters and/or the number of detectors at a particular location of the module, which can reduce the power consumption of the module.

[0472] As shown in FIG. 9A, the opaque frame **126** of the sensor or module **100** can include a plurality of light diffusing material(s) (or encapsulant) fill holes **144**. Light diffusing material(s) or encapsulant (for example, a flow of glass microspheres) can be injected into the plurality of chambers via the fill holes **144**, and be directed to the respective emitter or detector chambers as illustrated by the arrows in FIG. 9A along a plurality of the fill channels **146** (see FIG. 9B) which are interconnected with the fill holes **144**. The fill channels **146** can be located at a side of the opaque frame **126** facing away from the tissue of the wearer. As shown in FIG. 9B, the side of the opaque frame **126** facing away from the tissue of the wearer can further include a plurality of air vent channels **145**. Air can escape into the vent channels **145** as the diffusing material solution or encapsulant is injected into the respective chambers via the fill holes **144**, making it easier for the injected solution to flow into the respective chamber. As shown in FIG. 9B, the module **100** may not have air vent channels or fill channels between emitter and detector chambers to avoid light piping along such a channel. The encapsulant can be UV-cured after being injected into the respective chambers.

[0473] The opaque frame **126** may be configured such that the fill holes **144** and channels **146** allow the light diffusing materials to fill only the emitter chambers, or only the detector chambers, or both the emitter and detector chambers. Optionally, in addition or alternative to the light diffusing materials, the detector chamber can include light transmissive lens(es) or covers on the surface of the PCB that is not occupied by the detectors. The light transmissive lens(es) or covers may be polycarbonate. The light transmissive lens(es) or covers inside the detector chamber can help in focusing the reflected light onto the detectors inside the detector chamber.

[0474] FIGS. 27C and 27D illustrate another example opaque frame **2726** and light barrier construct **2720** of a sensor or module **2700**. The opaque frame **2726** and/or light barrier construct **2720** can include a plurality of light diffusing material(s) (or encapsulant) fill holes **2744**. Light diffusing material(s) or encapsulant (for example, a flow of glass microspheres) can be injected into the plurality of chambers via the fill holes **2744**, and be directed to the respective emitter or detector chambers as illustrated by the red arrows in FIGS. 27C and 27D along a plurality of the fill channels **2746** (see FIG. 27D) which are interconnected with the fill holes **2744**. The fill channels **2746** can be located at a side of the opaque frame **2726** facing away from the tissue of the wearer. The fill channels **2746** may travel through portions of the opaque frame **2726** and/or portions of the light barrier construct **2720**. As shown in FIG. 27D, the side of the opaque frame **2726** facing away from the tissue of the wearer can further include a plurality of air vent channels **2745**. Air can escape into the vent channels **2745** as the diffusing material solution or encapsulant is injected into the respective chambers via the fill holes **2744**, making it easier for the injected solution to flow into the respective chamber. As shown in FIG. 27D, each of the fill channels **2746** and/or the vent channels **2745** may connect more than one emitter chamber or more than one detector chamber. However, the module **2700** may not have air vent channels or fill channels extending between

emitter and detector chambers to avoid light piping along such a channel. The encapsulant can be UV-cured after being injected into the respective chambers.

[0475] The opaque frame **2726** may be configured such that the fill holes **2744** and channels **2746** allow the light diffusing materials to fill only the emitter chambers, or only the detector chambers, or both the emitter and detector chambers.

[0476] In FIG. **10**, a cross-sectional view of the sensor or module **100** illustrates some of the emitter and detector chambers. The chambers illustrated in FIG. **10** include a first emitter chamber **136a** enclosing a first emitter group **104a**, a second emitter chamber **136b** enclosing a second emitter group **104b**, a first detector chamber **140** enclosing one of first groups of detectors **106a** that surround the first emitter group **104a**, a second detector chamber **142** enclosing one of second groups of detectors **106b** that surround the second emitter group **104b**, and a third detector chamber **138** enclosing one of shared groups of detectors **106a/b** that surround both the first and second emitter groups **104a**, **104b** on opposite sides of the third detector chamber **138**.

[0477] As shown in FIG. **10**, light from the first emitter group **104a** can travel a shorter path, as indicated by the shorter arrows, to the first group of detectors **106a** or the shared group of detectors **106a/b**; and light from the first emitter group **104a** can travel a longer path, as indicated by the longer arrows, to the second group of detectors **106b**. The reverse is true for light from the second emitter group **104b**, which can travel a shorter path to the second group of detectors **106b** or the shared group of detectors **106a/b** and a longer path to the first group of detectors **106a**. As described herein, the different groups of emitters **104a**, **104b** and/or detectors **106a**, **106b**, **106a/b** can be run independently and/or simultaneously. Signals outputted by the different groups of detectors **106a**, **106b**, **106a/b** based on light emitted from the first emitter group **104a** and/or the second emitter group **104b** can provide different information due to the different light paths, which can travel through different areas of the tissue. The longer path penetrates deeper into the tissue and through a greater volume of the tissue to reach the “far” groups of detectors than the shorter path, which penetrates less deep into the tissue and travels through a smaller volume of tissue to reach the “near” group of detectors. The different information can be separated and/or combined to calculate a plurality of physiological parameters of the wearer of the sensor or module **100**, for example, an indication of the wearer's hydration status, which will be described in greater detail below.

[0478] FIG. **27E** is a front perspective view of an example aspect of a sensor or module **2700** including another example arrangement of emitter and detector chambers. The emitter and detector chambers may comprise one or more light blocks. The chambers illustrated in FIG. **27E** include a first emitter chamber **2736a** enclosing a first emitter group, a second emitter chamber **2736b** enclosing a second emitter group, a first group of detector chambers **2740**, a second group of detector chambers **2742**, and a third group of detector chambers **2738**. Each detector chamber can enclose one detector. The first and second emitter chambers **2736a**, **2736b** can be adjacent to each other. The first, second and third groups of detector chambers **2740**, **2742**, **2738** can extend around the first and second emitter chambers **2736a**, **2736b**.

[0479] As shown in FIG. **27E**, light from the first and second emitter groups in the first and second emitter chambers, respectively, can emit light that travel paths of different lengths for example to different detectors. Light from the first emitter group can travel a shorter path, as indicated by the shorter arrows, to the first group of detector chambers **2740**; and light from the first emitter group can travel an intermediate path, as indicated by the intermediate arrows, to the second group of detector chambers **2742**; and light from the first emitter group can travel a longer path, as indicated by the longer arrows, to the third group of detector chambers **2738**. The reverse is true for light from the second emitter group, which can travel a shorter path to the third group of detector chambers **2738**, and an intermediate path to the second group of detector chambers **2742**, and a longer path to the first group of detector chambers **2740**.

[0480] As described herein, the different emitters can be run independently and/or simultaneously.

For example, the emitters can be selectively activated (e.g., modulated) so that only one emitter (or subset of emitters) is emitting light at a given time. For example, in aspects wherein the first emitter group comprises four emitters, each of the four emitters of the first emitter group may be activated for a quarter cycle (e.g., a different quarter cycle than the other emitters) and off for the remaining three-quarters cycle. For example, a first emitter of the first emitter group may be activated to emit light during only a first quarter cycle, a second emitter of the first emitter group may be activated to emit light during only a second quarter cycle, a third emitter of the first emitter group may be activated to emit light during only a third quarter cycle and a fourth emitter of the first emitter group may be activated to emit light during only a fourth quarter cycle. The emitters of the second emitter group may operate in a similar manner as described.

[0481] As another example, in aspects wherein the first emitter group comprises four emitters, each of the four emitters of the first emitter group may be activated for an eighth of a cycle (e.g., a different eighth of the cycle than the other emitters) and off for the remaining seven-eighths cycle. An eighth of a cycle wherein no emitter is activated may occur between each of the cycles wherein an emitter is activated. For example, a first emitter of the first emitter group may be activated to emit light during only a first quarter cycle, a second emitter of the first emitter group may be activated to emit light during only a third quarter cycle, a third emitter of the first emitter group may be activated to emit light during only a fifth quarter cycle and a fourth emitter of the first emitter group may be activated to emit light during only a seventh quarter cycle. The emitters of the second emitter group may operate in a similar manner as described.

[0482] The above examples are not meant to be limiting. Alternative activation sequences for the emitters may be used to provide a time-multiplexed signal. In some aspects, the emitters can be selectively activated (e.g., modulated) so that two or more emitters are emitting light at a given time (e.g., during the same cycle or during overlapping cycles), for example in a manner similar to the examples given above.

[0483] The emitters may be modulated within an emitter group (e.g., first emitter group and second emitter group) or all of the emitters of the wearable device **10** may be modulated according to a single activation sequence. For example, the emitters of the first group may be modulated according to one activation sequence and the emitters of the second group may be modulated according to a second activation sequence. Alternatively, the emitters of the first and second emitters groups can all be modulated according to a single activation sequence.

[0484] In some aspects, the detectors may operate independently from and/or simultaneously with each of the other detectors. For example, each of the detectors may provide an individual signal to the module processor **108**.

[0485] Signals outputted by the different detectors of the different detector chambers **2740**, **2742**, **2738** based on light emitted from the first emitter group and/or the second emitter group can provide different information due to the different light paths, which can travel through different areas of the tissue. The longer path penetrates deeper into the tissue and through a greater volume of the tissue to reach the detectors of the “far” group of detector chambers than the intermediate and shorter paths. The shorter path penetrates less deep into the tissue and travels through a smaller volume of tissue to reach the detectors of the “near” group of detector chambers than the intermediate and longer paths. The different information can be separated and/or combined to calculate a plurality of physiological parameters of the wearer of the sensor or module **2700**.

[0486] For convenience, the terms “proximal” and “distal” are used herein to describe structures relative to the first emitter group or the second emitter group. For example, a detector may be proximal or distal to the first emitter group and may be proximal or distal to the second emitter group. The term “distal” refers to one or more detectors that are farther away from an emitter group than at least some of the other detectors. The term “proximal” refers to one or more detectors that are closer to an emitter group than at least some of the other detectors. The term “intermediate detector” refers to detectors that are closer to an emitter group than distal detectors and farther from

an emitter group than proximal detectors. The term “proximal detector” may be used interchangeably with “near detector” and the term “distal detector” may be used interchangeably with “far detector”.

[0487] A single detector may be both a proximal to one detector and distal to another detector. For example, a detector may be a proximal detector relative to the first emitter group and may be a distal detector relative to the second emitter group.

[0488] FIG. 11A illustrates schematically an example wearable device **10** disclosed herein. As described above, the device processor **14** can be connected to the module sensor **108** of the physiological parameter measurement module **100**, which includes the emitters, the detectors, the thermistors, and other sensors disclosed herein. The electrical connection between the device processor **14** and the sensor or module processor **108** can be establish optionally via a flex connector **32**. The sensor or module processor **108** can be coupled to the ECG electrodes **124**, **125**, optionally via an ECG flex connector **123**.

[0489] The device processor **14** can be connected to a display **12**, which can include the display screen and touch input from the wearer. The device processor **14** can include a battery **16**, and optionally one or more wireless charging coils **17** to enable wireless charging of the battery **16**. The device processor **14** can be connected to an antenna **19** for extending signals transmitted wirelessly, for example, to an external device as described with reference to FIG. 2. The device processor **14** can include connection to a first user interface (UI 1) **13a** and a second user interface (UI 2) **13b** on the device **10** to receive input from the wearer. As shown in FIG. 1D, example first and second user interface **13a**, **13b** can be in the form of buttons **13**. Additionally or alternatively, the device **10** can include a microphone. The device **10** can receive user inputs via the user interfaces, which can be the buttons, the microphone, and/or the touchscreen. The user inputs can command the device **10** to turn on and/or off certain measurements, and/or to control externally connected devices, such as an insulin pump, a therapeutics delivery device, or otherwise. The device processor **14** can be connected to a user feedback output **15** to provide feedback to the wearer, for example, in the form of vibration, an audio signal, and/or otherwise. The device processor **14** can optionally be connected to an accelerometer and/or a gyroscope **42** located on the device **10** that is different from the accelerometer **114** and gyroscope **112** on the physiological parameter measurement module **100**. The accelerometer and/or gyroscope **42** can measure position and/or orientation of the wearer for non-physiological parameter measurement functions, for example, for sensing that the wearer has woken up, rotating the display **12**, and/or the like.

[0490] FIG. 11B illustrates example components of the device processor **14** PCB board. As shown in FIG. 11B, the device processor **14** can include a Bluetooth co-processor **1400** and a system processor **1402**. The system processor **1402** can run the peripheral functions of the device **10**, receive user (that is, the wearer) input and communicate to the sensor or module processor **108**. The Bluetooth co-processor **1400** can focus on managing Bluetooth communication so as to allow the system processor **1402** to focus on the high memory utilization tasks, such as managing the display screen **12**. The Bluetooth co-processor **1400** can be activated when there is incoming and/or outgoing Bluetooth communication. Alternatively, the Bluetooth co-processor **1400** can be replaced by a different wireless co-processor configured to manage wireless communication using a different wireless communication protocol.

[0491] FIG. 11C illustrates example components of the module processor PCB board **116**. As shown in FIG. 11C, the sensor or module processor **108** can include a calculation processor **1080** and a system processor **1082**. The calculation processor **1080** can manage host communication with the device processor **14** via a host connector **1084**. The calculation processor **1080** can perform algorithm computations to calculate the physiological parameters based on the signals received from the ECG electrodes **124/125** and the optical sensor including the emitters **104**, the detectors **106**, and the temperature sensors **110**, and optionally from other sensors in communication with the sensor or module processor **108**. The calculation processor **1080** can have relatively large memory

suitable for running algorithm computations. The system processor **1082** can be in communication with a power management integrated circuit (PMIC) **1090**. The system processor **1082** can run the physical system of the sensor or module **100** (for example, including turning on and off the emitter LEDs, changing gain, setting current, reading the accelerometer **114** and/or the gyroscope **112**, and the like) and decimate data to a lower sampling rate. The system processor **1082** can focus on data processing, taking measurements and diagnostics, and basic functions of the sensor or module processor **108**. The system processor **1082** can allow the calculation processor **1082** to sleep (being inactive) most of the time, and only wake up when there is enough measurement data to perform calculations.

[0492] FIG. **11D** illustrates an example front-end analog signal conditioning circuitry **1088** of the module PCB **116** shown in FIG. **11C**. The entire front end circuitry **1088** can be located on a single application-specific integrated circuit (ASIC).

[0493] The front-end circuitry **1088** can include a transimpedance amplifier **1092** configured to receive analog signals from the optical sensor including the emitters **104**, the detectors **106**, and the temperature sensors **110**, which can be preprocessed (for example, via a low pass filter **1094** and a high pass filter **1096**) before being sent to an analog-digital converter **1098**. The analog-digital converter **1098** can output a digital signal based on the analog signals from the optical sensor including the emitters **104**, the detectors **106**, and the temperature sensors **110** to the system processor **1082** and the calculation processor **1080**. The front end circuitry **1088** can include a detector cathode switch matrix **1083** configured to activate the cathode of the detectors that are selected to be activated. The matrix **1082** can be further configured to deactivate (for example, by short-circuiting) anodes of the detectors that are selected to be deactivated in configurations in which the detectors share a common cathode and have different cathodes.

[0494] The front-end circuitry **1088** can include an ECG amplifier **1091** configured to receive analog signals from the ECG electrodes **124/125**, which can output the amplified analog signals to the analog-digital converter **1096**. The amplified analog signals can include an ECG differential between the positive and negative electrodes. The analog-digital converter **1098** can output a digital signal based on the analog signals from the ECG electrodes **124/125** to the system processor **1082** and the calculation processor **1080**.

[0495] The ECG electrodes **124** can include a negative electrode, a positive electrode, and a reference electrode. As shown in FIG. **12A**, the two electrodes **124** located on the sensor or module **100** can act as a reference electrode and a negative (or positive) electrode respectively. As shown in FIGS. **12B** and **12C**, a portion of the device housing **101** that surrounds the display screen **12** can function as another ECG electrode **125**. An electrically insulating material **127** can separate the ECG electrode **125** from the remainder of the housing **101** so that an electrical current between the ECG electrode **125** and the ECG electrodes **124** would travel through the wearer's body. When the wearer wants to make a measurement using the ECG sensor that includes the ECG electrodes **124**, **125**, the wearer can press on or touch the electrode **125** using the wearer's finger or another part of the wearer's body such that the wearer's skin makes contact with the electrode **125**.

[0496] In the illustrated examples, the ECG electrode **125** can be positive (or negative if one of the electrodes **124** serves as a positive electrode) electrode. As shown in FIG. **12C**, the electrode **125** is illustrated as being transparent to show one or more spring contacts **131** located underneath the electrode **125**. The shape, size, and/or number of the spring contacts **131** can vary from the example shown in FIG. **12C**. The spring contacts **131** can establish an electrical connection between the electrode **125** and the sensor or module processor **108** of the sensor or module **100**. For example, the spring contacts **131** can establish an electrical connection between the electrode **125** and the connector **132**. The spring contacts **131** can be biased toward the electrode **125** to ensure a firm electrical connection between the spring contacts **131** and the electrode **125**. Readings from the electrodes **124**, **125** can allow the sensor or module processor **108** to obtain the wearer's ECG signal and optionally to make physiological measurements based on the

obtained ECG, for example, the heart rate, the respiratory rate, and/or otherwise. The sensor or module processor **108** can communicate the ECG signals and/or ECG-related measurements to the wearable device processor **14**. The wearer's ECG waveform and/or the measurements made from the ECG can be displayed on the display screen **12**. FIG. **13A** illustrates an example LED driver circuitry **1086** of the module PCB **116** shown in FIG. **11C**. The entire LED driver circuitry **1086** can be located on the single ASIC with the front end circuitry **1088**. As described above, the system processor **1802** can output a control signal to turn on and off the emitter LEDs. As shown in FIG. **13A**, the LED driver circuitry **1086** can include an emitter switch matrix **1085** configured to drive any of the emitters (or emitter groups) that are selected to be turned on or turn off any of the emitters (or emitter groups) that are selected to be turned off.

[0497] FIG. **13B** illustrates an example emitter circuitry including eight different emitter LEDs **104**. The number of LEDs may vary and can be less than or greater than eight. The emitters of the physiological parameter measurement module can be configured to emit a plurality of (for example, three, four, or more) wavelengths. Each of the emitters can be configured to emit light of a different wavelength than the other emitters. Alternatively, one or more of the emitters can emit light of more than one wavelength. In the illustrated example, the emitter circuitry can include four drivers to drive the eight emitter LEDs. Alternatively, the module can include more than four LEDs per emitter group. Each LED Drive can drive an LED to emit light of a different wavelength. The device or the module can grant access of some of the LEDs to a third party device, for example, for measurement purposes. The LED drivers can selectively drive some but not all the LEDs.

[0498] The emitters can be configured to emit light of a first wavelength providing an intensity signal that can act as a reference signal. The first wavelength can be more absorbent by the human body than light of other wavelengths emitted by the emitters. The reference signal can be stronger and less likely to be affected by noise than the signals from other wavelengths emitted by the emitters. The reference signal can be used by the physiological parameter measurement sensor or module processor to extract information from the other signals, for example, information relevant to and/or indicative of the pulsing rate, harmonics, or otherwise. The physiological parameter measurement sensor or module processor can focus the analysis on the extracted information for calculating physiological parameters of the wearer. Including the reference signal can reduce power consumption and saving the battery life of the device. The first wavelength can be from about 525 nm to about 650 nm, or from about 580 nm to about 585 nm, or from about 645 nm to about 650 nm, or about 525 nm, or about 580 nm, or about 645 nm. The light providing the reference signal can have an orange or yellow color. Alternatively, the light providing the reference signal can have a green color.

[0499] The emitters can be configured to emit light having a second wavelength having a red color. The second wavelength can be from about 620 nm to about 660 nm. Light of the second wavelength can be more sensitive to changes in oxygen saturation (SpO₂) than light of other wavelengths emitted by the emitters. The second wavelength is preferably closer to 620 nm (for example, about 625 nm), which results in greater absorption by the body tissue of the wearer, and therefore a stronger signal and/or a steeper curve in the signal, than a wavelength that is closer to 660 nm. The physiological parameter measurement sensor or module processor **108** can extract information such as the pleth waveform from signals of the second wavelength.

[0500] The emitters can be configured to emit light having a third wavelength of about 900 nm to about 910 nm, or about 905 nm, or about 907 nm. The third wavelength can be in the infrared range. The sensor or module processor can use the third wavelength as a normalizing wavelength when calculating ratios of the intensity signals of the other wavelengths, for example, a ratio of the intensity signals of the second wavelength (red) to the third wavelength (infrared).

[0501] Additionally or optionally, the emitters can be configured to emit light having a fourth wavelength that is more sensitive to changes in water than the rest of the emitted wavelengths. The fourth wavelength can be in the infrared range and about 970 nm. The physiological parameter

measurement sensor or module processor can determine physiological parameters such as a hydration status of the wearer based at least in part on a comparison of the intensity signals of the fourth wavelength and a different wavelength detected by certain detectors. The detectors used for hydration monitoring can be located a predetermined distance away from the emitters (that is, being a “far” detector disclosed herein) so that light travels through a certain depth of the tissue before being detected by those detectors.

[0502] The emitters in the physiological parameter measurement sensor or module can be placed in two emitter groups. Each emitter group can include four emitter LEDs configured to emit the first, second, third, and fourth wavelengths described above. The emitters in the same emitter group can be located in the same emitter chamber disclosed herein. Each of the four drivers are configured to drive the emitters to emit one of the four wavelengths described above.

[0503] In some aspects, drivers may drive the emitters at varying intensities. The intensity at which the drivers drive the emitters may affect the amount of light that is outputted (e.g., lumens), the strength of the light signal that is outputted, and/or the distance that the outputted light travels. The drivers may drive the emitters at varying intensities according to modeling, logic and/or algorithms. The logic and/or algorithms may be based, at least in part, on various inputs. The inputs may include historical data, the amount of light that is attenuated, for example as the light penetrates and travels through the tissue of the wearer, or the amount of blood with which the light is interacting, or the type of blood (e.g., venous, arterial) or type of blood vessel (e.g., capillary, arteriole) with which the light is interacting and/or the heat being generated by the emitters. For example, the drivers may increase the intensity at which they drive the emitters based upon a determination that too much light is being attenuated in the tissue or that the light is not interacting with enough blood. As another example, the drivers may decrease the intensity at which they drive the emitters based upon a determination that the emitters have exceeded a threshold temperature. The threshold temperature may be a temperature which may be uncomfortable for human skin.

[0504] In some aspects, each of the drivers may be capable of driving a corresponding emitter at various intensities independently of the other drivers. In some aspects, each of the drivers may drive a corresponding emitter at various intensities in unison with each of the other drivers.

[0505] Additionally, various LEDs may be used in various aspects. For example, certain LEDs may be used which are capable of outputting more light with the same amount of power as other LEDs. These LEDs may be more expensive. In some aspects, less expensive LEDs may be used. In some aspects, a combination of various types of LEDs may be used.

[0506] FIG. 13C illustrates an example detector circuitry including fourteen detectors **106**. The total number of detectors on a module can vary. The fourteen detectors can form seven detector groups, each group including two detectors. The number of detectors in each group may vary. Detectors of the same detector group can be located in the same detector chamber disclosed herein. Each detector group can output one signal, which can be a combined signal of the two detectors in the same group. As shown in FIG. 13C, the detectors can share a common anode but have seven different cathodes, corresponding to the seven detector groups.

[0507] FIG. 13D illustrates an example thermistor circuitry. In the illustrated example, the physiological parameter measurement module can include two thermistors **110**. The two thermistors can be located in the two emitter chambers near the two emitter groups respectively.

Example Signal Processing of the Physiological Parameter Measurement Module

[0508] FIGS. 14A and 14B depict functional block diagrams of the operations of a conventional pulse oximeter carried out by the digital signal processing system. The signal processing functions described below are carried out by a digital signal processor (DSP) with a microcontroller providing system management. As shown in FIG. 14A, an analog signal from the detector(s) of the conventional pulse oximeter is digitized, filtered and normalized, and further processed using conventional pulse oximetry signal processing algorithms. Parallel signal processing engines-DST®, FST®, SST™, and MST™ are used to separate the arterial signal from sources of noise

(including the venous signal) to measure SpO2 and pulse rate accurately, even during motion. FIG. **14B** depicts a generalized functional block diagram for the operations performed on the 20 Khz sample data entering the digital signal processing system from an analog to digital converter (ADC). As illustrated in FIG. **14B**, the DSP first performs a demodulation, as represented in a demodulation module **400**. The processor performs decimation, as represented in a decimation module **402** on the resulting data from the demodulation. The processor calculates certain statistics, as represented in a statistics module **404**, and performs a saturation transform, as represented in a saturation transform module **406**, on the data resulting from the decimation operation. The processor forwards data subjected to the statistics operations and the data subjected to the saturation transform operations to saturation operations, as represented by a saturation calculation module **408** to output an oxygen saturation measurement and pulse rate operations, as represented in a pulse rate calculation module **410** to output a pulse rate value.

[0509] FIGS. **15A-15G** illustrate example signal processing of the physiological parameter measurement sensor or module disclosed herein. As shown in FIG. **15A**, the sensor or module processor can receive intensity signals from the detectors in response to detected reflected light of the first (reference signal or signal of green or yellow light), second (signal of red light), third (signal of infrared light), and fourth (signal of infrared light with a wavelength of 970 nm) wavelengths described above, and signals from the gyroscope and accelerometer. The sensor or module processor can output a plurality of physiological parameters based on the input signals from the sensors described above. The plurality of physiological parameters can include, for example, SpO2 (Sat), pulse rate (PR), perfusion index (PI), pleth variability index (PVI), respiration rate from the pleth (RRp), and a hydration index.

[0510] As shown in greater detail in FIG. **15B**, the sensor or module processor can process the intensity signal in response to detected light of the first, second, and third wavelengths in the unnormalized form and a normalized form (in normalization modules “Norm” **1500**, “Norm 1” **1502**, and “Norm 2” **1504**). As described above, the signal of the third wavelength can be used as the normalizing signal. The sensor or module processor can extract various information from the intensity signals in response to detected light of the first, second, and third wavelengths and signals from the accelerometer and the gyroscope, such as the PR (which can be output as the PR measurement), time domain (TD) saturation information, frequency domain (FD) saturation information, PI information, and PVI information, in a pulse rate determination module **1506**.

[0511] FIG. **15C** illustrates example processing of the raw signals from the accelerometer and the gyroscope to output the gyroscope and accelerometer signals. The sensor or module processor can combine each of the raw gyroscope and accelerometer signals (which can be raw signals from any axis of the gyroscope and/or accelerometer) with gyroscope/accelerometer time instants and pleth time instants signals in an interpolation module **1518** or interpolation **1** module **1520** respectively. The sensor or module processor can further process the outputs from the interpolation module **1518** or interpolation **1** module **1520** in a low pass filter and decimation module **1522** or low pass filter and decimation **1** module **1524** respectively to output a gyrol signal and an accelerometer **1** signal. The output gyre **1** and accelerometer **1** signals can be sent to the ASIC described above.

[0512] As shown in FIG. **15D**, the sensor or module processor can extract motion information from the gyroscope and accelerometer input and the normalized signals of the first, second, and third wavelengths in an interference mitigation (IM) and motion analysis module **1526**. As also shown in FIG. **15D**, the sensor or module processor can obtain time domain pulse rate (TDPR) information, TD saturation information, PI information, and PVI information in a time domain pulse rate determination module **1528** from the intensity signals of the first, second, and third wavelengths. The sensor or module processor can obtain frequency domain pulse rate (FDPR) information and FD saturation information in a frequency domain pulse rate determination module **1530** based on normalized signals of the first, second, and third wavelengths. The sensor or module processor can determine and output a pulse rate in a pulse rate decision logic **1532** based on the TDPR

information, FDPR information, interference mitigation (IM) PR information (output by the interference mitigation and motion analysis module **1526**), and motion information.

[0513] FIG. **15E** illustrates an example pulse rate determination decision logic. In this example, a decision logic stage 2 module **1534** can receive as input raw pulse rate calculations from individual pulse rate determination engines (for example, the time domain pulse rate determination module **1528**, the frequency domain pulse rate determination module **1530** and the interference mitigation and motion analysis module **1526** as shown in FIG. **15D**), pleth features including time domain and frequency domains from N channels (for example, N=4 or more) of pleth signals, and motion features obtained from a motion analysis module **1536**. The motion analysis module **1536** can assess the amount of motion, define the type of motion, and calculate a motion rate (for example, per minute) if the motion is determined to be periodic, and/or the like based on motion information from a 6DOF (degree-of-freedom) inertia measurement unit (IMU). The IMU can include the accelerometer and the gyroscope on the physiological parameter measurement module.

[0514] With continued reference to FIG. **15B**, the sensor or module processor can determine the oxygen saturation measurement based on the normalized signal of the third wavelength, the normalized signal of the second wavelength, the TD saturation information, the FD saturation information, the PR, and the motion information in an oxygen saturation determination module **1508**. FIG. **15F** illustrates an oxygen saturation determination module including a plurality of parallel signal processing engines, such as a Seed saturation module **1538**, an SST saturation module **1540**, a DST saturation module **1542**, an interference mitigation (IM) saturation module **1544**, and a signal/noise reference saturation module **1546**, configured to feed individual raw oxygen saturation (SpO₂) values to a decision logic **1548**. The decision logic **1548** can further receive as input the motion information and output a final oxygen saturation measurement based on the motion information and the raw oxygen saturation values determined by the parallel engines.

[0515] FIG. **15E** illustrates an example oxygen saturation determination decision logic. In this example, a saturation decision logic stage 2 module **1550** can receive as input raw oxygen saturation calculations from the parallel engines described above, pleth features, pulse rate, and motion features obtained from a motion analysis module **1552**. The pleth features received by the module **1550** can include the features in the pulse rate decision logic shown in FIG. **15E**.

Additionally, the pleth features received by the module **1550** can include features related to saturation, for example, the DC ratio of the second and third wavelengths. The motion analysis module **1552** can receive the same features as the pulse rate decision logic shown in FIG. **15E**.

[0516] With continued reference to FIG. **15B**, the sensor or module processor can determine the PI measurement based on the normalized signal of the third wavelength and the PI information in a perfusion index determination module **1510**. The sensor or module processor can determine the PVI measurement based on the PVI information in a pleth variability index determination module **1512**. The sensor or module processor can determine the RRp measurement based on the intensity signals of the first and second wavelength in a respiration rate determination module **1514**. The sensor or module processor can determine the hydration index in a hydration determination module **1516** based on the intensity signals (for example, from the “far detectors” disclosed herein) of the fourth wavelength, which is more sensitive to changes in water in the measurement site and another wavelength (for example, the third wavelength or about 905 nm) that is less sensitive to changes in water. The sensor or module processor can focus on the DC component of the signals for hydration status monitoring.

[0517] Various example physiological parameter measurement modules and wearable devices incorporating the same will be described below. Each of the example modules and devices can incorporate any of the features of the physiological parameter measurement module **100** and the device **10** described above, all of which are not repeated for brevity. Features of the example modules and devices disclosed herein can be incorporated into one another.

Examples of Physiological Parameter Measurement Modules with Double Emitter Groups

[0518] FIG. 16A illustrates schematically an example arrangement of an optical sensor, including emitters, detectors, and thermistors, on a sensor or module processor PCB 116. As shown in FIG. 16A, the PCB 116 can include a first group of emitters 104a and a second group of emitters 104b. Each group of emitters can include four emitters. The emitters in each group 104a, 104b can emit at least the first, second, third, and fourth wavelengths as described above. The first and second groups of emitters 104a, 104b can be located a distance from each other on a first side of a PCB 116. The PCB 116 can include a temperature sensor (such as a thermistor) 110 as described above located on the first side of the PCB 116. One temperature sensor 110 can be near the first group of emitters 104a. Another temperature sensor 110 can be near the second group of emitters 104b.

[0519] The PCB 116 can be elliptical in shape, although the shape of the PCB is not limiting. The two groups of the emitters 104a, 104b can be located on different parts of the first side of the PCB 116 divided along the minor diameter of the ellipse. Each of the two groups of the emitters 104a, 104b can be surrounded by a first light barrier and form an emitter chamber.

[0520] The first and second groups of emitters 104a, 104b can be surrounded by two rings of detectors 106a, 106b that are separated from the first and second groups of emitters 104a, 104b respectively by a distance. The two rings of detectors 106a, 106b can share a plurality of (for example, two or more) detectors 106a/b common to both rings. The detectors 106a/b common to both rings can be located along the minor axis of the ellipse. In the illustrated example, the PCB 116 can include fourteen detectors coupled to the PCB 116, but the total number of detectors can vary.

[0521] The detectors 106b can be the far detectors for the first group of emitters 104a and the detectors 106a, 106a/b can be the near detectors for the first group of emitters 104a. The detectors 106a can be the far detectors for the second group of emitters 104b and the detectors 106b, 106a/b can be the near detectors for the second group of emitters 104b. Accordingly, each detector 106a, 106b, 106a/b can receive two signals for each wavelength emitted by the first and second groups of emitters 104a, 104b respectively. As described above, signals outputted by the far and near detectors can provide different information due to the different light paths, which can travel through different areas of the tissue. In addition, the far detectors for each group of emitters 104a, 104b can detect the light emitted by the respective group of emitters 104a, 104b, for example, light of the fourth wavelength and another wavelength, and attenuated by tissue to provide an indication of the wearer's hydration status as described herein.

[0522] The detectors 106a, 106b, 106a/b can be separated or partitioned into seven detector regions. Each detector region can include two detectors, or any other number of detectors. Each detector region can form a detector chamber surrounded by light barriers. As described above, the sensor or module processor can process signals from a particular emitter and received at the detectors within the same detector region as one signal source. Accordingly, for each wavelength, the sensor or module processor can receive data from a total of fourteen signal sources, two from each detector region acting as the far and near detectors for the different groups of emitters respectively.

[0523] FIGS. 16B-16D illustrate an example physiological parameter measurement module 400 of a wearable device. The module 400 can incorporate any of the features of the module examples described herein.

[0524] As shown in FIG. 16B, the physiological parameter measurement module 400 can include a first group of emitters 404a and a second group of emitters 404b incorporating the arrangement shown in FIG. 16A. Each group of emitters can include four emitters (or optionally a different number of emitters, such as six or eight emitters). The emitters in each group 404a, 404b can emit at least the first, second, third, and fourth wavelengths as described above. Each of the two groups of the emitters 404a, 404b can be surrounded by a first light barrier 420 and form an emitter chamber.

[0525] The first and second groups of emitters 404a, 404b in the module 400 can be surrounded by

two rings of detectors **406a**, **406b** that are separated from the first and second groups of emitters **404a**, **404b** by the first light barrier **420**. The two rings of detectors **406a**, **406b** can share a plurality of (for example, two or more) detectors **406a/b** common to both rings. The detectors **406a**, **406b**, **406a/b** can have the same arrangement as the detectors shown in FIG. **16A**. In the illustrated example, the module **400** can include fourteen detectors, but the module **400** can also include a different total number of detectors.

[0526] As shown in FIGS. **16B** and **16D**, the detectors **406a**, **406b**, **406a/b** can be separated or partitioned into seven detector chambers by a portion of the first light barrier **420** and second light barriers **422**. Each detector region can include two detectors, or any other number of detectors. Along an outer perimeter of the module **400**, the detectors **406a**, **406b**, **406a/b** can be enclosed within a module side wall **424**. A sensor or module processor of the module **400** can process signals from a particular emitter and received at the detectors within the same detector region as one signal source as described above. The arrangement of emitters **104a**, **104b** and detectors **106a**, **106b**, **106a/b** and the light diffusing materials encapsulating the emitters **104a**, **104b** and/or detectors **106a**, **106b**, **106a/b** can improve the sensing coverage on the wearer's wrist, which has fewer capillaries per volume than the fingertip as described above. The aggregate light detecting area of the **106a**, **106b**, **106a/b** in FIG. **16B**, that is, the aggregate surface area of all the detector chambers, can occupy about 50% or more of the tissue-facing surface of the physiological parameter measurement module. The aggregate light detecting area in FIG. **16B** can be, for example, greater than about 100 mm.^{sup.2}, or greater than about 125 mm.^{sup.2}, or about 150 mm.^{sup.2}, or about 165 mm.^{sup.2}. The aggregate light emitting area in FIG. **16B**, that is, the aggregate surface area of both emitters chambers, can be, for example, greater than about 25 mm.^{sup.2}, or about 30 mm.^{sup.2}, or about 35 mm.^{sup.2}. Any other physiological parameter measurement module examples disclosed herein can have the same or substantially similar aggregate light detecting area and/or light emitting area as the module **400** shown in FIG. **16B**.

[0527] On the first side of the PCB **416**, the module **400** can be enclosed by a curved light transmissive cover **402** with a convex protrusion. As shown in FIG. **16C**, the cover **402** can have a continuous curvature. The first and second light barriers **420**, **422** are configured to be in contact with the first side of the PCB **416** at one end. At the other end, the height of the first and second light barriers **420**, **422**, and of the side wall **424** can generally follow the curvature of the cover **402**. The side wall **424** can be shorter than the second light barrier **422**. The height of the second light barrier **422** can increase from the perimeter of the module **400** toward a center of the module **400** until the second light barrier **422** merges with the first light barrier **420**, which is the highest among the light barriers. The light barriers **420**, **422** can extend to the tissue-facing surface of the cover **402** so that when the module **400** is pressed into the skin of the wearer of a device incorporating the module **400**, the tissue-facing surfaces of the first and second light barriers **420**, **422**, and of the side wall **424** can be configured to contact the skin of the wearer. The cover **402** can include individual lenses or covers such as shown in FIG. **7D** or a combination of individual emitter chamber covering lenses or covers and a lens or cover covering a plurality of detector chambers, such as shown in FIG. **7C**. The lenses or covers may be polycarbonate. The tissue-facing surface of the module **400** can include a continuous convex curvature.

[0528] The first and second light barriers **420**, **422** and the side wall **424** can optionally form a single light barrier construct. The single light barrier construct can be formed by any suitable manufacturing techniques and any suitable materials, for example, plastic, colored, or opaque sapphire glass, or others. The single light barrier construct can include at one end a recess that is shaped and sized to receive the PCB **416**, including the electronics on the PCB **416**. The first side of the PCB **416** can include the emitters **404a**, **404b**, detectors **406a**, **406b**, **406a/b**, temperature sensor **410**, and any other sensors, for example, the gyroscope, the accelerometer, and/or the like. The second side of the PCB **416** can include the sensor or module processor and other circuit hardware.

[0529] As described above, the module **400** can include a plurality of chambers such that light cannot travel between the chambers because of the various light barriers extending from the PCB **416** to the tissue-facing surface of the cover **402** as described herein. The light diffusing materials described above can be added above (for example, via the fill holes described herein) and around the emitters **404a**, **404b**, and/or optionally above and around the detectors **406a**, **406b**, **406a/b**, to improve distribution of emitted light and/or detected light after attenuation by the tissue. The light diffusing materials can include a flow of glass microsphere solution, which can be injected into the chambers after the module **400** has been assembled. After being injected into the respective chamber, the solution can be UV-cured. Air can escape via the vent openings disclosed herein as the diffusing material solution is injected into the respective chambers via the injection openings, making it easier for the glass microsphere solution to flow into the respective chamber. The cover **402** can also include glass microspheres. The light diffusing materials in the cover **402** and inside the emitter chambers and/or the first light barrier **420** can make the emitted light leave the emitter chambers enclosing the emitters **404a**, **404b** in a direction generally parallel to the height of the first light barrier **420**. The light diffusing materials in the cover **402** and the detector chambers can increase the amount of reflected light being directed to and detected by the detectors **406a**, **406b**, **406a/b**.

[0530] FIGS. **16E-16G** illustrate an example physiological parameter measurement modules **401** of a wearable device. The module **401** can include the same optical sensor arrangements as shown in FIGS. **16A-16D** and have any of the features of the module **400** in FIGS. **16B-16D** with the differences noted in the description of FIGS. **16E-16G**. The module **401** can have any of the features of the other physiological parameter measurement module examples described herein.

[0531] The module **401** can include a generally circular outer shape. The generally circular outer shape can be defined by an opaque frame **426** extending over of the PCB **416** from a first side of the PCB **416**. The opaque frame **426** can have a height such that a top side of the opaque frame **426** can be generally level with (or receding or protruding slightly from) a second side of the PCB **416**. As shown in FIG. **16G**, the PCB **416** can be generally circular in shape. The opaque frame **426** can be generally concentric with the PCB **416**. The opaque frame **426** and the PCB **416** are not transmissive to light. The opaque frame **426** in FIGS. **16E** and **16F** can include the first light barrier **420** and second light barriers **422** as an integral piece.

[0532] The module **401** can include one or more (for example, two or otherwise) ECG electrodes **424**. In the illustrated examples of FIGS. **16E-16G**, one of the ECG electrodes **424** can be a reference electrode and the other one of the ECG electrode **424** can be a negative or positive electrode. The opaque frame **426** can have indentations having the shape and size to accommodate the electrodes **424**, similar to the indentations on the opaque frame **126** shown in FIG. **7D**. As shown in FIG. **16F**, a bottom surface of the electrodes **424** can have a curvature that is generally continuous with the curvature of the opaque frame **426** and the light-transmissive cover **402**. As shown in FIG. **16G**, a top side of the electrodes **424** can have one or more posts **437** extending past openings in the opaque frame **426** into corresponding openings on the PCB **416**. The posts **437** of the electrodes **424** can establish an electrical connection with the corresponding openings of the PCB **416**. A plurality of screws (or other types of fasteners) can extend into the corresponding openings of the PCB **416** from the front side of the PCB **416** to secure the electrodes **424** to the module **401** by threadedly mating with the posts. When a wearer puts the wearable device incorporating the module **401** onto the wearer's wrist, the electrodes **424** can make contact with the wearer's skin. The electrodes **424** can have the same polarity as the electrodes **124** disclosed herein. As disclosed herein, the wearable device incorporating the module **401** can include another ECG electrode **125** located on the housing of the wearable device configured to make contact with the wearer's skin.

[0533] On the second side of the PCB **416**, which faces away from the cover **402**, the PCB **416** can be covered by melt plastic or other suitable electronics protective material **430** (similar to the

protective material **130** disclosed herein) except that a flex connector **432** can remain exposed. The flex connector **432** can be configured to connect the module **401** electrically to the wearable device incorporating the module **401**.

[0534] FIGS. **17A-17C** illustrate an example physiological parameter measurement modules **403** of a wearable device. The module **403** can include the same optical sensor arrangements as shown in FIGS. **16A-16G** and have any of the features of the module **400** in FIGS. **16B-16D** and any of the features of the module **401** in FIGS. **16E-16G** with the differences noted in the description of FIGS. **17A-17C**. The module **401** can have any of the features of the other physiological parameter measurement module examples described herein.

[0535] As shown in FIGS. **17A-17C**, the opaque frame **426** can include an opening fitted with the light transmissive cover **402**. The cover **402** extending over emitter chambers or detector chambers formed by the light barriers **420, 422, 423** and the PCB **415** can include a single lens or cover. The cover **402** can be elliptical in shape. The cover **402** can have a continuous convex curvature. As shown in FIG. **17C**, the light barriers **420, 422, 423** may not extend to the tissue-facing surface of the cover **402** and can extend to below the cover **402** such that when a wearer puts on a wearable device incorporating the module **402**, the wearer's tissue comes into contact with the cover **402** and the electrodes **424**, but not with any of the light barriers **420, 422, 423**.

[0536] FIGS. **18A-19C** illustrate other non-limiting examples of a physiological parameter measurement module with two emitter groups in two separate emitter chambers formed by a light barrier. In those configurations, the perimeter of the module can have a different shape. For example, FIG. **19A** illustrates schematically a module **300** having an outer shape of two circles partially overlapped with each other. The circle in the module **300** can have a radius, for example, between about 6 mm and about 12 mm, or between about 8 mm and about 10 mm. The module **300** can have any of the features of the other modules disclosed herein. The module **300** can include the substantially the same arrangement of emitters **300a, 300b** and detectors **306a, 306b, 306a/b** as the module **400, 401, 403** described above except that each emitter group **304a, 304b** includes three emitters. The module **300** can include a thermistor near each emitter group **304a, 304b**. The module **300** can have a length of, for example, between about 22 mm and about 28 mm, or between about 24 mm and about 26 mm.

[0537] FIG. **18B** illustrates a physiological parameter measurement module **301** including a variation of the arrangement of emitters and detectors of the module **300** in FIG. **18A**, and can include any of the features of the module **300** except for the differences described herein. The module **301** differs from the module **300** by not sharing detectors located between the two groups of emitters **304a, 304b**. The first group of emitters **304a** can be surrounded by a first ring of detectors **306a** on a first side of the minor axis **A2** and the second group of emitters **304b** can be surrounded by a second ring of detectors **306b** that are on a second side of the minor axis **A2**.

[0538] FIG. **19A** illustrates a physiological parameter measurement module **201** including a variation of the arrangement of emitters and detectors of the module **300** in FIG. **18A**. The physiological parameter measurement module **201** can have any of the features of the modules **300**, with the differences noted in the description of FIG. **19A**. The module **201** can have any of the features of the other modules disclosed herein. In the module **201**, the two overlapping circles of detectors **206a, 206b** are closer to each other than in the module **300**. The detectors **206a/b** can be further away from each other than in the module **300** and may not be located between or separating the two emitter groups **204a, 204b**. The module **201** can include two groups of emitters that are separated from each other by one light barrier. Each of the detectors in the module **201** can form its own detector chamber with one or more light barriers. The circle can have a radius, for example, between about 6 mm and about 12 mm, or between about 8 mm and about 10 mm. The module **300** can have a length of, for example, between about 18 mm and about 24 mm, or between about 20 mm and about 22 mm.

[0539] FIGS. **19B** and **19C** illustrate a variation of the module **201** in FIG. **19A** with the differences

noted in the description of FIGS. **19B** and **19C**. The module **200** in FIGS. **19B** and **19C** can have any of the features of the module examples described herein. In FIGS. **19B** and **19C**, a physiological parameter measurement module **200** can include two groups of emitters **204a**, **204b** surrounded by one ring of detectors **206**. The module **200** can have a width, for example, between about 16 mm and about 22 mm, or between about 18 mm and about 20 mm. The module **200** can have a length, for example, between about 20 mm and about 28 mm, or between about 22 mm and about 25 mm.

[0540] Each group of the emitters **204a**, **204b** can include three of emitters. Each group of the emitters **204a**, **204b** can emit at least the first, second, and third wavelength described above. Optionally, each emitter group **204a**, **204b** can include a fourth emitter configured to emit the fourth wavelength that is more sensitive to water. The emitters can be located at or near a center portion of a PCB **216** of the module **200**. The module **200** can include a temperature sensor located on the PCB **216** near each group of the emitters **204a**, **204b**.

[0541] The emitters can be covered by an inner lens or cover **202a**. In the illustrated example, the inner lens or cover **202a** can be generally elliptical. In other examples, the inner lens or cover may have any other shapes. The two groups of the emitters **204a**, **204b** can be located on two parts of the central portion of the PCB divided along the minor diameter of the ellipse. The two groups of the emitters **204a**, **204b** can be divided by an opaque divider barrier **228**, which can reduce mixing of light emitted by the two groups of the emitters **204a**, **204b**. As shown in FIG. **19C**, the divider barrier **228** can have a same or substantially the same height as the highest point of the inner lens or cover **202a** when assembled in the module **200**. The inner lens or cover **202a** can include two components divided by the divider barrier **228**.

[0542] The module **200** can include a plurality of detectors **206** (for example, about six, eight, ten, or more) that can be arranged on the PCB so that the detectors **206** are spaced apart around the emitters **204a**, **204b**. The emitters groups **204a**, **204b** and the detectors **206** can be separated by a first light barrier **220**. The first light barrier **220** can extend along and surround the inner lens or cover **202a**. The divider barrier **228** and the first light barrier **220** can form two emitter chambers **234a**, **234b**, each enclosing one of the two emitter groups **204a**, **204b**. The first light barrier **220** and the divider barrier **228** can also suppress light emitted by the emitters **204a**, **204b** at an angle so the light emitted by each group of emitters **204a**, **204b** can exit the inner lens or cover **202a** in a direction generally parallel to the height of the first light barrier **220**. The detectors **206** can be enclosed within a module side wall **224**. The module side wall **224** can define a perimeter of the module **200**. As shown in FIG. **19B**, the perimeter of the module **200** can have a generally elliptical outer shape. The detectors **206** can be further separated from one another by a plurality of divider barriers **226**, forming detector chambers **236**, each containing one detector **206**.

[0543] As shown in FIG. **19C**, the first light barrier **220** can protrude slightly from, that is, proud of the edge of the inner lens or cover **202a** and the other lenses or covers that will be described below. The detectors **206** can be covered by an outer lens or cover **202b**. The outer lens or cover **202b** can be generally concentric with the inner lens or cover **202a**. In the illustrated examples, the outer lens or cover **202b** can be an elliptical disc as shown in FIG. **19B**. In other examples such as those disclosed herein, the outer lens or cover can have other shapes. As shown in FIG. **19C**, the outer lens or cover **202b** can have a smaller curvature than the inner lens or cover **202a** such that the inner lens or cover **202a** protrudes more than if the inner lens or cover had the same curvature as the outer lens or cover **202b**.

[0544] As shown in FIG. **19C**, the side wall **224** can be shorter than the first light barrier **220**. The height of the side wall **224** can be configured such that the tissue-facing end of the side wall **224** is generally continuous with the curvature of outer lenses or covers **202b**. The divider barriers **226** can have a height lower than the first light barrier **220**. The height of the divider barriers **226** can be configured to accommodate the outer lens or cover **202b** such that when assembled, the outer lens or cover **202b** forms a substantially smooth surface with the module side wall **224**. The tissue-

facing ends of the first light barrier **220** and the side wall **224**, and the tissue-facing surfaces of the inner lens or cover **202a** and the outer lens or cover **202b** can form the tissue-facing surface of the module **200**. The slightly protruding first light barrier **220** and/or inner lens or cover **202a** can be pressed into the wearer's skin at a higher pressure than the remainder of the lens or cover or light barriers.

[0545] The light diffusing materials described above can be included in one or more of the chambers **234a**, **234b**, **236** of the module **200** to improve distribution of emitted lighted and/or detected light. As shown in FIG. **19B**, one or more of the lenses or covers **202a**, **202b** can include an injection opening **244** so that the light diffusing materials, which can include a flow of glass microsphere solution, can be injected into the respective chambers **234a**, **234b**, **236** after the module **200** has been assembled. After the injection, the solution can be UV-cured. The lenses or covers **202a**, **202b** can include one or more venting openings that are smaller than the injection openings **244**. Air can optionally escape via separate vent openings as the diffusing material solution is injected into the respective chambers **234a**, **234b**, **236** via the injection openings **244**. The inner lens or cover **202a** and the outer lens or cover **202b** can also include glass microspheres so as to act as light diffusers.

[0546] FIGS. **27A-27J** illustrate other non-limiting examples of a physiological parameter measurement module **2700** of a wearable device. The module **2700** can incorporate any of the features of the module examples described herein. FIG. **27F** illustrates an example arrangement of an optical sensor, including emitters, detectors, and thermistors, on a sensor or module processor substrate **2716**. The substrate **2716** can include a printed circuit board (PCB). As shown in FIG. **27F**, the substrate **2716** can include a first group of emitters **2704a** and a second group of emitters **2704b**. Each of the first and second groups of emitters **2704a**, **2704b** can include four emitters (or optionally a different number of emitters, such as three, six or eight emitters or any other number of emitters as required or desired). Each of the emitters of the first and second groups of emitters **2704a**, **2704b** may comprise an LED.

[0547] The emitters in the first and second group of emitters **2704a**, **2704b** can include operational and/or structural features discussed above, for example with reference to FIGS. **13A-13B**. For example, the first and second group of emitters **2704a**, **2704b** can be configured to emit a plurality of (for example, three, four, or more) wavelengths. In some aspects, each of the emitters (e.g., within a group) can be configured to emit light of a different wavelength than the other emitters (e.g., of that group). Alternatively, one or more of the emitters can emit light of more than one wavelength. The emitters in the first and second group of emitters **2704a**, **2704b** can emit at least the first, second, third, and fourth wavelengths as described above with reference to FIG. **13B**.

[0548] Each of the emitters in the first group of emitters **2704a** may be located within close proximity to each of the other emitters in the first group of emitters **2704a**. For example, each of the emitters of the first group of emitters **2704a** may be located on the PCB between 0.2 mm and 2 mm from each of the other emitters in the first group of emitters **2704a**. For example, each of the emitters of the first group of emitters **2704a** may be located on the PCB about 0.5 mm from each of the other emitters of the first group of emitters **2704a**. In some aspects, each of the emitters in the first group of emitters **2704a** are positioned such that no more than a certain distance is between each of the emitters in the first group of emitters **2704a**, such as 0.5 mm, 1 mm, 1.5 mm and/or 2 mm, or any other distance as required or desired. Each of the emitters in the second group of emitters **2704b** may be located within close proximity to each of the other emitters in the second group of emitters **2704b**, for example as described above with reference to the first group of emitters **2704a**.

[0549] Each of the two groups of the emitters **2704a**, **2704b** can be surrounded by a first light barrier and form emitter chambers for the group of emitters **2704a** and group of emitters **2704b**, respectively. The first and second groups of emitters **2704a**, **2704b** can be located a distance from each other on a first side of a substrate **2716**. The substrate **2716** can include one or more

temperature sensor(s) (such as a thermistor) **2710** as described above located on the first side of the substrate **2716**. One temperature sensor **2710** can be near the first group of emitters **2704a** within the respective emitter chamber. Another temperature sensor **2710** can be near the second group of emitters **2704b** within the respective emitter chamber.

[0550] The substrate **2716** can be circular in shape, although the shape of the PCB is not limiting. The two groups of the emitters **2704a**, **2704b** can be located on different parts of the first side of the substrate **2716** divided along a center line of the circle. Each of the two groups of the emitters **2704a**, **2704b** can be surrounded by a first light barrier and form an emitter chamber.

[0551] The first and second groups of emitters **2704a**, **2704b** can be surrounded by detectors **2706**. As described in greater detail with reference to FIG. 27A, the detectors **2706** can be positioned on the substrate **2716** in a substantially circular or annular arrangement. The detectors **2706** may surround and/or enclose the first and second groups of **2704a**, **2704b**. Each of the detectors may be a similar or same distance from a geometric center of the substrate **2716** and/or the sensor or module **2700**. In some aspects, the detectors **2706** may be rectangular including longer sides and shorter sides. The detectors **2706** may be positioned on the substrate **2716** such that a long side of each detector is orthogonal to a radius of the substrate **2716**. In the illustrated example, the substrate **2716** includes six detectors coupled to the substrate **2716**, but the total number of detectors can vary.

[0552] The detectors **2706b** can be the far detectors for the first group of emitters **2704a**. The detectors **2706a** can be the near detectors for the first group of emitters **2704a**. The detectors **2706a** can be the far detectors for the second group of emitters **2704b**. The detectors **2706b** can be the near detectors for the second group of emitters **2704b**. The detectors **2706c** can be the intermediate detectors for the first and second groups of emitters **2704a**, **2704b**. Accordingly, each detector **2706a**, **2706b**, **2706c** can receive two signals for each wavelength emitted by the first and second groups of emitters **2704a**, **2704b** respectively. As described above, signals outputted by the far, near and intermediate detectors can provide different information due to the different light paths, which can travel through different areas of the tissue.

[0553] In some aspects, the sensor or module processor may evaluate the various signals outputted by the detectors for example by comparing the signal quality of the detectors. The sensor or module processor may select less than all of the detector signals for processing for each of the far, near and intermediate detectors. For example, the sensor or module processor may rely on signals from one or two detectors from the four possible far detectors, and one or two detectors from the four possible near detectors, and one or two detectors from the four possible intermediate detectors.

[0554] In addition, the far detectors for each group of emitters **2704a**, **2704b** can detect the light emitted by the respective group of emitters **2704a**, **2704b**, for example, light of the fourth wavelength and another wavelength, and attenuated by tissue to provide an indication of the wearer's hydration status as described herein.

[0555] The detectors **2706a**, **2706b**, **2706c** can be separated or partitioned into six detector regions. Each detector region can include one detector, or any other number of detectors. Each detector region can form a detector chamber surrounded by light barriers (as shown in FIG. 26E). As described above, the sensor or module processor can process signals from a particular emitter and received at the detectors within the same detector region as one signal source.

[0556] The module **2700** can include individual lenses or covers or a combination of individual emitter chamber covering lenses or covers and a lens or cover covering a plurality of detector chambers. The lenses or covers may be polycarbonate. The tissue-facing surface of the module **2700** can include a continuous convex curvature.

[0557] FIG. 27I illustrates another example arrangement of an optical sensor, including emitters, detectors, and thermistors, on a sensor or module processor substrate **2716'**. The substrate **2716'** of FIG. 27I can include structural and/or operational features similar to those discussed above with reference to FIG. 27F. As shown in FIG. 27I, each of the first and second groups of emitters

2704a', **2704b'** can include five emitters (or optionally a different number of emitters as required or desired). Each of the emitters of the first and second groups of emitters **2704a'**, **2704b'** may comprise an LED and can be configured to emit light at various wavelengths such as any of the wavelengths discussed herein, for example, a first wavelength of about 525 nm to about 650 nm (such as about 525 nm or about 580 nm or about 645 nm), a second wavelength from about 620 nm to about 660 nm (such as about 625 nm), a third wavelength from about 650 nm to about 670 nm (such as about 660 nm), a fourth wavelength from about 900 nm to about 910 nm, and a fifth wavelength at about 970 nm.

[0558] As shown in FIG. 27I, the substrate **2716'** can include spring contacts **2755'** for facilitating physical and/or electrical connection between the substrate **2716'** and electrodes (e.g., electrodes **2724** shown in FIG. 27B, for example).

[0559] As shown in FIGS. 27G-27H, on the second side of the substrate **2716**, which faces away from the cover, the substrate **2716** can be covered by melt plastic or other suitable electronics protective material **2730** (similar to the protective material **130** disclosed herein) except that a flex connector **2732** can remain exposed. The flex connector **2732** can be configured to connect the module **2700** electrically to the wearable device incorporating the module **2700**.

[0560] FIG. 27J illustrates a perspective cross-sectional view of an example physiological parameter measurement sensor or module **2700**. As shown, the sensor or module **2700** can include a first circuit board (e.g., substrate **2716**) and a second circuit board (e.g., instrument PCB **2768**). The sensor or module **2700** can include an induction coil **2765**. The induction coil **2765** can be electrically and/or physically connected to the substrate **2716**, for example via solder and/or a flex connector connection **2762**. The induction coil can provide a voltage to the substrate **2716**, for example via the solder and/or flex connector connection **2762**. The substrate **2716** can include charging circuitry which can be configured to receive and/or control a voltage received from the induction coil **2765**. The substrate **2716** can be configured to transfer voltage received from the induction coil **2765** to the instrument PCB **2768**, for example via a flex connector **2761**. The instrument PCB **2768** can be configured to transfer voltage received from the substrate **2716** to a battery or to other components of the sensor or module **2700** as required or desired.

Examples of Physiological Parameter Measurement Modules with Inner and Outer Detector Groups and Examples of Wearable Devices Incorporating the Same

[0561] FIGS. 20A-20D illustrate an example physiological parameter measurement module **600** of a wearable device. The module **600** can have any of the features of the module examples described herein, with the differences noted in the description of FIGS. 20A-20D. The physiological parameter measurement module **600** can include a single emitter group having a plurality of emitters **604**, such as four emitters as shown in FIG. 20A, six emitters, or eight emitters. The emitters **604** of the module **600** can emit at least the first, second, third, and fourth wavelengths as described above. The emitters **604** can be located at or near a center portion of a PCB **616** of the module **600**. The module **600** can include a temperature sensor **610** located on the PCB **616** near the emitters **604**.

[0562] The module **600** can include a plurality of detectors **606** that can be arranged on the PCB **616** as an inner group of detectors **606** and an outer group of detectors **606**. The inner group **606c** of detectors **606**, which can include, for example, about ten (or a different number of) detectors **606**, can surround the emitters **604** and be spaced apart from one another.

[0563] The outer group of detectors **606** can be located further away from the emitters **604** than the inner group of detectors **606**. The outer group of detectors **606** can be separated into a first outer group **606a** and a second outer group **606b** of detectors **606**. As shown in FIG. 20A, the module **600** can have a first axis **A1** and a second axis **A2**. The outer groups **606a**, **606b** of detectors **606** can be located further away from the emitters **204** than the inner group of detectors **606** generally along the first axis **A1**. The two outer groups **606a**, **606b** of detectors **606** are on opposite sides of the inner group of detectors along the first axis **A1**. The first and second outer groups **606a**, **606b** of

detectors **606** can be generally symmetrical about the first axis **A1** and the second axis **A2**. Each of the first or second outer groups **606a**, **606b** of detectors **606** can include about five (or a different number) of detectors **606** that are spaced apart from one another generally along the second axis **A2**. The outer groups **606a**, **606b** of detectors **606** can be arranged to be generally concentric with the inner group **606c** of detectors **606**.

[0564] The module **600** can be longer in the first axis **A1** than in the second axis **A2**. The module **600** can have a dimension of about 25.4 mm (1 inch) along the first axis **A1**. The module can have a dimension of about 19.1 mm (0.75 inch) along the second axis **A2**. As shown in FIG. 20A, when a watch incorporating the module **600** is worn on the wrist of a wearer, the first axis **A1** can be generally parallel to the width of the wrist and generally perpendicular to the direction of blood flow along the wrist (that is, along a direction between the hand and the forearm) and the second axis **A2** can be generally perpendicular to the width of the wrist and generally parallel to the direction of blood flow along the wrist. The distribution of the detectors **606** along the first axis **A1** can improve detection of the light attenuated by the pulsing arterial blood in the capillaries as the detectors **606** are arranged to cover a greater cross-section of the blood flow through the wrist. Similarly, in the other example modules described herein, such as the sensor or module **100**, **400**, **401**, **403**, **300**, **301**, **200**, **201**, the physiological parameter measurement module is incorporated in the wearable device such that the longer side of the module is generally perpendicular to the direction of the blood flow along the wrist (see, for example, FIG. 1B) when the wearable device is worn on the wrist.

[0565] As shown in FIG. 20A, the emitters **604** can be covered by an inner lens or cover **602a**. In the illustrated example, the inner lens or cover **602a** can be generally circular. In other examples such as disclosed herein, the inner lens or cover may not be generally circular, but can have other shapes, for example, elliptical, rectangular, square, diamond, or otherwise. The inner group **606c** of detectors **606** can be covered by a first outer lens or cover **602b**. The first outer lens or cover **602b** can be generally concentric with the inner lens or cover **602a**. In the illustrated example, the first outer lens or cover **602b** can be disc shaped. The first and second outer groups **606a**, **606b** of detectors **606** can be covered by a second outer lens or cover **606c** and a third outer lens or cover **606d** respectively. The second and third outer lenses or covers **606c**, **606d** can be symmetrical about the second axis **A2**. As shown in FIG. 20B, the first, second, and third outer lenses or covers **602b**, **602c**, **602d** can have substantially the same curvature. The inner lens or cover **602a** can be more curved than the outer lenses or covers **602b**, **602c**, **602d** such that the inner lens or cover **602a** protrudes more than if the inner lens or cover **602a** had same curvature as the outer lenses or covers **602b**, **602c**, **602d**.

[0566] The inner group **606c** of detectors **606** and the emitters **604** can be separate by a first light barrier **620**. The first light barrier **620** can extend along and surround the inner lens or cover **602a**, forming an emitter chamber. The first and second outer groups **606a**, **606b** of detectors **606** can be separated from the inner group **606c** of detectors **606** by a second light barrier **622**. The second light barrier **622** can be shorter than the first light barrier **620**. The first and second outer groups **606a**, **606b** of detectors **606** can be enclosed within a module side wall **624** enclosing a perimeter of the module **600**. The perimeter of the module **600** can be elliptical or any other shape. The side wall **624** can be shorter than the second light barrier **622**. The height of the first and second light barriers **620**, **622**, and of the side wall **624** can generally follow or be substantially continuous with the curvature of the first, second, and third outer lenses or covers **602b**, **602c**, **602d**. The first and second light barriers **620**, **622**, and of the side wall **624** can have a height so as to be configured to contact the skin of the wearer. Accordingly, the tissue-facing surface of the module **600** can be defined by the tissue-facing side of the first and second light barriers **620**, **622**, and of the side wall **624** and tissue-facing surfaces of the inner lens or cover **602a** and the first, second, and third outer lenses or covers **602b**, **602c**, **602d**.

[0567] In the illustrated example, the inner group **606c** of detectors **606** can be separated by a third

light barrier **626** and a fourth light barrier **628** (see FIGS. **20C** and **20D**). The third and fourth light barriers **626**, **628** can have a height lower than the first light barrier **620** or the second light barrier **622**. The height of the third and fourth light barriers **626**, **628** can be configured to accommodate the first outer lens or cover **602b** such that when assembled, the first outer lens or cover **602b** forms a substantially smooth surface with the second and third outer lenses or covers **602c**, **602d**. The first outer lens or cover **602b** can sit on top of the third and fourth light barriers **626**, **628**.

[0568] The first light barrier **620** can protrude slightly from, that is, sit proud of the edge of the inner lens or cover **602a** and the outer lenses or covers **602b**, **602c**, **602d**. The slightly protruding first light barrier **620** and/or inner lens or cover **602a** can be pressed into the wearer's skin at a higher pressure than the remainder of the lenses or covers or light barriers. The first light barrier **620** can also reduce mixing of the emitted and reflected light and/or suppress light emitted by the emitters **604** at an angle so that the emitted light exits the inner lens or cover **602a** generally in a direction parallel to the height of the first light barrier **620**.

[0569] As shown in FIGS. **20C** and **20D**, the first, second, third, and fourth light barriers **620**, **622**, **626**, **628** and the side wall **624** can optionally form a single light barrier construct **630**. The single light barrier construct **630** can be formed by any suitable manufacturing techniques. The single light barrier construct **630** can include at one end a recess **632** (see FIG. **20C**) that is configured to receive the PCB **616** (and the emitters **604**, detectors **606**, temperature sensor **610**, and any other sensors, for example, the gyroscope, the accelerometer, and/or the like, and the sensor or module processor, which are located on the PCB **616**). The single light barrier construct **630** can receive the lenses, including the inner lens or cover **602a**, the first, second, and third outer lenses or covers **602b**, **602c**, **602d** at another end that is opposite to the end including the recess **632**.

[0570] The module housing can include a plurality of chambers such that light cannot travel between the chambers because of the various light barriers described herein. As described above, the first chamber can be enclosed by the inner lens or cover **602a**, the first light barrier **620**, and a portion of the PCB **616**. The first chamber **634** enclose the emitters **604**. A second chamber and a third chamber can be enclosed by the first outer lens or cover **602b**, the first light barrier **620**, the second light barrier **622**, the third light barrier **626**, the fourth light barrier **628**, and a portion of the PCB **616**. The second and third chambers can enclose the inner group **606c** of detectors **606**, with half of the inner group **606c** of detectors enclosed by each of the second and third chambers. A fourth chamber can be closed by the second outer lens or cover **602c**, the second light barrier **622**, the side wall **624**, and part of the PCB **616**. A fifth chamber can be enclosed by the third outer lens or cover **602d**, the second light barrier **622**, the side wall **624**, and part of the PCB **616**. The fourth and fifth chambers can enclose the first and second outer groups **606a**, **606b** of detectors **606** respectively.

[0571] Light from the emitters **604** can travel a shorter path to the inner group **606c** of detectors **606** and a longer path to the first and second outer groups **606a**, **606b** of detectors **606**. The inner group **606c** of detectors **606** and the first and second outer groups **606a**, **606b** of detectors **606** can be run independently and/or simultaneously. Signals outputted by the inner and outer groups **606a**, **606b** of detectors **606** can provide different information due to the different light paths, which can travel through different areas of the tissue. The longer path penetrates deeper into the tissue and through a greater volume of the tissue to reach one of the outer groups **606a**, **606b** of detectors **606** than the short path, which penetrates less deep into the tissue and travels through a smaller volume of tissue to reach one of the inner group **606c** of detectors **606**. The different information can be separated and/or combined to calculate a plurality of physiological parameters of the wearer of the module **600**, for example, an indication of the wearer's hydration status, which will be described in greater detail below.

[0572] The light diffusing materials described above can be included in one or more chambers of the module **600** to improve distribution of emitted light and/or detected light after attenuation by the tissue. As shown in FIG. **20A**, one or more of the lenses or covers **602a**, **602b**, **602c**, **602d** can

include an injection opening **644** so that the light diffusing materials, which can include a flow of glass microsphere solution, can be injected into the respective chambers after the module **600** has been assembled. After being injected into the respective chamber, the solution can be UV-cured. The lenses or covers **602a**, **602b**, **602c**, **602d** can include one or more venting openings **645** that are smaller than the injection openings **644**. Each of the lenses or covers can include at least one venting opening **645**. Air can escape via the vent openings **645** as the diffusing material solution is injected into the respective chambers via the injection openings **644**, making it easier for the glass microsphere solution to flow into the respective chamber. The inner lens or cover **602a** and/or the outer lenses or covers **602b**, **602c**, **602d** can also include glass microspheres. The light diffusing materials in the inner lens or cover **602a** and the UV-cured material in the first chamber **634** and/or the first light barrier **620** can make the emitted light leave the first chamber **634** in a direction generally parallel to the height of the first light barrier **620**. The light diffusing materials in the outer lenses or covers **602b**, **602c**, **602d** and the UV-cured material in the other chambers **636**, **638**, **640**, **642** can increase the amount of reflected light being directed to the detectors **606**.

[0573] The module **600** shown in FIGS. **20A-20D** can be incorporated in a wearable device disclosed herein, such as a watch **900** shown in FIGS. **20E-20J**. The watch processor **914** and power source can be enclosed within the watching housing **901**. The watch housing **901** can include a connection port opening **950** configured to allow access to a connection port **952** that is in electrical communication with the watch processor **914** and/or the power source. The connection port opening **950** can be located at one end of the watch housing **901** transverse to the first axis **A1** of the module **600**. The connection port **952** can allow for charging of the power source and/or data transfer to and from the watch processor **914**. Optionally, as shown in FIGS. **20F** and **20I**, the watch **900** can include a cable connector **945** extending outward from the watch housing **901**. The cable connector **945** can be located adjacent to or near the connection port opening **950**.

[0574] The watch **900** can include a display screen **912** positioned at a first side of the watch housing **901**. The watch housing **901** has a second side that is opposite the first side. The second side of the watch housing **901** can include an opening sized to retain the physiological parameter measurement module **600** while still allowing the tissue-facing surface of the module **600** to be exposed. The second side of the watch housing **901** can be removably attached to the first side of the watch housing **901** without using external fasteners or alternatively via one or more fasteners. An electrical connection can be established between the physiological parameter measurement module PCB and the watch circuit, for example, using a flex connector as disclosed herein.

[0575] The watch housing **901** can include strap coupling extensions **948** on opposing sides of the watch **900** along the length of the housing **901** (that is, along the first axis **A1** of the module **600**). The extensions **948** can include a bar **946** for coupling to any suitable watch straps.

[0576] FIGS. **21A-21C** and **22A-22C** illustrate alternative lens or cover curvatures of the physiological parameter measurement module **600** of FIGS. **20A-20D** and can incorporate any of the features of the module **600** of FIGS. **20A-20D** except the differences described below. As shown in FIGS. **21A-21C**, the first outer lens or cover **602b** of the module **601** can be more convex (that is, protrude more) than the inner lens or cover **602a** the second and third outer lenses or covers **602c**, **602d**. The curvatures of the tissue-facing side of the second light barrier **622** and of the side wall **624** can be substantially continuous with the curvature of the second and third outer lenses or covers **602c**, **602d**. The second light barrier **622** can be shorter than the first light barrier **620**. The first light barrier **620** can be higher than an outer edge of the inner lens or cover **602a**, which can facilitate separation of light emitted by the emitters **604** and light being detected by the detectors **606** before the light is attenuated by the wearer's body tissue. In the FIGS. **22A-22C**, the module **603** can be different from the module **601** in FIGS. **21A-21C** in that the inner lens or cover **602a** can have the same height as the first light barrier **620** and the first outer lens or cover **602b**. The inner lens or cover **602a** can have a generally flat surface or a slight curvature that can be substantially continuous from the curvature of the first outer lens or cover **602b**. The module **601**,

603 in FIGS. **21A-22C** can facilitate pressing the first outer lens or cover **602b** or the first outer lens or cover **602b** and the inner lens or cover **602a** into the skin of the wearer more than the remainder of the tissue-facing surface of the module **600**.

[0577] FIGS. **23A-23E** illustrate a watch **700** that can incorporate the physiological parameter measurement module **600**. The watch **700** can have any of the features of the watch **900** with the differences noted in the description of FIGS. **23A-23E**. As shown in FIGS. **23A-23E**, the watch housing **701** of the watch **700** can include a flap **750** on a side of the housing **701** along a length of the watch housing **701**, which is along the first axis **A1** of the physiological parameter measurement module (see FIG. **23E**). The flap **750** can be opened to give access to a connection port (such as the connection port in the watch **900**) in electrical communication with the watch processor **714** and/or the power source **716**. The connection port can allow for charging of the power source **716** and/or data transfer to and from the watch processor **714**. The flap **750** can be closed when the connection port **752** is not in use.

[0578] The watch **700** can include a display screen positioned at a first side of the watch housing **701**. The watch housing **701** has a second side that is opposite the first side. The second side of the watch housing **701** can include an opening sized to retain the physiological parameter measurement module **600** while still allowing the tissue-facing surface of the module **600** to be exposed. The second side of the watch housing **701** can be removably attached to the first side of the watch housing **701** via one or more screws **718** or other fasteners. When fully assembled, the watch **700** can have a thickness or height, for example, between 10 mm to about 15 mm, or between 12 mm to about 14 mm.

[0579] The watch housing **701** can include suitable strap connections configured to couple to watch strap(s). The strap connections in the watch housing **701** can be different from the strap connections shown in the watch **900**. In an example, a plurality of strap openings can be at opposite ends of the watch and the watch housing can additionally and/or alternatively include strap slots on the same opposite ends of the watch as the strap openings. In this example, the strap slots can be configured to slidably receive ends of watch straps that include a shape corresponding to the shape of the strap slots. The strap openings can be configured to receive spring-biased buttons near the ends of the watch straps to releasably retain the straps after the ends of the watch straps are received into the strap slots. Alternatively, the watch may not include strap openings. The strap(s) coupled to the watch examples disclosed herein can be configured to allow adjusting of tightness around the wearer's wrist, for example, using a buckle connector, a Velcro connector, and/or the like.

Hydration Monitoring by Wearable Devices Incorporating Examples Physiological Parameter Measurement Modules with “Near” and “Far” Detectors or Detector Groups

[0580] The physiological parameter measurement module examples disclosed herein can monitor a hydration status of the wearer. This is because water in the body tissue can allow a greater portion of the light of the third (or first or second) wavelength disclosed herein to go through (that is, acting as a light pipe), but can bulk absorb the light of the fourth wavelength disclosed herein. The physiological parameter measurement processor can compare intensity signals of the fourth wavelength and another wavelength that is less sensitive to changes in water from the same detector(s). When the wearer's hydration status is in a normal range such that the wearer is not considered dehydrated in a medical sense, the signals of the fourth wavelength and the other wavelength can show opposite trends, that is, one is increasing when the other one is decreasing. When the wearer becomes dehydrated in a medical sense, the opposite trends can become less distinct, for example, by falling below a threshold.

[0581] Hydration monitoring can be performed when the physiological parameter measurement module, such as the sensor or module **100**, is configured such that at least some of the detectors **106** are located further away (far detector) from one of the emitters **104** (or emitter groups) than the other detectors **106** (near detector), such as illustrated in FIG. **10**. In configurations where there are

two emitter groups, each detector **106** or detector region (which can include more than one detector **106** placed enclosed in the same detector chamber) can act as a near (or shallow) detector or detector region for the group of emitters that are closer to that detector **106** or detector region and as a far (or deeper) detector or detector region for the group of emitters that are further away from that detector **106** or detector region.

[0582] The physiological parameter measurement module **400**, **401**, **403** illustrates an example configuration for hydration monitoring of the wearer. The detectors **406a** can be the far detectors for the second group of emitters **404b** and the detectors **406b**, **406a/b** can be the near detectors for the second group of emitters **404b**. The detectors **406b** can be the far detectors for the first group of emitters **404a** and the detectors **406a**, **406a/b** can be the near detectors for the first group of emitters **404a**. The physiological parameter measurement modules **300**, **301** illustrate similar detector arrangements in configurations (except that in the module **301**, there are no shared detectors between the two groups of emitters **304a**, **304b**) where the modules **300**, **301** include a fourth emitter in at least one of the emitter groups configured to emit light of the fourth wavelength.

[0583] The physiological parameter measurement modules **200**, **201** illustrate additional example detectors configurations that can include “near” detectors for one emitter group and “far” detectors for another emitter group, in configurations where the modules **200**, **201** include a fourth emitter configured to emit light of the fourth wavelength. For example, the detectors **206** on the far side of each group of emitters **204a**, **204b** can act as “far” detectors for detecting the light emitted by the respective group of emitters **204a**, **204b**, for example, light of the fourth wavelength and another wavelength, and attenuated by tissue to provide an indication of the wearer's hydration status

[0584] The physiological parameter measurement module **600** illustrates an example configuration for hydration monitoring of the wearer, with the inner group **606c** of detectors **606** acting as the “near” detectors and the outer groups **606a**, **606b** of the detectors acting as the “far” detectors.

[0585] In the above-described configurations, each detector or detector region can provide two measurements calculated from the signals received from the closer emitter group and the signals from the further emitter group respectively. Signals detected at the far detectors can provide indication of the hydration status of the wearer as light travels through a deeper portion of the tissue of the wearer to reach the far detectors than to reach the near detectors). Signals detected at the near detectors can optionally be used as reference or for comparison with the signals detected at the far detectors when the physiological parameter measurement sensor or module processor determines the wearer's hydration status. The sensor or module processor of the physiological parameter measurement module disclosed herein can compare intensity signals of the fourth wavelength and another wavelength (for example, the third wavelength or about 905 nm) that is less sensitive to changes in water from one of the “far” detectors. The module processor can focus on the DC component, or the DC bulk absorption measurement of the signals detected by the “far” detectors for hydration status monitoring. At the DC level, water can act as a light block (that is, less transmissive of light) for the fourth wavelength and as a lens or cover (that is, more transmissive of light) for the other wavelength.

[0586] Additionally and/or alternatively, any of the modules disclosed herein can monitor the wearer's hydration status by monitoring the wearer's PVI values. The module can determine a baseline PVI value of the wearer, and can output a notification that the wearer is dehydrated or hydrated based on fluctuations in the PVI value from the baseline.

[0587] The module can further combine the hydration status monitoring by the optical detectors and other sensors (such as a sweat sensor or a skin impedance sensors) in outputting a final hydration status indication of the wearer. The module can calculate an average, a weight average or otherwise of raw hydration index values calculated based on signals from the different sensors, and/or rely on the different hydration monitoring sensors for redundancy.

[0588] As a person's hydration status is not expected to change rapidly, the physiological parameter measurement module can optionally make a measurement of the hydration status less frequently

than making measurements related to the wearer's pulse rate or SpO2 or other parameters. For example, the physiological parameter measurement sensor or module processor can make a measurement of hydration status every 5 minutes, or longer, and/or upon (for example, only upon) a request by the wearer, such as when the wearer presses a button (a physical button and/or a touch button on the display) on the device or otherwise instructs the device using voice commands, hand gestures, and/or the like.

Examples of Generally Circular Physiological Parameter Measurement Modules and Examples of Wearable Devices Incorporating the Same

[0589] A physiological parameter measurement module can alternatively include an inner portion of emitters and an outer ring of detectors as shown in FIGS. 24A-24B and FIGS. 25A-25B. The sensor or module **1000** in FIGS. 24A-24B and the module **1100** in FIGS. 25A-25B can have any of the features of the module examples described herein, with the differences noted in the description of FIGS. 24A-24B and 25A-25B. Such a physiological parameter measurement module can have a generally circular outer shape. The sensor or module **1000** in FIGS. 24A-24B can be smaller than the module **1100** in FIGS. 25A-25B. For example, the sensor or module **1000** can have an outer diameter between about 12 mm and about 16 mm, or between about 14 mm and about 15 mm. For example, the module **1100** can have an outer diameter between about 16 mm and about 22 mm, or between about 18 mm and about 20 mm.

[0590] The physiological parameter measurement module **1000**, **1100** can each include a single emitter group having a plurality of emitters **1004**, **1104**, such as three emitters. The emitters **1004**, **1104** of the sensor or module **1000**, **1100** can emit at least the first, second, and third wavelengths as described above. The emitters **1004**, **1104** can be located at or near a center portion of a PCB of the sensor or module **1000**, **1100**. The sensor or module **1000**, **1100** can include a temperature sensor located on the PCB near the emitters **1004**, **1104**.

[0591] The sensor or module **1000**, **1100** can include a plurality of detectors **1006**, **1106** (for example, about six, eight, or more) that can be arranged on the PCB so that the detectors **1006**, **1106** are spaced apart around the emitters **1004**, **1104**. The emitters **1004**, **1104** and the detectors **1006**, **1106** can be separated by a first light barrier **1020**, **1120**. The first light barrier **1020**, **1120** can surround the emitters **1004**, **1104**. The first light barrier **1020**, **1120** can also suppress light emitted by the emitters **1004**, **1104** at an angle so that the emitted light exits the inner lens or cover **1002a**, **1102a** in a direction generally parallel to the height of the first light barrier **1020**, **1120**.

[0592] The emitters **1004**, **1104** can be covered by an inner lens or cover **1002a**, **1102a**. In the illustrated example, the inner lens or cover **1002a**, **1102a** can be generally circular. The detectors **1006**, **1106** can be covered by an outer lens or cover **1002b**, **1102b**. The outer lens or cover **1002b**, **1102b** can be generally concentric with the inner lens or cover **1002a**, **1102a**. In the illustrated examples, the outer lens or cover **1002b**, **1102b** can be a disc when viewed directly above from the sensor or module **1000**, **1100**. In other examples such as those disclosed herein, the outer lens or cover can have other shapes, for example, being elliptical or otherwise. The outer lens or cover **1002b**, **1102b** can have a smaller curvature than the inner lens or cover **1002a**, **1102a** such that the inner lens or cover **1002a**, **1102a** protrudes more than if the inner lens or cover had the same curvature as the outer lens or cover **1002b**, **1102b**. As shown in FIGS. 24B and 25B, the first light barrier **1020**, **1120** can protrude slightly from, that is, proud of the outer edge of the inner lens or cover **1002a**, **1102a**. The slightly protruding first light barrier **1020**, **1120** and/or inner lens or cover **1002a**, **1102a** can be pressed into the wearer's skin at a higher pressure than the remainder of the light barriers or lenses or covers of the sensor or module **1000**, **1100**.

[0593] The detectors **1006**, **1106** can be enclosed within a module side wall **1024**, **1124** that defines a perimeter of the sensor or module **1000**, **1100**. The perimeter can be generally circular or of any other shape. The side wall **1024**, **1124** can be shorter than the first light barrier **1020**, **1120**. The height of the side wall **1024**, **1124** can be such that the tissue-facing end of the side wall **1024**, **1124** is generally continuous with the curvature of outer lenses or covers **1002b**, **1102b**. In the illustrated

example, the detectors **1006, 1106** can be separated from one another by a plurality of generally opaque divider barriers **1026, 1126**. The divider barriers **1026, 1126** can have a height lower than the first light barrier **1020, 1120**. The height of the divider barriers **1026, 1126** can be configured to accommodate the outer lens or cover **1002b, 1102b** such that when assembled, the outer lens or cover **1002b, 1102b** forms a substantially smooth surface with the module side wall **1024, 1124**. The outer lens or cover **1002b, 1102b** can sit on top of the divider barriers **1026, 1126**. The tissue-facing end of the first light barrier **1020, 1120** and the side wall **1024, 1124**, and the tissue-facing surfaces of the inner lens or cover **1002a, 1102a** and the outer lens or cover **1002b, 1102b** can be configured to contact the skin of the wearer and form the tissue-facing surface of the sensor or module **1000, 1100**.

[0594] The first light barrier **1020, 1120**, the side wall **1024, 1124**, and the divider barriers **1026, 1126** can optionally form a single light barrier construct. The single light barrier construct can receive the PCB of the sensor or module **1000, 1100**, and the emitters **1004, 1104**, detectors **1006, 1106**, temperature sensor, and any other sensors, for example, the gyroscope, the accelerometer, and/or the like, and the sensor or module processor that are located on the PCB. The single light barrier construct can receive the lenses, including the inner lens or cover **1002a, 1102a** and the outer lens or cover **1002b, 1102b** on another end that is opposite the end receiving the PCB. As shown in FIGS. 25A and 25B, the light barrier construct of the module **1100** or the PCB can additionally include a plurality of (for example, four or otherwise) extension prongs **1152**. The plurality of extension prongs **1152** can be generally equally spaced around the side wall **1124**.

[0595] The sensor or module **1000, 1100** can include a plurality of chambers such that light cannot travel between the chambers because of the various light barriers described herein. A first chamber **1034, 1134** can be enclosed by the inner lens or cover **1002a, 1102a**, the first light barrier **1020, 1120**, and a portion of the PCB. The first chamber **1034, 1134** can enclose the emitters **1004, 1104**. A plurality of second chambers **1036, 1136** can be enclosed by the outer lens or cover **1002b, 1102b**, the first light barrier **1020, 1120**, the divider barriers **1026, 1126**, the side wall **1024, 1124**, and part of the PCB. Each of the second chambers **1036, 1136** can enclose one detector **1006, 1106**. [0596] The light diffusing materials described above can be included in one or more of the chambers **1034, 1134, 1036, 1136** of the module housing to improve distribution of emitted light and/or detected light. The inner lens or cover **1002a, 1102a** and the outer lens or cover **1002b, 1102b** can also include glass microspheres as described above.

[0597] The watch **1200** in FIGS. 25C-25H is illustrated as incorporating the module **1100** shown in FIGS. 25A-25B. However, any of the example watches disclosed herein can incorporate the physiological parameter measurement module **1000, 1100** shown in FIGS. 24A-24B or FIGS. 25A-25B. The watch **1200** can have any of the features of the wearable devices disclosed herein, such as the watch **700, 900**, all of which are not repeated for brevity. The watch processor **1214** and power source can be enclosed within the watching housing **1201**. The watch housing **1201** can include a connection port opening **1250** configured to allow access to a connection port **1252** in electrical communication with the watch processor **1214** and/or the power source. The opening **1250** can be on one side of the watch **1200** perpendicular to the first axis A1 of the module **1100**, closer to the strap coupling mechanisms. The connection port **1252** can allow for charging of the power source and/or data transfer to and from the watch processor **1214**. Optionally, as shown in FIGS. 25D, 25F, and 25H, the watch **1200** can include a cable connector **845** extending outward from the watch housing **1201**. The cable connector **1245** can be located adjacent to or near the connection port opening **1250**.

[0598] The watch **1200** can include a display screen **1212** positioned at a first side of the watch housing **1201**. The watch housing **1201** has a second side that is opposite the first side. The second side of the watch housing **1201** can include an opening sized to retain the physiological parameter measurement module **1100** while still allowing the tissue-facing surface of the module **1100** to be exposed. The extension prongs **1152** of the module **1100** can be received into corresponding

structures, for example, recesses, on the second side of the watch housing **1201**, which can prevent rotation of the module **1100** when being installed in the watch **1200**. The second side of the watch housing **1201** can be removably attached to the first side of the watch housing **1201** without using external fasteners or via one or more fasteners as described above. An electrical connection can be established between the physiological parameter measurement module circuit and the watch circuit. Optionally, the electrical connection can include a flex circuit.

[0599] The watch housing **1201** can include strap coupling extensions **1248** on opposite sides of the watch **1200** along the first axis **A1** of the module **1100**. The extensions **1248** can include a bar **1246** for coupling to any suitable watch straps.

Example Second Sensor Connection on Physiological Parameter Measurement Modules for Preventing Opioid Overdose

[0600] The physiological parameter measurement module examples disclosed herein can include an optional connector **118** (see FIG. 7A) for receiving a second sensor, which can be a plethysmograph sensor or other suitable sensors. The connector **118** can be oriented such that the second sensor can extend from a housing of the device **10** with reduced or no impingement of the tissue at the device/tissue interface, resulting in less or no effect of the connection of a second sensor to the connector **118** on the blood flow through the device measurement site. The second plethysmograph sensor can include any suitable plethysmograph sensors, for example, a fingertip sensor configured to monitor opioid overdose as described in U.S. Pub. No. 20190374173, the entirety of which is incorporated herein by reference and should be considered part of the disclosure. FIG. 1C illustrates a non-limiting example of the second sensor **119** that is a fingertip sensor. The second sensor **119** can extend from a wearable device as shown in FIG. 1C or any of the wearable device examples disclosed herein.

[0601] Alternative to the connection to a wearable device as shown in FIG. 1C, the connector from the watch disclosed herein can extend from an opening on a tissue-facing side of the device housing, for example, on a raised platform **703**, **903** (FIGS. 201 and 23A). The connector can be coupled to the PCB **616** via a cable, which can optionally have a length configured to extend around the raised platform **703**, **903**, for example, in a groove of the raised platform **703**, **903**, or otherwise. Having the cable extending around the raised platform **703**, **903** can allow adjustment of the slack of the cable when the connector connects to the second sensor. Having the connector extending from an opening on the raised platform **703**, **903** can also avoid the connector and/or the cable impinging on the tissue at the watch/tissue interface as described above. The connector can alternatively be located at other suitable locations on the watch **700**, **900**.

[0602] The second plethysmograph sensor can have a higher measurements accuracy than the physiological parameter measurement module disclosed herein. The wearer can disconnect and/or deactivate the second sensor while the wearer is awake and/or moving about. The wearer can connect and activate the second sensor, for example, when going to sleep or resting. The sensor or module processor can ignore signals from the detectors of the module when the second sensor is activated so that the sensor or module processor can output physiological parameters based on the readings from the second sensor. Alternatively, the sensor or module processor can output physiological parameters based on a combination of the readings from the second sensor and the detectors of the module. The wearer can have the flexibility of choosing to use the physiological parameter measurement module and/or the second sensor, depending on the wearer's need.

[0603] The second plethysmograph sensor can aid in detection of opioid overdose in a wearer who uses opioid (for example, for medical reasons), in particular, by detecting low saturation of oxygen in the blood of the wearer. Depressed breathing is the most dangerous side effect of opioid overdose. Lack of oxygen to the brain can not only result in permanent neurologic damage, but may also be accompanied by the widespread failure of other organ systems, including the heart and kidneys. If a person experiencing an opioid overdose is left alone and asleep, the person could easily die as the respiratory depression worsens. The second plethysmograph sensor can be

configured to detect depressed breathing by detecting decreased oxygen saturation in the blood of the wearer. The wearable device can be configured to automatically notify a first responder and/or the wearer's family or guardian in response to detecting opioid overdose of the wearer.

[0604] Optionally, the device processor of the wearable device can be in communication (for example, via Bluetooth or NFC communication, or via the network) with a processor of a drug delivery apparatus that is wearable by the wearer and configured to deliver one or more doses of a therapeutic drug, such as opioid. The drug delivery apparatus can include a delivery device that includes a dose of a therapeutic drug stored in a reservoir, a drug delivery channel, a dispensing device to dispense the therapeutic drug from the reservoir through the drug delivery channel, and activation circuitry to activate the dispensing device. The processor of the drug delivery apparatus can receive the parameters measured by the second plethysmograph sensor of the wearable device disclosed herein. The processor of the drug delivery apparatus can store memory-storing instructions and be configured to execute the instructions to at least compare the received parameters from the wearable device to a threshold that is indicative of opioid overdose. The processor of the drug delivery apparatus can determine whether an overdose event is occurring or likely to occur based on the comparison and send at least one activation signal to the drug delivery apparatus to dispense at least one dose of the therapeutic drug based on the determination.

[0605] Alternatively, the sensor or module processor of the physiological parameter measurement module can perform the comparison of the parameters measured by the second plethysmograph sensor to the predetermined opioid overdose threshold. Optionally, a microneedle patch may be used for providing a medication that can counteract opioid overdose. The wearer can apply the microneedle patch containing the medication to the skin when the wearable device outputs an alert that the wearer's physiological parameters (for example, SpO₂) has exceeded a threshold (which may be indicative of opioid overdose).

[0606] Alternatively or additionally, the second sensor can be any other suitable noninvasive sensor disclosed herein. Alternatively or additionally, the physiological parameter measurement module examples disclosed herein can connect to a second sensor via wireless connection, for example, using Bluetooth technology. The module can receive measured parameters from the connected second sensor and/or process the sensor data received from the second sensor to calculate additional physiological parameters.

Example Microneedle Patch

[0607] In addition and/or alternative to delivering medication to prevent opioid overdose as described herein, a microneedles patch can be used for other purposes in combination with the wearable device. Microneedles have been used in recent years as a painless alternative to hypodermic needles to deliver drugs to the body. Microneedles on a patch can be placed on an arm or leg, or other parts of the body, which then create small holes in the skin's outermost layer, allowing the drugs coated on each needle to diffuse into the body. Microneedles can be made from silicon, metals, synthetic polymers, or natural, biodegradable materials such as silk and chitin.

[0608] Because of the small size, microneedles are minimally invasive and cause less pain compared to larger needles (for example, hypodermic needles). Additionally, the microneedle patch are easier to apply by the wearer than a hypothermal needle. In comparison, larger needles may require correct injection depth and injection angle to ensure that the drugs are injected at a right location.

[0609] FIG. 26A illustrates schematically a microneedle **3100** of a microneedle patch that has penetrated the tissue surface **2** of the wearer. Depending on its height, the microneedle **3100** may have varying injection depths. For example, the microneedle **3100** may puncture just the epidermis (including the stratum corneum, which is the outer layer of the epidermis) **42**. In other examples, the microneedle **102** may puncture the epidermis **42** and dermis **44**, with a tip of the microneedle **3102** terminating in the dermis **44**. In other examples such as shown in FIG. 26A, the microneedle **3100** may puncture the epidermis **42** and dermis **44**, with the tip **3102** end in the subcutaneous

tissues **46**.

[0610] Depending on the use, the microneedles **3100** with different heights may be used for delivery of medication and/or irrigation fluid **3104** into different parts of the wearer's tissue. The microneedles **3100** can be used to deliver a broad range of drugs, biotherapeutics, and vaccines. The microneedles **3100** can be hollow with internal reservoirs to store and deliver drugs or irrigation fluid **3104**. Alternatively, the microneedles **3100** can be solid and coated with drugs **3104**, and optionally other surfactant/thickening agents. Optionally, the microneedle **3100** can be dissolvable and encapsulate the drug in a nontoxic polymer that can dissolve once inside the skin.

[0611] Alternatively or additionally, the microneedles **3100** can be used to extract a tissue fluid sample **3104** (for example, the interstitial fluid of the wearer) for detection and/or analysis of analytes in the sample **3104**. Optionally, the microneedle **3100** can irrigate the tissue of the wearer with a fluid before extracting the fluid (which, for example, may have equilibrated with the chemical composition of the wearer's bodily fluid sample) back into the microneedles **3100**. The microneedles **3100** can be hollow and can extract a fluid sample via surface tension. The analyte detection and/or analysis can provide information such as the hydration status, glucose concentration, hemoglobin concentration, and/or orthogonal information about the fluid. The analyte detection and/or analysis can provide additional information related to, for example, sodium, potassium, glucose, chloride, bicarbonate, blood urea nitrogen, magnesium, creatinine, LDL cholesterol, HDL cholesterol, triglyceride, pH, and the like.

[0612] A microneedle patch may be located under one of the straps or the body of the wearable device, or be applied remotely (anywhere else on the wearer's body) from the wearable device without contacting the device. A plurality of microneedle patches can be applied to the wearer at different locations on the wearer's body. As shown in FIGS. **26B** and **26C**, the microneedles **3100** may be connected to a patch body **3106**, forming a microneedle patch **3108**. The patch body **3106** may be circular, oval, rectangular, square, triangular, tear-drop shaped, or of any other shape. The size of the patch body **3106** is not limiting. A surface of the patch body **3106** that is not connected to the microneedles **3100** can include an adhesive layer for releasably attach the patch **3108** to the wearable device. The adhesive layer may be covered by a back layer, which can be peeled off before applying the patch **3108** to the wearable device.

[0613] As shown in FIG. **26B**, the microneedle patch **3108** can be placed on the body of the device **10**. The patch **3108** can be applied under the skin-facing surface of the physiological parameter measurement sensor or module **100**. The microneedles **3100** of the microneedle patch **3108** can face the skin of the wearer of the device **10** when the device **10** is worn. Accordingly, when the device **10** is worn, for example, on the wrist of the wearer with the straps wrapped around the wearer's wrist, the microneedles **3100** can puncture the skin on the wrist.

[0614] Additionally or alternatively, the microneedle patch **3108** may be integrated or releasably secured to an inner side of the adjustable strap **30** of the wearable device **10**, such as shown in FIG. **26C**. The microneedles **3100** can be pointing toward the skin around the wrist of the wearer when the device **10** is worn. When the strap **30** is wrapped around the wrist of the wearer, the microneedle patch **3108** may come in contact with the skin around the wrist of the wearer and the microneedles **3100** can penetrate the skin of the wearer.

[0615] As shown in FIG. **26D**, the microneedle patch **3108** can communicate with the wearable device **10**, using the wearable device **10** as a platform or hub to detect and/or analyze analytes in the fluid sample collected in the microneedles patch **3108**. The patch **3108** can optionally include a sensor **3110**, for example, an electrochemical sensor (with electrodes built into the microneedles), a colorimetric sensor, or otherwise. Alternatively, the patch **3108** can be brought to an external sensor for analyte detection and analysis. The patch **3108** can include an antenna **3112**, which may be an NFC antenna or otherwise. The sensor **3100** can output a signal via the antenna **3112**. The wearable device can receive the signal from the sensor **3100** via the antenna **19**. The device processor **14** (or optionally the sensor or module processor of the physiological parameter measurement sensor or

module on the device **10**) can process the signal from the sensor **3100** to determine the presence and/or concentration of certain analyte(s) in the fluid sample.

Examples Device Tightness Monitoring Systems and Methods

[0616] A desired tightness and/or pressure of the device on the body can be indicated by the skin interfacing with the wearable device moving with the device when the device is moved. If there is insufficient tightness and/or pressure of the device on the body of the wearer, ambient light entering the device-skin interface can result in noises in the signals detected by the detectors, and therefore inaccurate measurements made by the device. If the device is worn too tight (and/or the pressure exerted by the device on the body is too high), blood pulsation and circulation at the wrist can be restricted, which can lead to a decrease in oxygen saturation readings of the wearer of the device. Optionally, the device can output a warning that the device is worn too tight (which can include a message displayed on the device to the wearer to loosen the straps) when the device has determined that the wearer's oxygen saturation readings are decreasing by a certain percentage, at a certain rate, and/or at a certain rate within a predetermined amount of time.

[0617] The device **10** can include an optional strain gauge **20** (see FIG. 7A) to measure a pressure of the device **10** on the wearer. The strain gauge **20** can be located in a device housing **101** between the physiological parameter measurement module **100** and other components of the device **10**, for example, the power source **16**, the device processor **14**, or otherwise. For example, the strain gauge **20** can be flanged between the physiological parameter measurement module **100** and the device processor **14**. When the device **10** is worn on the wearer, for example, on the wrist, the pressure exerted by the module, particularly by the convex protrusion of the cover **102** against the tissue can be transmitted to and measured by the strain gauge **20**. The strain gauge **20** can also be incorporated in the other wearable device examples disclosed herein.

[0618] Readings from the strain gauge **20** can be communicated to the device processor **14**, which can process the readings and output an indication of the pressure asserted by the device **10** on the wearer to be displayed on the display **12**. The indication can be in a variety of suitable forms, for example, using different colors to indicate whether the pressure is too low, appropriate, or too high for obtaining accurate or reliable measurements using the physiological parameter measurement module **100**. In one example, the device **10** can display a green light when the pressure on the wearer is suitable for using the physiological parameter measurement module **100** and display a red or other colored light for a pressure that is too high or too low than the desired pressure or pressure range. The physiological parameter measurement module **100** may not be activated unless the readings from the strain gauge **20** indicate that the desired pressure or pressure range has been achieved. Optionally, the device processor can also deactivate the physiological parameter measurement module **100**, and/or any other sensors on or attached to the device **10**, in response to not detecting any readings from the strain gauge **20**, indicating that the device **10** is not worn on the wearer. Automatically turning on and/or off the sensors on or attached to the device **10** can reduce power consumption and increase battery life of the device **10**.

[0619] Optionally, the wearable device **10** can include a motor to adjust tightness of the straps based on a monitored tightness of the straps and/or pressure exerted by the sensor or module **100** on the wearer's skin.

Example Additional Features of the Wearable Device

[0620] The wearable device examples disclosed herein can provide protection of the wearer's safety by sending an alert to a first responder (for example, a hospital emergency room, a firefighter, 911, security at the facility where the wearer is located, or otherwise) and/or the wearer's family or guardian when the wearer is in danger, for example, when the wearer is drowning. The wearable device can include a swim mode, which the wearer can activate when going swimming. The physiological parameter measurement module of the wearable device can monitor one or more parameters to determine that the wearer is likely drowning (such as drowning of a child in water), for example, by determining that the wearer's respiratory rate has become irregular (such as

showing fluctuations greater than a predetermined number per minute), or the wearer's SpO2 value declines by a predetermined amount, or otherwise. Alternatively, the module processor can determine that wearer is likely drowning based on the gyroscope and/or accelerometer readings, which can further be combined with the parameters monitored by the other sensors. In response to determining that the wearer is likely drowning, the module can send a notification to the processor of the wearable device, which can send an alert to a first responder and/or the wearer's family or guardian. Additionally or alternatively, the wearable device can include a distress button that the wearer can push in an emergency, such as when the wearer is drowning, has sustained a fall (which can alternatively or additionally be determined using the gyroscope and/or accelerometer readings, which can further be combined with the parameters monitored by the other sensors) while being alone, or otherwise.

[0621] The physiological parameters (for example but not limited to, SpO2, PR, PI, PVI, RR, Hydration, ECG-related parameters, etc.) measured by the module disclosed herein can be reliable enough for healthcare or medical purposes, for example, in hospitals. The module can be configured to take measurements at the same time every day. The wearable device (or the physiological parameter measurement module of the device) can further include a hospital patient ID tag on a near-field communication (NFC) or Bluetooth chip, or a watch strap or band. Essential patient information, such as the patient's name, admission date, reason for admission, blood type, drug allergies, etc. can be stored on the memory device of the watch or the physiological parameter measurement module. The patient ID tag cannot be easily removed and/or may include special tools like theft prevention devices, for example, requiring the patient to cut the watch strap off. Alternatively, the wearable device can display the patient information (for example, name, admission date, etc.) on the screen when the patient is admitted to the hospital. The patient ID tag can be either disposable after the patient is discharged or reusable after disinfection. The physiological parameter measurement module can be removed and replaced when the patient ID tag (for example, the watch band) is changed. If the wearable device is worn by a caregiver, the caregiver can use the wearable device for communications with other caregivers (for example, to share critical, substantially real-time information about patients, update changes in patient status, and/or the like), replacing the need for specialized communication tools, for example, Vocera®, Spok®, etc.

Additional Example Aspects and Implementations of a Sensor or Module

[0622] FIG. 27A is a front view of an example aspect of a sensor or module **2700**. The sensor or module **2700** includes an opaque frame **2726**, one or more electrodes **2724**, one or more detector chambers **2788**, one or more emitter chambers **2778**, and a light barrier construct **2720**.

[0623] The opaque frame **2726** can include one or more materials configured to prevent or block the transmission of light. In some aspects, the opaque frame **2726** may form a single integrated unit. In some aspects, the opaque frame **2726** may be formed of a continuous material. The light barrier construct **2720** can include one or more materials configured to prevent or block the transmission of light. In some aspects, the light barrier construct **2720** may form a single integrated unit. In some aspects, the light barrier construct **2720** may be formed of a continuous material. In some aspects, the light barrier construct **2720** and the opaque frame **2726** may form a single integrated unit. In some aspects, the light barrier construct **2720** and the opaque frame **2726** may be separably connected.

[0624] The light barrier construct **2720** may include one or more light barriers, such as light barriers **2720a**, **2720b**, **2720c**, **2720d**, which are provided as non-limiting examples. In some aspects, light barriers may also be referred to as light blocks herein. The light barriers may form one or more portions of the light barrier construct **2720**. The light barrier construct **2720** (or light barrier portions thereof) may prevent light from passing therethrough. The light barrier construct **2720** may include spaces between various light barriers which may define one or more chambers (e.g., detector chambers **2788**, emitter chambers **2778**). In some aspects, the one or more chambers

(e.g., detector chambers **2788**, emitter chambers **2778**) may be enclosed by the light barrier construct **2720** or light barrier portions thereof, a surface of a substrate (e.g., PCB), and a lens or cover. In some aspects, light may only enter the chambers through the lens or cover.

[0625] An example of a light barrier is provided with reference to example light barrier **2720a**. Light barrier **2720a** forms a portion of light barrier construct **2720**. Light barrier **2720a** may prevent (e.g., block) light from passing therethrough between adjacent chambers. For example, light barrier **2720a** may prevent light from passing through the light barrier construct **2720** between an emitter chamber **2778** and a detector chamber **2788**. Light barrier **2720a**, or portions thereof, may include a width **2771**. In some aspects, width **2771** may be less than about 1.85 mm. In some aspects, width **2771** may be less than about 1.9 mm. In some aspects, width **2771** may be less than about 1.95 mm. In some aspects, width **2771** may be about 1.88 mm. In some aspects, the width **2771** may be less (e.g., smaller) than length **2779**. In some aspects, width **2771** may be less than about 55% of length **2779**. In some aspects, width **2771** may be less than about 60% of length **2779**. In some aspects, width **2771** may be less than about 65% of length **2779**. In some aspects, width **2771** may be about 58.9% of length **2779**.

[0626] Another example of a light barrier is provided with reference to example light barrier **2720b**. Light barrier **2720b** forms a portion of light barrier construct **2720**. Light barrier **2720b** may prevent (e.g., block) light from passing therethrough between adjacent chambers. For example, light barrier **2720b** may prevent light from passing through the light barrier construct **2720** between an emitter chamber **2778** and a detector chamber **2788**. Light barrier **2720b**, or portions thereof, may include a width **2772**. In some aspects, width **2772** may be less than about 1.35 mm. In some aspects, width **2772** may be less than about 1.40 mm. In some aspects, width **2772** may be less than about 1.45 mm. In some aspects, width **2772** may be about 1.37 mm. In some aspects, the width **2772** may be substantially similar to width **2771**. In some aspects, the width **2772** may be less (e.g., smaller) than width **2771**. In some aspects, width **2772** may be less than about 70% of width **2771**. In some aspects, width **2772** may be less than about 75% of width **2771**. In some aspects, width **2772** may be less than about 80% of width **2771**. In some aspects, width **2772** may be about 72.9% of width **2771**.

[0627] Another example of a light barrier is provided with reference to example light barrier **2720c**. Light barrier **2720c** forms a portion of light barrier construct **2720**. Light barrier **2720c** may prevent (e.g., block) light from passing therethrough between adjacent chambers. For example, light barrier **2720c** may prevent light from passing through the light barrier construct **2720** between adjacent detector chamber **2788**.

[0628] Another example of a light barrier is provided with reference to example light barrier **2720d**. Light barrier **2720d** forms a portion of light barrier construct **2720**. Light barrier **2720d** may prevent (e.g., block) light from passing therethrough between adjacent chambers. For example, light barrier **2720d** may prevent light from passing through the light barrier construct **2720** between adjacent emitter chambers **2778**. In some aspects, light barrier **2720d** may have a width **2775** separating adjacent emitter chambers of less than about 1.30 mm. In some aspects, width **2775** may be less than about 1.25 mm. In some aspects, width **2775** may be less than about 1.20 mm. In some aspects, width **2775** may be about 1.20 mm. In some aspects, width **2775** may be substantially similar to width **2772**. In some aspects, width **2775** may be less (e.g., smaller) than width **2772**. In some aspects, width **2775** may be less than about 95% of width **2772**. In some aspects, width **2775** may be less than about 90% of width **2772**. In some aspects, width **2775** may be less than about 85% of width **2772**. In some aspects, width **2775** may be about 87.6% of width **2772**.

[0629] The emitter chambers **2778** are positioned within a central region of the sensor or module **2700**. The emitter chambers **2778** may be positioned adjacent to one another across a centerline of the sensor or module **2700** as described in greater detail with reference to FIG. **27K**, for example. The emitter chambers **2778** may be positioned adjacent the center point C.sub.1. Each of the emitter chambers **2778** may be a similar size and/or shape. The emitter chambers **2778** may be

separated, at least in part, by light barrier **2720d** of the light barrier construct **2720**. In some aspects, as shown in this example, the light barrier **2720d** may form an entire distance between emitter chambers **2778**. For example, emitter chambers **2778** may be separated by only the light barrier **2720d** such that other components (e.g., detectors, detector chambers, etc.) are not positioned between the emitter chambers **2778**.

[0630] A portion of the emitter chambers **2778** may extend a length **2779** away from center point C.sub.1. In some aspects, length **2779** may be less than about 3.15 mm. In some aspects, length **2779** may be less than about 3.20 mm. In some aspects, length **2779** may be less than about 3.25 mm. In some aspects, length **2779** may be about 3.19 mm. In some aspects, the length **2779** may be greater (e.g., larger) than a width of a light barrier separating an emitter chamber from a detector chamber such as width **2771**. In some aspects, length **2779** may be greater than about 165% of width **2771**. In some aspects, length **2779** may be greater than about 170% of width **2771**. In some aspects, length **2779** may be greater than about 175% of width **2771**. In some aspects, length **2779** may be about 169.7% of width **2771**.

[0631] As shown in this example aspect, the detector chambers **2788** are arranged in a substantially circular pattern. Each of the detector chambers **2788** houses a detector **2706** positioned on a substrate (e.g., PCB) in a substantially circular or annular pattern. The detectors **2706** may be positioned in a central region of each of the respective detector chambers **2778**. The detector chambers **2788** are arranged along a ring defined by ring L.sub.1. In some aspects, such as shown in this example aspect, detectors **2706** of respective detector chambers **2788** may also be arranged along a same ring along which the detector chambers **2788** are arranged (such as in aspects where detectors are positioned in a central region of respective chambers). The ring L.sub.1 may intersect a central region of the detector chambers **2788**. In this example aspect, the ring L.sub.1 encloses an entirety of the emitter chambers emitter chambers **2778** such that the emitter chambers **2778** are positioned within an interior region (e.g., a central region) of the ring L.sub.1 defined by the detector chambers **2788**. In some aspects, each of the detector chambers **2788** (and corresponding detectors **2706** within respective detector chambers **2788**) may be positioned at a substantially similar or same distance away from the center point C.sub.1 (e.g., center of sensor or module **2700**). In some aspects, the detectors **2706** may be rectangular including longer sides and shorter sides. The detectors **2706** may be positioned on a substrate of the sensor or module **2700** such that a long side of each detector is orthogonal to a radius extending away from center point C.sub.1 (e.g., radius r.sub.1, radius r.sub.2, radius r.sub.3). Advantageously, orienting the detectors **2706** on the sensor or module **2700** in an annular arrangement with a long side of the detectors **2706** orthogonal the center point C.sub.1 may improve an accuracy of physiological measurements by ensuring that light from emitters travels along a known path length from emitters to the detectors **2706** and may also reduce processing requirements of the sensor or module **2700** by reducing the amount of variables (e.g., number of light path lengths) required to process in order to determine physiological data.

[0632] The electrodes **2724** can include a reference electrode and a negative electrode (and/or a positive electrode). In some aspects, a wearable device such as a watch incorporating the sensor or module **2700** can include another ECG electrode (e.g., a positive electrode) located on the housing of the wearable device configured to make contact with the wearer's skin. In some configurations, a surface of the electrodes **2724** may be flush with a surface of the opaque frame **2726**.

[0633] The electrodes **2724** are positioned within or along a portion of the opaque frame **2726** such as shown in FIG. 27B for example. In some aspects, the electrodes **2724** can be substantially semicircular. In some aspects, the electrodes **2724** can be substantially semiannular. In the example aspect shown, each of the electrodes **2724** forms a substantial half annulus. Advantageously, an annular shaped electrode may improve contact with the skin of a wearer (e.g., by contacting a diverse area of skin) while simultaneously reducing the amount of surface area of the electrode. In some aspects, each of the electrodes **2724** may be a similar size and/or shape. In some aspects, the

electrodes **2724** may be various sizes and/or shapes. In this example aspect, the electrodes **2724** are positioned within the sensor or module **2700** (e.g., within the opaque frame **2726**) along ring defined by L.sub.2. In various aspects described herein, the ring L.sub.2 may include various radii which may advantageously provide improved contact between the electrodes **2724** and the skin of a wearer of the device.

[0634] The opaque frame **2726** includes one or more gaps (e.g., g.sub.1, g.sub.2) between electrodes **2724**. The gaps g.sub.1, g.sub.2, (or other portions of the opaque frame **2726**) may electrically insulate each of the electrodes **2724** from one another. Each of the electrodes **2724** includes substantially straight edge along a portion of respective gaps g.sub.1, g.sub.2. In some aspects, the gaps g.sub.1, g.sub.2, may be a similar or a same size. In some aspects, the gaps g.sub.1, g.sub.2, may be a different size than each other. In some aspects, the gaps g.sub.1, g.sub.2, may be less than about 1.6 mm. In some aspects, the gaps g.sub.1, g.sub.2, may be less than about 1.65 mm. In some aspects, the gaps g.sub.1, g.sub.2, may be less than about 1.7 mm. In some aspects, the gaps g.sub.1, g.sub.2, may be about 1.62 mm. As discussed above, in some implementations the frame **2726** includes recesses **2824** sized and/or shaped to receive the ECG electrodes **2724**. In some implementations, each of such recesses **2824** includes first and second ends, the first ends of the recesses **2824** are separated from one another by gap g.sub.1, and the second ends of the recesses **2824** are separated from one another by gap g.sub.2.

[0635] The ring L.sub.1 may be concentric with an outer perimeter of the sensor or module **2700**. The ring L.sub.2 may be concentric with an outer perimeter of the sensor or module **2700**. The ring L.sub.2 may be concentric with a ring defined by positions of the detector chambers **2788** such as ring L.sub.1. Center point C.sub.1 may define a geometric center of ring L.sub.1. Center point C.sub.1 may define a geometric center of ring L.sub.2. Center point C.sub.1 may define a geometric center of an outer perimeter of the sensor or module **2700**. In some aspects, such as shown in FIG. **27A**, each of L.sub.1, L.sub.2, and an outer perimeter of the sensor or module **2700** are concentric with each other and share a same geometric center shown as C.sub.1.

[0636] The ring L.sub.1 may include a radius r.sub.1. In some aspects, radius r.sub.1 may be less than about 6.25 mm. In some aspects, radius r.sub.1 may be less than about 6.50 mm. In some aspects, radius r.sub.1 may be less than about 6.75 mm. In some aspects, radius r.sub.1 may be about 6.34 mm. In some aspects, the radius r.sub.1 may be less (e.g., smaller) than radius r.sub.2. In some aspects, radius r.sub.1 may be less than about 55% of r.sub.2. In some aspects, radius r.sub.1 may be less than about 60% of r.sub.2. In some aspects, radius r.sub.1 may be less than about 65% of r.sub.2. In some aspects, radius r.sub.1 may be about 59% of r.sub.2. In some aspects, the radius r.sub.1 may be less (e.g., smaller) than radius r.sub.3. In some aspects, radius r.sub.1 may be less than about 40% of r.sub.3. In some aspects, radius r.sub.1 may be less than about 45% of r.sub.3. In some aspects, radius r.sub.1 may be less than about 50% of r.sub.3. In some aspects, radius r.sub.1 may be about 41.7% of r.sub.3.

[0637] The ring L.sub.2 may include a radius r.sub.2. In some aspects, radius r.sub.2 may be less than about 10.5 mm. In some aspects, radius r.sub.2 may be less than about 10.75 mm. In some aspects, radius r.sub.2 may be less than about 11.0 mm. In some aspects, radius r.sub.2 may be about 10.73 mm. In some aspects, the radius r.sub.2 may be less (e.g., smaller) than radius r.sub.3. In some aspects, radius r.sub.2 may be less than about 65% of r.sub.3. In some aspects, radius r.sub.2 may be less than about 70% of r.sub.3. In some aspects, radius r.sub.2 may be less than about 75% of r.sub.3. In some aspects, radius r.sub.2 may be about 70.6% of r.sub.3.

[0638] In some aspects, the sensor or module **2700** (e.g., an outer perimeter of the sensor or module **2700**) may include a radius r.sub.3. In some aspects, radius r.sub.3 may be less than about 14.5 mm. In some aspects, radius r.sub.3 may be less than about 15.0 mm. In some aspects, radius r.sub.3 may be less than about 15.50 mm. In some aspects, radius r.sub.3 may be less than about 16.0 mm. In some aspects, radius r.sub.3 may be about 15.19 mm.

[0639] FIGS. **27K-27L** illustrate an example physiological parameter measurement sensor or

module **2700'** and example light paths between emitters and detectors of the module **2700'**. [0640] FIG. **27K** illustrates an example arrangement of emitter and detector chambers of the sensor or module **2700'**. As shown, the sensor or module **2700'** can include a first emitter chamber **2736a'** enclosing a first emitter group comprising one or more emitters, a second emitter chamber **2736b'** enclosing a second emitter group comprising one or more emitters, one or more first detector chambers **2740'**, one or more second detector chambers **2742'**, and one or more third detector chambers **2738'**. In some aspects, each detector chamber may enclose one detector.

[0641] The first emitter group of the first emitter chamber **2736a'** may comprise the same number and type of emitters as the second emitter group of the second emitter chamber **2736b'**. In other words, each emitter of the first emitter group may correspond to an emitter of the same type (e.g., same wavelength) of the second emitter group. The emitters of the first emitter group may be arranged in a configuration that mirrors the emitters of the second emitter group across a centerline **2750** of the sensor or module **2700'** as shown in FIG. **27K**. For example, each emitter of the first group of emitters may be located a distance away from a centerline **2750** of the sensor or module **2700'** that is a same distance that a corresponding emitter of the second group of emitters is located away from the centerline **2750** of the sensor or module **2700'**. For example, the first and second emitter groups may each include an emitter that emits light of a first wavelength and that are positioned at locations that are mirror images of each other across a centerline **2750** of the sensor or module **2700'**. Additionally, the first and second emitter groups may each include an emitter that emits light of a second wavelength and that are positioned at locations that are mirror images of each other across a centerline **2750** of the sensor or module **2700'**. Each of the emitters of the first emitter group may correspond to an emitter of the second emitter group located at a mirror image position, and vice versa.

[0642] The one or more second detector chambers **2742'** may be bisected by a centerline **2750** of the sensor or module **2700'**. Each of the detectors of the respective one or more second detector chambers **2742'** may be bisected by a centerline **2750** of the sensor or module **2700'**. In other words, the one or more second detector chambers **2742'** and the respective detectors and the sensor or module **2700'** may each share a same (e.g., parallel) centerline **2750**. The sensor or module **2700'** may be oriented (e.g., rotated) with respect to the tissue of a wearer in any orientation. In an example implementation where the sensor or module **2700'** is worn on a wrist of a user, the sensor or module **2700'** may be rotated in any direction with respect to the wrist or forearm of the wearer. In one example configuration, the sensor or module **2700'** may be oriented with respect to the forearm (or other body part) of a wearer such that the centerline **2750'** of the sensor or module is perpendicular to a line extending along a length of the forearm of the wearer (e.g., from the elbow to the wrist). Advantageously, such a configuration may improve physiological measurements by facilitating light emitted from the emitter chambers and detected at the detector chambers (e.g., light travelling from emitter chamber **2736a'** to detector chamber **2738'**) to penetrate into soft tissue of the wearer (e.g., blood vessels) rather than other tissues such as bone. In another example configuration, the sensor or module **2700'** may be oriented with respect to the forearm (or other body part) of a wearer such that the centerline **2750'** of the sensor or module is parallel to a line extending along a length of the forearm of the wearer (e.g., from the elbow to the wrist).

Advantageously, such a configuration may improve physiological measurements by facilitating light emitted from the emitter chambers and detected at the detector chambers (e.g., light travelling from emitter chamber **2736a'** to detector chamber **2742'**) to penetrate into soft tissue of the wearer (e.g., blood vessels) rather than other tissues such as bone.

[0643] As shown in FIG. **27K**, emitters of the first and second emitter groups that correspond to each other (e.g., emit the same wavelength and mirror each other) may each emit light that travels along respective paths to the detectors of the one or more second detector chambers **2742'**. The respective paths of light from the corresponding emitters may be of equal length. This may be because the corresponding emitters are each positioned an equal distance away from a detector of a

chamber **2742'**. The corresponding emitters may each be an equal distance away from a detector of a chamber **2742'** because they are positioned at mirror images of each other across a centerline **2750** of the sensor or module **2700'** that bisects the one or more second detector chambers **2472'** and respective detectors.

[0644] The one or more second detector chambers **2742'** and their respective detectors may be used, at least in part, for calibration, for example to characterize the emitters, by providing known information such as a known ratio. For example, information corresponding to a wavelength detected at a detector of a chamber **2742'** from an emitter of the first group of emitters may be similar or the same as information corresponding to that wavelength detected at the detector of the chamber **2742'** from an emitter of the second group of emitters and a comparison (e.g., subtracting, dividing, etc.) of the information resulting from the first and second groups of emitters may yield a known number such as zero or one because the corresponding emitters from the first and second emitter groups may be an equal distance from the detector of chamber **2742'** and light emitted therefrom may travel a same distance to the detector of chamber **2742'**. As an example of normalization, ratios of wavelengths detected at detectors of chambers **2738'**, **2740'** may be normalized (e.g., divided by) ratios of wavelengths detected at detectors of chambers **2742'**. In instances where the information resulting from detection of light from the first and second groups of emitters is not the same or is substantially different (e.g., as a result of emission intensity variations or other such discrepancies) the information may be adjusted or normalized (e.g., calibrated) to account for such differences. This normalization or on-board calibration or characterization of the emitters may improve accuracy of the physiological measurements and provide for continuous calibration or normalization during measurements. In some aspects, a processor may be configured to calibrate or normalize the physiological parameter measurement of the sensor continuously. In some aspects, a processor may be configured to calibrate or normalize the physiological parameter measurement of the sensor while the optical physiological sensor measures physiological parameters of the wearer.

[0645] FIG. **27L** illustrates an example arrangement of emitter and detector chambers of the sensor or module **2700'**. As shown, the sensor or module **2700'** can include a first emitter chamber **2736a'**, a second emitter chamber **2736b'**, one or more first detector chambers **2740'**, one or more second detector chambers **2742'**, and one or more third detector chambers **2738'**, for example as discussed elsewhere herein.

[0646] The first and second emitter chambers **2736a'**, **2736b'** may be located at non-equal distances away from each of the chambers of the one or more detector chambers **2738'**, **2740'**. Thus, with respect to each detector chamber of the chambers **2738'**, **2740'**, the first and second emitter chamber **2736a'**, **2736b'**, may each be a “near” or “far” emitter chamber. In other words, each detector of the detector chambers **2738'**, **2740'** may detect light, of any given wavelength, from both a “near” emitter and a “far” emitter, with the near and far emitters being included in either the first or second emitter group, respectively.

[0647] As an example, as shown in FIG. **27L**, light of a given wavelength may travel along a path from an emitter in the first emitter group to the detector of detector chamber **2738'** and light of the same wavelength may travel along a path from an emitter in the second emitter group to the same detector. The light from the first emitter group may travel along a longer path than light from the second emitter group before reaching the detector of chamber **2738'**. Thus, for any detector of detector chambers **2738'** or **2740'**, the detector may receive light of a given wavelength from both a near (e.g., proximal) emitter and a far (e.g., distal) emitter. This may not be the case for detectors of chambers **2742'** because the first and second emitter groups may each be located a same distance away from any given detector of detector chambers **2742'**, as described herein.

[0648] For convenience, the terms “proximal” and “distal” may be used herein to describe structures relative to any of the detector chambers or their respective detectors. For example, an emitter may be proximal to a detector chamber of the first detector chambers and distal to a

detector of the second detector chambers. The term “distal” refers to one or more emitters that are farther away from a detector chamber than at least some of the other emitters. The term “proximal” refers to one or more emitters that are closer to a detector chamber than at least some of the other emitters. The term “proximal emitter” may be used interchangeably with “near emitter” and the term “distal emitter” may be used interchangeably with “far emitter”.

[0649] A single emitter may be both proximal to one detector and distal to another detector. For example, an emitter may be a proximal emitter relative to a detector of the first detector chambers and may be a distal emitter relative to a detector of the second detector chambers.

[0650] Light of a given wavelength that is detected at a detector may provide different information depending on the length of the path it has travelled from the emitter (e.g., along a long path from a distal emitter or along a short path from a proximal emitter). For example, light that has travelled along a long path from a distal emitter may penetrate deeper into the tissue of a wearer of the device and may provide information pertaining to pulsatile blood flow or constituents. The use of a proximal and distal emitter for each wavelength may improve accuracy of the measurement, for example information pertaining to light that has travelled along a long path from a distal emitter may be normalized by (e.g., divided by) information pertaining to light that has travelled along a short path from a proximal emitter.

[0651] FIGS. 27M-27P illustrate an example physiological parameter measurement sensor or module 2700' and example light barriers or light blocks between emitter and detector chambers of the module 2700'.

[0652] FIG. 27M is a front view of an example aspect of a sensor or module 2700'. The sensor or module 2700' includes an opaque frame 2726', one or more electrodes 2724', one or more detector chambers 2788', one or more emitter chambers 2778', and a light barrier construct 2720'.

[0653] The opaque frame 2726' can include one or more materials configured to prevent or block the transmission of light. In some aspects, the opaque frame 2726' may form a single integrated unit. In some aspects, the opaque frame 2726' may be formed of a continuous material. The light barrier construct 2720' can include one or more materials configured to prevent or block the transmission of light. In some aspects, the light barrier construct 2720' may form a single integrated unit. In some aspects, the light barrier construct 2720' may be formed of a continuous material. In some aspects, the light barrier construct 2720' and the opaque frame 2726' may form a single integrated unit. In some aspects, the light barrier construct 2720' and the opaque frame 2726' may be separably connected.

[0654] The light barrier construct 2720' may include one or more light barriers, such as light barriers 2720a', 2720b', 2720c', 2720d', which are provided as non-limiting examples. In some aspects, light barriers may also be referred to as light blocks herein. The light barriers may form one or more portions of the light barrier construct 2720'. The light barrier construct 2720' (or light barrier portions thereof) may prevent light from passing therethrough. The light barrier construct 2720' may include spaces between various light barriers which may define one or more chambers (e.g., detector chambers 2788', emitter chambers 2778'). In some aspects, the one or more chambers (e.g., detector chambers 2788', emitter chambers 2778') may be enclosed by the light barrier construct 2720' or light barrier portions thereof, a surface of a substrate (e.g., PCB), and a lens or cover. In some aspects, light may only enter the chambers through the lens or cover.

[0655] An example of a light barrier is provided with reference to example light barrier 2720a'. Light barrier 2720a' forms a portion of light barrier construct 2720'. Light barrier 2720a' may prevent (e.g., block) light from passing therethrough between adjacent chambers. For example, light barrier 2720a' may prevent light from passing through the light barrier construct 2720' between an emitter chamber 2778' and a detector chamber 2788'. Light barrier 2720a', or portions thereof, may include a width 2771'. In some aspects, width 2771' may be less than about 3.30 mm. In some aspects, width 2771' may

be less than about 3.20 mm. In some aspects, width 2771' may be about 3.24 mm. In some aspects, the width 2771' may be greater (e.g., larger) than length 2779. In some aspects, width 2771' may be less than about 165% of length 2779. In some aspects, width 2771' may be less than about 160% of length 2779. In some aspects, width 2771' may be less than about 155% of length 2779. In some aspects, width 2771' may be about 160% of length 2779. Advantageously, a greater width 2771' (e.g., a wider light barrier separating the emitter chambers 2778' and detector chambers 2788') may cause light emitted from the emitter chambers 2778' to travel a greater distance before reaching the detector chambers 2788'. Light that travels a greater distance may penetrate deeper into the tissue of the wearer which may improve accuracy of a physiological measurement.

[0656] Another example of a light barrier is provided with reference to example light barrier 2720b'. Light barrier 2720b' forms a portion of light barrier construct 2720'. Light barrier 2720b' may prevent (e.g., block) light from passing therethrough between adjacent chambers. For example, light barrier 2720b' may prevent light from passing through the light barrier construct 2720' between an emitter chamber 2778' and a detector chamber 2788'. Light barrier 2720b', or portions thereof, may include a width 2772'. In some aspects, width 2772' may be less than about 1.65 mm. In some aspects, width 2772' may be less than about 1.60 mm. In some aspects, width 2772' may be less than about 1.55 mm. In some aspects, width 2772' may be about 1.59 mm. In some aspects, the width 2772' may be less (e.g., smaller) than width 2771'. In some aspects, width 2772' may be less than about 60% of width 2771'. In some aspects, width 2772' may be less than about 55% of width 2771'. In some aspects, width 2772' may be less than about 50% of width 2771'. In some aspects, width 2772' may be about 49% of width 2771'. Advantageously, a greater width 2772' may cause light emitted from the emitter chambers 2778' to travel a greater distance before reaching the detector chambers 2788'. Light that travels a greater distance may penetrate deeper into the tissue of the wearer which may improve accuracy of a physiological measurement

[0657] Another example of a light barrier is provided with reference to example light barrier 2720c'. Light barrier 2720c' forms a portion of light barrier construct 2720'. Light barrier 2720c' may prevent (e.g., block) light from passing therethrough between adjacent chambers. For example, light barrier 2720c' may prevent light from passing through the light barrier construct 2720' between adjacent detector chamber 2788'.

[0658] Another example of a light barrier is provided with reference to example light barrier 2720d'. Light barrier 2720d' forms a portion of light barrier construct 2720'. Light barrier 2720d' may prevent (e.g., block) light from passing therethrough between adjacent chambers. For example, light barrier 2720d' may prevent light from passing through the light barrier construct 2720' between adjacent emitter chambers 2778'. In some aspects, light barrier 2720d' may have a width 2775' separating adjacent emitter chambers of less than about 1.40 mm. In some aspects, width 2775' may be less than about 1.35 mm. In some aspects, width 2775' may be less than about 1.30 mm. In some aspects, width 2775' may be about 1.28 mm. In some aspects, width 2775' may be less (e.g., smaller) than width 2771'. In some aspects, width 2775' may be less than about 50% of width 2771'. In some aspects, width 2775' may be less than about 45% of width 2771'. In some aspects, width 2775' may be less than about 40% of width 2771'. In some aspects, width 2775' may be less than about 35% of width 2771'. In some aspects, width 2775' may be about 39.5% of width 2771'.

[0659] The emitter chambers 2778' are positioned within a central region of the sensor or module 2700'. The emitter chambers 2778' may be positioned adjacent to one another across a centerline of the sensor or module 2700' as described in greater detail with reference to FIG. 27K, for example. The emitter chambers 2778' may be positioned adjacent the center point C'.sub.1. Each of the emitter chambers 2778' may be a similar size and/or shape. The emitter chambers 2778' may be separated, at least in part, by light barrier 2720d' of the light barrier construct 2720'. In some aspects, as shown in this example, the light barrier 2720d' may form an entire distance between emitter chambers 2778'. For example, emitter chambers 2778' may be separated by only the light

barrier **2720d'** such that other components (e.g., detectors, detector chambers, etc.) are not positioned between the emitter chambers **2778'**.

[0660] A portion of the emitter chambers **2778'** may extend a length **2779** away from center point C'.sub.1. In some aspects, length **2779** may be less than about 2.15 mm. In some aspects, length **2779** may be less than about 2.10 mm. In some aspects, length **2779** may be less than about 2.05 mm. In some aspects, length **2779** may be less than about 2.0 mm. In some aspects, length **2779** may be about 2.02 mm. In some aspects, the length **2779** may be less (e.g., smaller) than a width of a light barrier separating an emitter chamber from a detector chamber such as width **2771'**. In some aspects, length **2779** may be less than about 70% of width **2771'**. In some aspects, length **2779** may be less than about 65% of width **2771'**. In some aspects, length **2779** may be less than about 60% of width **2771'**. In some aspects, length **2779** may be about 62.3% of width **2771'**.

[0661] As shown in this example aspect, the detector chambers **2788'** are arranged in a substantially circular pattern. Each of the detector chambers **2788'** houses a detector **2706** positioned on a substrate (e.g., PCB) in a substantially circular or annular pattern. The detectors **2706** may be positioned in a central region of each of the respective detector chambers **2778'**. The detector chambers **2788'** are arranged along a ring defined by ring L'.sub.1. In some aspects, such as shown in this example aspect, detectors **2706** of respective detector chambers **2788'** may also be arranged along a same ring along which the detector chambers **2788'** are arranged (such as in aspects where detectors are positioned in a central region of respective chambers). The ring L'.sub.1 may intersect a central region of the detector chambers **2788'**. In this example aspect, the ring L'.sub.1 encloses an entirety of the emitter chambers emitter chambers **2778'** such that the emitter chambers **2778'** are positioned within an interior region (e.g., a central region) of the ring L'.sub.1 defined by the detector chambers **2788'**. In some aspects, each of the detector chambers **2788'** (and corresponding detectors **2706** within respective detector chambers **2788'**) may be positioned at a substantially similar or same distance away from the center point C'.sub.1 (e.g., center of sensor or module **2700'**). In some aspects, the detectors **2706** may be rectangular including longer sides and shorter sides. The detectors **2706** may be positioned on a substrate of the sensor or module **2700'** such that a long side of each detector is orthogonal to a radius extending away from center point C'.sub.1 (e.g., radius r'.sub.1, radius r'.sub.2, radius r'.sub.3). Advantageously, orienting the detectors **2706** on the sensor or module **2700'** in an annular arrangement with a long side of the detectors **2706** orthogonal the center point C'.sub.1 may improve an accuracy of physiological measurements by ensuring that light from emitters travels along a known path length from emitters to the detectors **2706** and may also reduce processing requirements of the sensor or module **2700'** by reducing the amount of variables (e.g., number of light path lengths) required to process in order to determine physiological data.

[0662] The electrodes **2724'** can include a reference electrode and a negative electrode (and/or a positive electrode). In some aspects, a wearable device such as a watch incorporating the sensor or module **2700'** can include another ECG electrode (e.g., a positive electrode) located on the housing of the wearable device configured to make contact with the wearer's skin. In some configurations, a surface of the electrodes **2724'** may be flush with a surface of the opaque frame **2726'**.

[0663] The electrodes **2724'** are positioned within or along a portion of the opaque frame **2726'** such as shown in FIG. 27B for example. In some aspects, the electrodes **2724'** can be substantially semicircular. In some aspects, the electrodes **2724'** can be substantially semi-annular. In the example aspect shown, each of the electrodes **2724'** forms a substantial half annulus.

Advantageously, an annular shaped electrode may improve contact with the skin of a wearer (e.g., by contacting a diverse area of skin) while simultaneously reducing the amount of surface area of the electrode. In some aspects, each of the electrodes **2724'** may be a similar size and/or shape. In some aspects, the electrodes **2724'** may be various sizes and/or shapes. In this example aspect, the electrodes **2724'** are positioned within the sensor or module **2700'** (e.g., within the opaque frame **2726'**) along ring defined by L'.sub.2. In various aspects described herein, the ring L'.sub.2 may

include various radii which may advantageously provide improved contact between the electrodes **2724'** and the skin of a wearer of the device. In some implementations, frame **2726'** includes recesses that **2824'** are sized and/or shaped to accommodate the ECG electrodes **2724'**. In some implementations, recesses **2824'** have a depth (for example, measured from a plane of the frame **2726'**) that is substantially equal to a thickness of the ECG electrodes **2724'**. In some implementations, recesses **2824'** have a size and/or shape that matches a size and/or shape of the ECG electrodes **2724'**. For example, in some implementations in which the ECG electrodes have a semi-annular shape, the recesses **2824'** can have a semi-annular shape.

[0664] The opaque frame **2726'** includes one or more gaps (e.g., $g'.sub.1$, $g'.sub.2$) between electrodes **2724'**. The gaps $g'.sub.1$, $g'.sub.2$, (or other portions of the opaque frame **2726'**) may electrically insulate each of the electrodes **2724'** from one another. Each of the electrodes **2724'** includes a curved edge along a portion of respective gaps $g'.sub.1$, $g'.sub.2$. In some aspects, the gaps $g'.sub.1$, $g'.sub.2$, may be a similar or a same size. In some aspects, the gaps $g'.sub.1$, $g'.sub.2$, may be a different size than each other. In some aspects, the gaps $g'.sub.1$, $g'.sub.2$, may be less than about 0.6 mm. In some aspects, the gaps $g'.sub.1$, $g'.sub.2$, may be less than about 0.65 mm. In some aspects, the gaps $g'.sub.1$, $g'.sub.2$, may be less than about 0.7 mm. In some aspects, the gaps $g'.sub.1$, $g'.sub.2$, may be about 0.62 mm. As discussed above, in some implementations the frame **2726'** includes recesses **2824'** sized and/or shaped to receive the ECG electrodes **2724'**. In some implementations, each of such recesses **2824'** includes first and second ends, the first ends of the recesses **2824'** are separated from one another by gap $g'.sub.1$, and the second ends of the recesses **2824'** are separated from one another by gap $g'.sub.2$ (see FIG. 27M). In some implementations, such as that illustrated in at least FIG. 27M, ends of the recesses **2824'** and/or ends of ECG electrodes **2724'** have a rounded shape.

[0665] The ring $L'.sub.1$ may be concentric with an outer perimeter of the sensor or module **2700'**. The ring $L'.sub.2$ may be concentric with an outer perimeter of the sensor or module **2700'**. The ring $L'.sub.2$ may be concentric with a ring defined by positions of the detector chambers **2788'** such as ring $L'.sub.1$. Center point $C'.sub.1$ may define a geometric center of ring $L'.sub.1$. Center point $C'.sub.1$ may define a geometric center of ring $L'.sub.2$. Center point $C'.sub.1$ may define a geometric center of an outer perimeter of the sensor or module **2700'**. In some aspects, such as shown in FIG. 27M, each of $L'.sub.1$, $L'.sub.2$, and an outer perimeter of the sensor or module **2700'** are concentric with each other and share a same geometric center shown as $C'.sub.1$.

[0666] The ring $L'.sub.1$ may include a radius $r'.sub.1$. In some aspects, radius $r'.sub.1$ may be less than about 6.5 mm. In some aspects, radius $r'.sub.1$ may be less than about 6.45 mm. In some aspects, radius $r'.sub.1$ may be less than about 6.40 mm. In some aspects, radius $r'.sub.1$ may be less (e.g., smaller) than radius $r'.sub.2$. In some aspects, radius $r'.sub.1$ may be less than about 60% of $r'.sub.2$. In some aspects, radius $r'.sub.1$ may be less than about 55% of $r'.sub.2$. In some aspects, radius $r'.sub.1$ may be less than about 50% of $r'.sub.2$. In some aspects, radius $r'.sub.1$ may be about 50.9% of $r'.sub.2$. In some aspects, the radius $r'.sub.1$ may be less (e.g., smaller) than radius $r'.sub.3$. In some aspects, radius $r'.sub.1$ may be less than about 40% of $r'.sub.3$. In some aspects, radius $r'.sub.1$ may be less than about 45% of $r'.sub.3$. In some aspects, radius $r'.sub.1$ may be less than about 50% of $r'.sub.3$. In some aspects, radius $r'.sub.1$ may be about 42% of $r'.sub.3$.

[0667] The ring $L'.sub.2$ may include a radius $r'.sub.2$. In some aspects, radius $r'.sub.2$ may be less than about 13 mm. In some aspects, radius $r'.sub.2$ may be less than about 12.75 mm. In some aspects, radius $r'.sub.2$ may be less than about 12.5 mm. In some aspects, radius $r'.sub.2$ may be about 12.59 mm. In some aspects, the radius $r'.sub.2$ may be less (e.g., smaller) than radius $r'.sub.3$. In some aspects, radius $r'.sub.2$ may be less than about 80% of $r'.sub.3$. In some aspects, radius $r'.sub.2$ may be less than about 85% of $r'.sub.3$. In some aspects, radius $r'.sub.2$ may be less than about 90% of $r'.sub.3$. In some aspects, radius $r'.sub.2$ may be about 82.7% of $r'.sub.3$.

[0668] In some aspects, the sensor or module **2700'** (e.g., an outer perimeter of the sensor or

module **2700'**) may include a radius $r'.sub.3$. In some aspects, radius $r'.sub.3$ may be less than about 15 mm. In some aspects, radius $r'.sub.3$ may be less than about 15.0 mm. In some aspects, radius $r'.sub.3$ may be less than about 15.25 mm. In some aspects, radius $r'.sub.3$ may be less than about 15.5 mm. In some aspects, radius $r'.sub.3$ may be about 15.22 mm.

[0669] FIG. **27N** is a side cutaway view of an example aspect of a sensor or module **2700'**. The sensor or module **2700'** includes a barrier construct **2720'**, an outer surface **2791'**, and a substrate **2716'**. The outer surface **2791'** may include light barrier construct portions, lens portions, opaque frame portions, and/or electrode portions. The outer surface **2791'** of the sensor or module **2700'** may face and/or contact the skin of a wearer and may include a generally convex shape. A central region of the sensor or module **2700'** may have a height **2793'**. For example, the height of the light barrier construct **2720'** at a central region of the sensor or module **2700'** may correspond to height **2793'**. The height **2793'** may be a maximum distance the outer surface **2791'** extends perpendicularly away from the substrate **2716'** (e.g., toward the skin of a wearer). An outer region (e.g., along a perimeter of the substrate **2716'**) of the sensor or module **2700'** may have a height **2795'**. For example, the height of the light barrier construct **2720'** and/or opaque frame **2726'** at an outer region of the sensor or module **2700'** may correspond to height **2795'**. The height **2795'** may be a minimum distance the outer surface **2791'** extends perpendicularly away from the substrate **2716'** (e.g., toward the skin of a wearer).

[0670] In some aspects, height **2793'** may be less than about 2.95 mm. In some aspects, height **2793'** may be less than about 2.90 mm. In some aspects, height **2793'** may be less than about 2.85 mm. In some aspects, height **2793'** may be less than about 2.80 mm. In some aspects, height **2793'** may be about 2.85 mm. In some aspects, height **2793'** may be less than about 2.70 mm. In some aspects, height **2793'** may be less than about 2.65 mm. In some aspects, height **2793'** may be less than about 2.60 mm. In some aspects, height **2793'** may be less than about 2.55 mm. In some aspects, height **2793'** may be about 2.58 mm.

[0671] In some aspects, height **2795'** may be less than about 1.40 mm. In some aspects, height **2795'** may be less than about 1.35 mm. In some aspects, height **2795'** may be less than about 1.30 mm. In some aspects, height **2795'** may be less than about 1.25 mm. In some aspects, height **2795'** may be about 1.29 mm. In some aspects, height **2795'** may be less than about 1.90 mm. In some aspects, height **2795'** may be less than about 1.85 mm. In some aspects, height **2795'** may be less than about 1.80 mm. In some aspects, height **2795'** may be less than about 1.75 mm. In some aspects, height **2795'** may be about 1.78 mm.

[0672] In some aspects, the height **2793'** may be greater (e.g., larger) than height **2795'**. In some aspects, height **2793'** may be less than about 230% of height **2795'**. In some aspects, height **2793'** may be less than about 225% of height **2795'**. In some aspects, height **2793'** may be less than about 220% of height **2795'**. In some aspects, height **2793'** may be less than about 215% of height **2795'**. In some aspects, height **2793'** may be about 221% of height **2795'**. In some aspects, height **2793'** may be less than about 155% of height **2795'**. In some aspects, height **2793'** may be less than about 150% of height **2795'**. In some aspects, height **2793'** may be less than about 145% of height **2795'**. In some aspects, height **2793'** may be less than about 140% of height **2795'**. In some aspects, height **2793'** may be about 145% of height **2795'**.

[0673] Advantageously, a greater height **2793'** (and/or greater ratio of height **2793'** to **2795'**) (for example, a taller light barrier at a central region of the sensor or module **2700** may cause light emitted from emitter chambers to travel a greater distance before reaching the detector chambers. Light that travels a greater distance may penetrate deeper into the tissue of the wearer which may improve accuracy of a physiological measurement. A smaller height **2793'** (and/or smaller ratio of height **2793'** to **2795'**) may reduce discomfort to the wearer wearing the wearable device **10** or may reduce obstruction to blood flow of the wearer by reducing the amount of pressure the wearable device places on the wearer. The height **2793'** and/or height **2795'** may be selected to balance the above-mentioned considerations such as increasing the depth which light penetrates into the tissue

and reducing discomfort or blood flow obstruction of the wearer.

[0674] FIG. 27O and FIG. 27P illustrate two example aspects of a sensor or module 2700' with different light barrier construct configurations. FIGS. 27O and 27P also show an example light path from an emitter chamber to a detector chamber. The light barrier construct 2720' (or portions thereof) shown in the example aspect of FIG. 27O may be taller (e.g., extending away from a surface of the substrate 2716) and/or wider than the light barrier construct 2720 (or portions thereof) shown in the example aspect of FIG. 27P. The greater height and/or width of the light barrier construct 2720' in the aspect of FIG. 27O may cause the light emitted from an emitter chamber 2778' to travel a greater distance before reaching a detector chamber and thus penetrate deeper into the tissue of the wearer than in the aspect of FIG. 27P. Thus, adjusting the height and/or width of the light barrier construct may affect the path the light travels from the emitter chamber to the detector chamber which may affect an accuracy of a physiological measurement. The height and/or width of the light barrier construct may be adjusted, according to various aspects, as required or desired.

[0675] FIG. 27Q illustrates a cutaway side view of an example sensor or module 2700' showing light transmissive lens(es) or cover(s) 2702' and light diffusing material. The light diffusing materials can be included in one or more of the emitter or detector chambers to improve distribution of emitted lighted and/or detected light. The diffusing materials or encapsulant, can include, for example, microspheres or glass microspheres. The encapsulant can eliminate air gaps between the surface of the light transmissive cover 2702' and the emitters and/or the detectors. The encapsulant can be included around the emitters to more evenly spread the emitted light, causing the emitted light to appear to be emitted from an entire emitter chamber rather than from a point source (that is, a single LED emitter) if the encapsulant were absent. The light transmissive lens(es) or cover(s) 2702' may include polycarbonate.

Example Graphical User Interfaces of External Devices

[0676] As discussed herein and as shown in FIG. 2, the wearable device 10 can be in communication, for example wirelessly, to an external device. FIG. 28 shows a block diagram illustrating an example aspect of the wearable device 10 in communication with an external device 2802. The communication may be wireless, such as, but not limited to, Bluetooth and/or near-field communication (NFC) wireless communication. As shown in FIG. 2, the wearable device 10 may be in communication with any number and/or types of external devices 2802 which may include a patient monitor 202 mobile communication device 204 (for example, a smartphone), a computer 206 (which can be a laptop or a desktop), a tablet 208, a nurses' station system 201, glasses such as smart glasses configured to display images on a surface of the glasses and/or the like. The external device 2802 may include a health application 2804. "External device" and "computing device" may be used interchangeably herein.

[0677] A user may operate the external device 2802 as described herein. A wearer may wear the wearable device 10. In some implementations, the user of the external device 2802 and the wearer of the wearable device 10 are different people. In some implementations, the user of the external device 2802 and the wearer of the wearable device 10 are the same person. The terms "user" and "wearer" and "patient" may be used interchangeably herein and may all refer to a person wearing the wearable device 10 and/or a person using the health application 2804 and their uses in any of the given examples are not meant to be limiting of the present disclosure.

[0678] The wearable device 10 may communicate information such as physiological data of the wearer/user to the external device 2802. The external device 2802 may display the physiological parameters received from the wearable device 10, as described herein.

[0679] The external device 2802 may control operation of the wearable device 10, for example via a wireless connection as described herein. For example, the external device 2802 may cause the wearable device 10 to start or stop taking measurements of a wearer's physiological parameters. In some aspects, the wearable device 10 may continuously measure and communicate a wearer's

physiological parameters to the external device **2802**. In some aspects, the external device **2802** may continuously display the wearer's physiological parameters received from the wearable device **10**. In some aspects, the wearable device **10** may measure and communicate physiological parameters to an external device **2802** for a finite amount of time, such as 1 minute, upon receiving user input at the external device **2802** communicated to the wearable device **10**.

[0680] FIGS. **29**, **30A-30H**, **31A-31D**, **32**, **33A-33C**, **34A-34C** and **35A-35B** illustrate example graphical user interfaces of a health application **2804**, according to some aspects of the present disclosure. In various aspects, aspects of the user interfaces may be rearranged from what is shown and described below, and/or particular aspects may or may not be included. The health application **2804** can execute on the external device **2802** to present the graphical user interfaces of FIGS. **29**, **30A-30H**, **31A-31D**, **32**, **33A-33C**, **34A-34C** and **35A-35B**. As described herein, the health application **2804** can receive a respective client configuration package that causes the presentation of the graphical user interfaces of FIGS. **29**, **30A-30H**, **31A-31D**, **32**, **33A-33C**, **34A-34C** and **35A-35B**. The graphical user interfaces of FIGS. **29**, **30A-30H**, **31A-31D**, **32**, **33A-33C**, **34A-34C** and **35A-35B** may have similar user interface elements and/or capabilities.

[0681] FIG. **29** illustrates an example dashboard user interface **2900** of the health application **2804**. The dashboard user interface **2900** can display current physiological parameters **2902** of a wearer such as pulse rate, SpO₂, RRp, PVi, Pi and the like. In addition to the presentation of current wearer physiological parameter(s) **2902**, the dashboard user interface **2900** can present indicator(s) associated with one or more of the physiological parameters **2902** that visually indicate a status of the parameters **2902** and various status ranges for each parameter **2902**. The indicator(s) may be color coded or otherwise show a severity or status of a physiological parameter **2902**. The dashboard user interface **2900** may additionally display a history of wearer statistics/information such as workout history information, sleep information, activity levels, steps taken, and/or calories burned.

[0682] The dashboard user interface **2900** may additionally display one or more navigation selectors **2904** configured for selection by a user. The one or more navigation selectors **2904** may include a home navigation selector, an activity navigation selector, a workout navigation selector, a vitals navigation selector, a sleep navigation selector, a history navigation selector, a share navigation selector and/or a settings navigation selector. Selection of the navigation selectors **2904** may cause the health application **2804** to display any of the graphical user interfaces described herein associated with the selected navigation selector **2904**. The navigation selectors **2904** may be displayed in any of the graphical user interfaces described herein.

[0683] FIGS. **30A-30D** illustrate example spot check monitoring user interfaces of the health application **2804**. The wearable device **10** may be configured to perform spot checks. A spot check may be a discrete period of time during which the wearable device **10** performs physiological measurements of a wearer. For example, during a spot check the wearable device **10** may perform physiological measurements for one minute. A user/wearer may initiate a spot check at any desirable time, for example via the health application **2804** of the external device **2802**.

[0684] FIG. **30A** illustrates an example spot check monitoring user interface **3000** of the health application **2804**. A user/wearer may initiate a spot check by selecting a start spot check selectable component **3002**. Upon selection of the start spot check selectable component **3002** via the health application **2804**, the external device **2802** may cause the wearable device **10** to commence measuring physiological parameters as described herein. In some aspects, upon selection of the start spot check selectable component **3002** via the health application **2804**, the external device **2802** may check the power level of the wearable device **10** (for example, by checking a charge and/or voltage level of the power source **16**) prior to causing the wearable device to start the spot check to determine if the wearable device **10** has sufficient power to perform the spot check. Upon determining that the wearable device **10** has sufficient power, the external application may cause the wearable device **10** to perform the spot check. Upon determining that the wearable device **10**

does not have sufficient power, the external application may not cause the wearable device **10** to perform the spot check and/or may cause the spot check to be performed only for certain physiological parameters. In some aspects, upon receiving a command from the external device **2802** to perform a spot check, the wearable device **10** may check its power level (for example, by checking a charge and/or voltage level of the power source **16**) prior to commencing physiological measurements for the spot check to determine if it has sufficient power to perform the spot check. Upon determining that it has sufficient power, the wearable device **10** may commence performing the spot check. Upon determining that it does not have sufficient power, the wearable device **10** may not perform the spot check.

[0685] In some aspects, the wearable device **10** may perform a spot check for a predetermined amount of time, for example three minutes, two minutes, one minute, 30 seconds, 15 seconds or any amount of time as required or desired. In some aspects, a user may select the predetermined amount of time for which the spot check is to be performed, for example before or after selecting the selectable component **3002**. In some aspects, the wearable device **10** may perform a spot check until it receives a command to stop performing the spot check. For example, the external device **2802** may cause the wearable device **10** to stop performing a spot check upon receiving a user command, for example via a selectable component (e.g., similar to selectable component **3002**) of the health application **2804**.

[0686] FIG. **30B** illustrates an example spot check monitoring user interface **3010** of the health application **2804**. The spot check monitoring user interface **3010** may display the substantially real-time measurements of physiological parameters as measured by the wearable device **10** during the spot check such as pulse rate, SpO₂, RRp, PVi, Pi and the like. As described herein, substantially real-time measurements, waveforms, parameters and/or the like may refer to physiological measurements that are displayed (e.g., via user interfaces of the health application) at the same time or substantially the same time (e.g., neglecting any small delays such as those that are imperceptible to humans such as delays arising from electrical conduction or transmission) that they are obtained by the wearable device **10**. The spot check monitoring user interface **3010** may display the substantially real-time waveforms **3012** of any of the physiological parameters such as pulse waveform and/or plethysmograph waveform obtained during the spot check.

[0687] FIG. **30C** illustrates an example spot check monitoring user interface **3020** of the health application **2804**. The spot check monitoring user interface **3020** may display substantially real-time measurements of wearer physiological parameters such as those described with reference to FIG. **30B**. The spot check monitoring user interface **3020** can further include historical data. In particular, the spot check monitoring user interface **3020** can include a visualization(s) that present historical trends of wearer physiological parameters. As shown, the visualization(s) can include one or more graphs with an x-axis of time and a y-axis of parameter values.

[0688] FIG. **30D** illustrates an example spot check monitoring user interface **3030** of the health application **2804**. The spot check monitoring user interface **3030** may be displayed when the wearable device **10** has completed the spot check. In some aspects, as depicted in FIG. **30D**, the user interface **3030** may display physiological parameters obtained during the spot check in a similar format to that shown in user interface **3020** of FIG. **30C**. In some aspects, the user interface **3030** may display physiological parameters obtained during the spot check in a similar format to that shown in user interface **3010** of FIG. **30B**.

[0689] In some aspects, the spot check monitoring user interface **3030** may be displayed for a predetermined length of time, such as one minute, after the spot check has terminated, and after the predetermined length of time the health application **2802** may display another user interface such as any of the user interfaces described herein, such as the dashboard user interface **2900** or the spot check monitoring user interface **3000**. In some aspects, a user may navigate to another user interface such as any of the user interfaces described herein, such as the dashboard user interface **2900** or the spot check monitoring user interface **3000**, upon selection of the selectable component

3032. In some aspects, the health application **2804** may display the user interface **3030** until a user selects the selectable component **3032**.

[0690] FIGS. **30E-30H** illustrate example continuous monitoring user interfaces of the health application **2804**. The wearable device **10** may be configured to perform continuous measurements of physiological parameters of a wearer. For example, the wearable device may continuously measure a wearer's physiological parameters indefinitely, for example as long as the device is powered on or until a user selects to cease the continuous operation mode and/or until a user selects operation of a non-continuous monitoring mode of operation such as the spot check as described herein.

[0691] In some aspects, the external device **2802** may check the power level of the wearable device **10** (for example by checking a charge and/or voltage level of the power source **16**) prior to causing the wearable device to enter a continuous monitoring mode and/or during continuous monitoring to determine if the wearable device **10** has sufficient power to perform continuous monitoring. Upon determining that the wearable device **10** has sufficient power, the external application may cause the wearable device **10** to initiate continuous monitoring and/or to continue continuous monitoring. Upon determining that the wearable device **10** does not have sufficient power, the external application may not cause the wearable device **10** to initiate continuous monitoring and/or may cause the wearable device **10** to discontinue continuous monitoring, and/or may cause the wearable device **10** to initiate a mode of operation that is different than continuous monitoring such as a modified continuous monitoring mode or a spot check mode. In some aspects, upon receiving a command, for example from the external device **2802** to perform continuous monitoring, the wearable device **10** may check its power level (for example by checking a charge and/or voltage level of the power source **16**) prior to commencing and/or during continuous monitoring to determine if it has sufficient power to perform continuous monitoring. Upon determining that it has sufficient power, the wearable device **10** may commence performing continuous monitoring and/or may continue performing continuous monitoring. Upon determining that it does not have sufficient power, the wearable device **10** may not initiate continuous monitoring and/or may discontinue continuous monitoring, and/or may cause the wearable device **10** to initiate a mode of operation that is different than continuous monitoring such as a modified continuous monitoring mode or a spot check mode.

[0692] FIG. **30E** illustrates an example continuous monitoring user interface **3040** of the health application **2804**. User interface **3040** may be displayed when the wearable device **10** is operating in continuous monitoring mode and is simultaneously unable to obtain physiological measurements (or the physiological measurements obtained are determined to fall below an accuracy threshold), for example when the wearable device **10** not being worn (or not being worn properly) by a wearer.

[0693] FIG. **30F** illustrates an example continuous monitoring user interface **3050** of the health application **2804**. The continuous monitoring user interface **3050** may display the substantially real-time measurements of physiological parameters as measured by the wearable device **10** during continuous monitoring in a similar manner as explained with reference to the display of physiological parameters during a spot check in example user interface **3010** of FIG. **30B**.

[0694] FIG. **30G** illustrates an example continuous monitoring user interface **3060** of the health application **2804**. The continuous monitoring user interface **3060** may display substantially real-time measurements of wearer physiological parameters such as those described with reference to FIG. **30F** or FIG. **30B**. The continuous monitoring user interface **3060** can further include historical data. In particular, the continuous monitoring user interface **3060** can include a visualization(s) that present historical trends of wearer physiological parameters. As shown, the visualization(s) can include one or more graphs with an x-axis of time and a y-axis of parameter values.

[0695] FIG. **30H** illustrates an example continuous monitoring user interface **3070** of the health application **2804**. The continuous monitoring user interface **3070** may display information **3702** relating to any of the physiological parameters discussed herein such as those displayed in any of

the graphical user interfaces discussed herein. The information **3072** may help a user/wearer understand the physiological parameter. In the example of FIG. **30H**, the continuous monitoring user interface **3070** displays information **3072** relating to SpO.sub.2 physiological parameter. The continuous monitoring user interface **3070** may display information relating to a physiological parameter in response to a user selection of the displayed corresponding physiological parameter. In some aspects, as depicted in FIG. **30H**, a user may select a physiological parameter from a displayed format similar to that shown in user interface **3050** of FIG. **30F** to display information **3072**. In some aspects, a user may select a physiological parameter from a displayed format similar to that shown in user interface **3060** of FIG. **30G** to display information **3072**.

[0696] FIGS. **31A-31D** illustrate example measurement settings user interfaces of the health application **2804**. As discussed with reference to the example measurement settings interfaces, a user may adjust the settings of the health application **2804** with regards to at least the physiological measurements and/or may adjust the mode of operation of the wearable device **10**.

[0697] FIG. **31A** illustrates an example measurement settings user interface **3100** of the health application **2804**. The measurement settings user interface **3100** may allow a user to select a mode of operation of the wearable device **10**. For example, a user may select, via user interface **3100**, whether the wearable device operates in continuous monitoring mode or spot check mode. A user may also navigate, via selection in the user interface **3100**, to additional measurement settings user interfaces such as user interface **3110**.

[0698] FIG. **31B** illustrates an example measurement settings user interface **3110** of the health application **2804**. User interface **3110** may display various physiological parameters, as shown. A user may select any of the displayed physiological parameters to edit and/or adjust the settings associated with the selected physiological parameter. User selection of the physiological parameters may navigate a user to additional measurement settings user interfaces such as user interfaces **3120** or **3130**.

[0699] FIG. **31C** illustrates an example measurement settings user interface **3120** of the health application **2804**. User interface **3120** may allow a user to set one or more threshold levels relating to a physiological parameter such as any of the physiological parameters measured by the wearable device **10** and/or displayed by the graphical user interfaces as discussed herein. In some aspects, when the physiological parameter value as measured by the wearable device **10** reaches the threshold level, the external device **2802** may generate a notification. For example, if a threshold level is breached, then the external device **2802** can cause an emergency notification to be generated. The emergency notification may be displayed on the graphical user interface of the external device **2802** and/or communicated to another device such as the wearable device **10**, another external device **2802** and/or a third-party device. In some aspects, the generation of physiological parameter notification(s) can be activated or deactivated by the user.

[0700] The example measurement settings user interface **3120** displays a slidable component **3122** relating to SpO.sub.2 threshold. A user may slide the slidable component **3122** to adjust and/or select the threshold SpO.sub.2 limit. As shown, a user has set the SpO.sub.2 threshold limit to 88%. As discussed herein, in some aspects, when the wearable device **10** measures an SpO.sub.2 level of the wearer to be 88% or less than 88%, the health application **2804** of the external device **2802** may generate a notification and/or alert.

[0701] FIG. **31D** illustrates an example measurement settings user interface **3130** of the health application **2804**. The example measurement settings user interface **3130** displays a slidable component **3134** relating to RRp threshold. A user may slide the slidable component **3134** to adjust and/or select an upper and lower RRp threshold limit. As shown, a user has set the RRp lower threshold limit to 6 and has set the RRp upper threshold limit to 30. As discussed herein, in some aspects, when the wearable device **10** measures an RRp level of the wearer to be less than 6 or greater than 30, the health application **2804** of the external device **2802** may generate a notification and/or alert.

[0702] The example measurement settings user interface **3130** displays a toggle component **3132** relating to RRp display. A user may select the toggle component **3132** to enable or disable the display of RRp physiological parameter measurements in any of the user interfaces discussed herein, for example user interfaces **29** or **30A-30H**. A toggle component or other means similar to the toggle component **3132** shown in user interface **3130** may be displayed in other user interfaces relating to other physiological parameters. For example, a similar toggle component may be displayed in user interface **3120** to allow a user to enable or disable the display of SpO₂ physiological parameter measurement in the user interfaces discussed herein.

[0703] FIG. **32** illustrates an example activity user interface **3200** of the health application **2804**. The activity user interface **3200** may display a history of activity level of the wearer for any number of activity categories **3202** such as the number of steps taken, distance travelled, and/or calories burned. The activity user interface **3200** may display the activity levels of the wearer throughout a time period such as a 12 hour time period, a 24 hour time period, a week time period or any other time period. For example, the activity user interface **3200** can include a visualization(s) that presents historical trends of wearer activity levels **3202**. As shown, the visualization(s) can include one or more graphs with an x-axis of time and a y-axis of activity level. The activity user interface **3200** may display goals of the wearer/user for any of the activity categories **3202** and may display a visualization **3204** corresponding to the goal and/or a wearer's status in achieving the goal.

[0704] FIG. **33A** illustrates an example workout user interface **3300** of the health application **2804**. The user may select a selectable component **3302** of the workout user interface **3300** to cause the wearable device to begin monitoring wearer physiological parameters and/or to cause the health application **2804** to begin tracking/logging wearer physiological parameters during the workout. The selectable component **3302** may correspond to the type of workout such as running, swimming, high-intensity interval training (H.I.I.T), strength training and the like. The workout user interface **3300** may display information **3304** relating to a previous workout.

[0705] FIG. **33B** illustrates an example workout user interface **3310** of the health application **2804**. The workout user interface **3310** may be displayed by the health application **2804** during and/or after a workout of the wearer. The workout user interface **3310** may display physiological parameters of the wearer, such as received by the wearable device **10**, and other information relating to the workout such as time duration of the workout. The workout user interface **3310** may display historical and/or substantially real-time physiological parameters of the wearer. The workout user interface **3310** may display a visualization(s) **3306** of any of the wearer physiological parameters. For example, the workout user interface **3310** can include a visualization(s) that present historical trends of wearer physiological parameters during the workout. As shown, the visualization(s) can include one or more graphs with an x-axis of time and a y-axis of parameter values.

[0706] FIG. **33C** illustrates an example workout user interface **3320** of the health application **2804**. The workout user interface **3320** may be displayed by the health application **2804** after a workout of the wearer. The workout user interface **3320** may display information **3308** relating to the workout of the wearer.

[0707] FIG. **34A** illustrates an example sleep user interface **3400** of the health application **2804**. The sleep user interface **3400** may display a sleep quality index which may be calculated based, at least in part, on one or more of the physiological parameters of the wearer received from the wearable device **10**. The sleep user interface **3400** may display a visualization(s) **3402** of the sleep quality index. For example, the sleep user interface **3400** can include a visualization(s) that present historical trends of the sleep quality index during a time the wearer is asleep. As shown, the visualization(s) can include one or more graphs with an x-axis of time and a y-axis of sleep quality index.

[0708] FIG. **34B** illustrates an example sleep user interface **3410** of the health application **2804**.

The sleep user interface **3410** may display a history of events occurring during a time the wearer is asleep. The events may be major or minor events as determined by, for example, a magnitude of the event. For example, the history of events may display all the times during which the wearer was asleep wherein a physiological parameter, such as SpO₂, exceeded a threshold. The threshold(s) may be set by the user/wearer as discussed with reference to FIGS. **31C-31D**.

[0709] FIG. **34C** illustrates an example sleep user interface **3420** of the health application **2804**. The sleep user interface **3420** may display physiological parameters of the wearer, such as current physiological parameters and/or average physiological parameters of the wearer during a time the wearer was asleep. The sleep user interface **3420** may display a visualization(s) **3404** of the physiological parameters. For example, the sleep user interface **3420** can include a visualization(s) that present historical trends of the physiological parameters during a time the wearer is asleep. As shown, the visualization(s) can include one or more graphs with an x-axis of time and a y-axis of parameter value. The visualization may further include an indication of a threshold(s) and may show all times wherein the physiological parameter exceeded the threshold(s). The threshold(s) may correspond to the threshold(s) discussed with reference to FIG. **34B** and may be set by the wearer as discussed with reference to FIGS. **31C-31D**.

[0710] FIG. **35A** illustrates an example history user interface **3500** of the health application **2804**. The history user interface **3500** may display historical data, for example historical data of the wearer physiological parameters received from the wearable device **10**. The history user interface **3500** may display historical data over any range of time, for example historical data for the day and/or historical data for the week. The history user interface **3500** may display historical data grouped according to any time frame, for example historical data grouped by day and/or historical data grouped by hour. As shown in FIG. **35A**, the history user interface **3500** may display historical data for the current day and any number of preceding days.

[0711] FIG. **35B** illustrates an example history user interface **3510** of the health application **2804**. The history user interface **3510** may display historical data, such as historical physiological parameter values and/or activity levels of the wearer. The history user interface **3510** can include a visualization(s) that present historical trends of wearer physiological parameters and/or activities, such as calories burned or steps taken. As shown, the visualization(s) can include one or more graphs with an x-axis of time and a y-axis of values, such as physiological parameter values or activity levels. The history user interface **3510** may display the visualization(s) such that the x-axis of time of one graph corresponds with the x-axis of time of any of the other graphs. The history user interface **3510** may further display historical data value(s) of the visualization graph(s) corresponding to the time of the of the visualization graph(s) upon selection by a user such as by adjusting a slider along the x-axis of the graphs as shown in FIG. **35B**.

[0712] Appendix A that is attached to this specification is hereby incorporated by reference in its entirety and is made a part of this specification. Appendix A shows additional example graphical user interfaces of the health application **2804**. A1 illustrates example user interfaces of the wearable device **10** for connecting the wearable device **10** with an external device **2802**. A2 illustrates an example user interface for initiating the health application **2804** on an external device **2802**. A3-A5 illustrate example user interfaces for creating an account on the health application **2804**. A6 illustrates an example user interface of a successful account creation for health application **2804**. A7 illustrates example user interfaces of successfully pairing the wearable device **10** with an external device **2802**. A8 illustrates example user interfaces for setting goals in the health application **2804**. A9 illustrates example user interfaces of unsuccessfully logging in to the health application **2804**.

[0713] A10 illustrates example user interfaces of a homepage of the health application **2804**. A11-A13 illustrate example user interfaces of a dashboard of the health application **2804**. A14 illustrates example user interfaces for spot check monitoring of wearer physiological parameters with the wearable device **10**. As shown in A14, a user may select a component via the user interface to start

a physiological measurement with the wearable device **10**. The wearable device **10** may measure physiological parameters and display the measurements to the user interface for a predetermined amount of time (e.g., one minute) and/or until the user selects a component via the user interface to stop the physiological measurement. A15 illustrates an example user interface for displaying physiological parameters measured continuously with the wearable device **10**. A16 illustrates example user interfaces for adjusting physiological measurement settings. A17 illustrates an example user interface for displaying physiological parameter measurements.

[0714] A18-A19 illustrates example user interfaces for displaying activity of a user. A20-A21 illustrate example user interfaces for displaying workout information of a user. A22-A24 illustrate example user interfaces for displaying sleep information of a user. A25-A26 illustrates example user interfaces for setting sleep settings for a user. A27-A29 illustrate example user interfaces for displaying a history of a user's physiological parameter measurements and activity.

[0715] A30-A36 illustrate example user interfaces for setting the settings of the wearable device **10**.

[0716] A37 illustrates example user interfaces of a dashboard of the health application **2804**. A38 illustrates example user interfaces of a history of physiological parameters of a user. A39 illustrates an example user interface for sharing information, such as physiological parameters, from the external device **2802** to another device. A40-A41 illustrate example user interfaces for navigating between interfaces of the health application **2804** using an interactive navigation bar displayed at the bottom of the user interface. A42 illustrates example user interfaces of a dashboard and account homepage of the health application **2804**. A43 illustrates example user interfaces for navigating between interfaces of the health application **2804** using an interactive navigation bar displayed at the bottom of the user interface. A44 illustrates example user interfaces of a dashboard of the health application **2804**.

Example Graphical User Interfaces of the Wearable Device

[0717] As discussed herein, for example, with reference to FIG. **11A**, the wearable device **10** can include a display **12**, which can include the display screen and touch input from the wearer. For example, the display **12** may comprise a capacitive touchscreen configured to receive touch input from the wearer to control functionality of the wearable device **10**. The wearable device **10** can also include user interfaces on the device **10** to receive input from the wearer. As shown in FIG. **1D**, example user interfaces can include buttons **13**.

[0718] FIG. **36A** shows an example wearable device **10** including a display **12** and buttons **13**. FIGS. **36B-36E** show example screens of the display **12** of the wearable device **10**. The display **12** may be configured to display many different screens. For example, in some aspects, the display **12** may display a screen with various physiological parameter information (such as values and trends) and in other aspects, the display **12** may display a screen with no physiological parameter information. In some aspects, the display **12** may display a screen with non-physiological related information such as date, time and other notifications.

[0719] The display **12** may display various screens in a cyclical manner. For example, the display **12** may display a screen with a first physiological parameter information (such as value and trend) and then may display a screen with a second physiological parameter information and then may display a screen with a third physiological parameter information and so forth. In some aspects, the order of the screens the display **12** displays is constant. The display **12** may cycle through displaying various screens automatically and/or manually. For example, the display **12** may change the screen that is displayed automatically, without user input, every 1 minute, every 30 seconds, every 10 seconds, every 5 seconds, every 3 seconds, every 2 seconds, every 1 second or any other time frame that is required or desired. In some aspects, a wearer/user may select the time frame (e.g., frequency) in which the display **12** will change the screen. In some aspects, a wearer/user may manually change the screen that is displayed by the display **12**. For example, a wearer may touch the display **12** to provide touch input to the wearable device **10** and/or may press a button **13**

to provide input to the wearable device **10**. Upon receiving said input from the wearer, the display **12** may change the screen that is displayed. The display **12** may not change the screen again until another user input, as described, is received.

[0720] In some aspects, the display **12** may cycle through displaying various screens automatically and manually, as simultaneous modes of operation. For example, the display **12** may cycle through the various screens automatically without user input (e.g., every 3 seconds, for example) unless or until a wearer/user provides input (via, the display **12** and/or buttons **13**) upon which the display **12** will change the screen.

[0721] In some aspects, the display **12** may cycle through displaying the various screens automatically or manually, but not both modes of operation at the same time. For example, the display **12** may cycle through the various screens automatically (e.g., every 3 seconds, for example) regardless of user input that is received. As another example, the display **12** may cycle through the various screens manually, only upon receiving user input, regardless of time that has elapsed. In some aspects, the display **12** may toggle between cycling through the displaying the various screens automatically and manually, for example upon adjustment of user settings of the wearable device **10**.

[0722] FIGS. **36A-36E** show example screens of the display **12** of the wearable device **10**. FIG. **36A** shows a screen without physiological parameter information (such as values and trends). FIG. **36B** shows a screen with SpO₂ physiological parameter value of the wearer. FIG. **36C** shows a screen with RR physiological parameter value of the wearer. FIG. **36D** shows a screen with the number of steps taken by the wearer. FIG. **36E** shows a screen with PR physiological parameter value of the wearer. The display **12** may cycle through displaying the various screens of FIGS. **36A-36E** automatically and/or manually, as described above.

[0723] FIGS. **37A-37E** show example screens of the display **12** of the wearable device **10**. The display **12** may cycle through displaying the various screens of FIGS. **37A-37E** automatically and/or manually, as described above.

[0724] FIG. **38A** shows an example screen of the display **12** of the wearable device **10**. The display **12** displays a screen with time, and RRp physiological parameter value and trend and PR physiological parameter value and trend. In some aspects, the display **12** may cycle through displaying physiological information (such as values and trends) of other parameters (such as SpO₂, steps taken etc.) automatically and/or manually as described above with reference to FIGS. **37A-37E**, for example. In some aspects, a user/wearer may select physiological information to be displayed as a default by display **12**, for example by adjusting the settings of the wearable device **10**.

[0725] FIG. **38B** shows an example screen of the display **12** of the wearable device **10**. The display **12** displays a screen with time, and SpO₂ physiological parameter value and trend and PR physiological parameter value and trend. In some aspects, the display **12** may automatically display physiological information (such as value and trend) relating to a parameter if that parameter has exceeded a threshold. For example, if a physiological parameter value exceeds a high or low threshold, the display **12** may automatically display that physiological parameter value and trend. This may be in place of physiological information of another parameter already displayed. For example, as shown in FIG. **38B**, the wearer may have an SpO₂ parameter value of 89. This value may be below a low limit threshold for SpO₂ (for example, as set by a user/wearer as described with reference to FIG. **31C**, for example). Upon determining that the SpO₂ has fallen below a low limit threshold, the wearable device **10** may cause the display **12** to display the physiological information (such as value and trend) for SpO₂. This may replace physiological information of another parameter already displayed that has not exceeded a threshold. In this example, the display **12** replaces RRp (which has not exceeded any thresholds) in the screen (shown in FIG. **38A**) with SpO₂ in the screen (shown in FIG. **38B**). Physiological parameters that have exceeded a threshold that are displayed in the screen by display **12** may differ in color or some other respect to

physiological parameters that have not exceeded a threshold.

[0726] FIG. 39 is a flowchart illustrating an example process 3900 for performing monitoring operation(s) of one or more physiological parameters of a user/wearer. The process 3900 may be performed by a system that includes a wearable device 10 and a computing device 2802. The computing device 2802 may be, for example, any of the devices discussed with reference to FIG. 2, such as a patient monitor 202, a mobile communication device 204, a computer 206, a tablet 208, a nurses' station system 210 or any other device as required or desired. As discussed herein, the wearable device 10 and the computing device 2802 may be in communication with each other, for example, via Bluetooth technology or near-field communication (NFC) technology.

[0727] At block 3902, the wearable device 10 and the computing device 2802 may establish communication with each other. For example, the wearable device 10 may be paired with the computing device using Bluetooth or NFC.

[0728] At block 3904, the computing device 2802 may receive a user input. For example, a user can provide the user input via an interactive user interface of the computing device 2802. Example user input can include a user selection to initiate a monitoring operation, such as sport check or continuous monitoring. Additionally or alternatively, the wearable device 10 can receive user input, such as a user selection to initiate a monitoring operation, such as sport check or continuous monitoring.

[0729] At block 3906, the computing device 2802 may check to determine whether it has received user input corresponding to an instruction to perform a monitoring operation. In response to receiving the user input for a monitoring operation, the computing device 2802 may generate a signal to communicate to the wearable device 10, such as by sending a wireless communication to the wearable device 10. The signal may instruct the wearable device 10 to perform one or more operations according to the user input. For example, via a user interface of the computing device 2802, a user may select a spot check physiological monitoring operation. In response to the user input, the computing device 2802 may generate a signal to wirelessly communicate to the wearable device 10 to instruct the wearable device 10 to perform a spot check monitoring operation.

[0730] If the user input is not associated with a monitoring operation, the process 3900 can return to block 3904 to wait for additional input. In some aspects, the computing device 2802 can perform some other operation associated with the non-monitoring operation user input, as described herein.

[0731] The wearable device 10 may check to determine whether it has received a signal instructing it to perform a monitoring operation. If no signal has been received to perform a monitoring operation, the wearable device 10 may continue to determine whether it has received a signal at block 3906. If the wearable device 10 has received a signal, for example from the computing device 2802, instructing it to perform a monitoring operation, the process 3900 may continue to step 3908.

[0732] At block 3908, it may be determined whether the wearable device 10 has sufficient power to perform the requested monitoring operation. One or more techniques can be used to determine whether a battery has sufficient charge to initiate the particular monitoring operation. For example, the wearable device 10 may determine a current charge status of its battery. In some aspects, the wearable device 10 may include an estimate power consumption rate for respective operations and/or sensors. Additionally or alternatively, the wearable device 10 or another computing device 2802 may log power consumption for particular operations or sensors of wearable device(s) to determine an estimated power consumption rate. A determination of sufficient power for an operation can be based on the particular operation. For example, a spot check operation can use less power than a continuous monitoring operation. Moreover, an estimation of power consumption can be based on the particular operation. The estimation of power consumption can include an estimated duration of the operation. For example, a spot check operation can last a predetermined period of time, such as 1 second, 5 seconds, or 15 seconds, and the estimated power consumption can be based on that duration and the particular sensor(s) involved in the operation. Similarly, a

continuous monitoring operation can have an estimated duration that is used for power consumption purposes, such as an estimated runtime for thirty minutes, one hour, four hours, or eight hours. The wearable device **10** or another computing device **2802** can estimate the power consumption for the operation by multiplying the estimated power consumption rate with the predetermined or estimated duration for the particular operation. The estimated power consumption can then be compared to the current battery charge level. For example, if the estimated power operation is greater than the current battery charge or is estimated to leave the battery at less than five percent or ten percent capacity, the wearable device **10** or another computing device **2802** can determine that there is insufficient power for the requested monitoring operation. If the wearable device **10** has sufficient power to perform the requested monitoring operation, the process proceeds to block **3912** to perform the monitoring operation.

[0733] If the wearable device **10** does not have sufficient power, the process **3900** proceeds to block **3910** to provide a lower power warning.

[0734] At block **3910**, a low power warning can be provided. For example, the wearable device **10** may transmit a signal (such as a message) to the computing device **2802** that the wearable device **10** has insufficient power for the monitoring operation. The signal may cause the computing device **2802** to display a lower power warning message to a user.

[0735] In some aspects, if the wearable device **10** does not have sufficient power to perform the requested monitoring operation, the wearable device may continue to step **3912** to perform a monitoring operation that is different than the requested monitoring operation. For example, if the wearable device **10** does not have sufficient power to perform the requested operation, the wearable device **10** may determine a modified monitoring operation that may be performed, such as a modified continuous monitoring operation and/or a modified spot check monitoring operation. The determination may be based at least in part on the power of the monitoring device (e.g., remaining battery life). For example, a user may request a continuous monitoring operation and upon determining that there is insufficient power to perform the continuous monitoring operation (e.g., at step **3908**) the wearable device **10** may instead perform a spot check operation (which may require less power than continuous monitoring) for an amount of time that is commensurate with the amount of power (e.g., remaining battery life) of the wearable device **10**. As another example, a user may request a spot check monitoring operation to be performed for an amount of time and upon determining that there is insufficient power to perform the spot check monitoring operation (e.g., at step **3908**) for the duration of the length of time the wearable device **10** may instead perform a spot check operation for an abbreviated length of time (which may require less power than the request spot check monitoring) which may be commensurate with the amount of power (e.g., remaining battery life) of the wearable device **10**.

[0736] At block **3912**, a monitoring operation may be performed. For example, the wearable device **10** may perform the monitoring operation. As described herein, example monitoring operations can include one or more of a spot-check monitoring operation, a continuous monitoring operation, and/or any other monitoring operation as discussed herein. The wearable device **10** may perform the monitoring operation on the wearer/user for one or more of the physiological parameters, such as, but not limited to, SpO2, RR, PR, RRp, etc.

[0737] The wearable device **10** may transmit physiological parameter measurement data to the computing device **2802**. In some aspects, the wearable device **10** may transmit measurement data during the monitoring operation (such as while the wearable device **10** measures the physiological parameters and/or during a continuous monitoring operation).

[0738] At block **3914**, one or more user interfaces may be presented. For example, the computing device **2802** may present the physiological parameter measurement data received from the wearable device **10** in a user interface. Similarly, the wearable device **10** can present the physiological parameter measurement data in a user interface on the wearable device **10**. In response to receiving physiological parameter measurement data from the wearable device **10**, the

computing device **2802** may display the user interfaces described herein, such as with reference to FIGS. **30B**, **30C**, **30F** and/or **30G**. The computing device **2802** may display the physiological parameter measurement data in substantially real-time, for example, as the wearable device **10** performs the monitoring operation and measures the one or more physiological parameters. [0739] In some implementations, the wearable device **10** may generate user interface data to display physiological parameter measurement data to a wearer/user of the wearable device **10**. For example, the wearable device **10** may display the user interfaces described herein on a display screen **12** of the wearable device **10**, such as with reference to FIGS. **38A-38B**. The wearable device **10** may display the physiological parameter measurement data in substantially real-time, for example, as the wearable device **10** performs the monitoring operation and measures the one or more physiological parameters.

Additional Implementation Details

[0740] FIG. **40** is a block diagram that illustrates example components of a computing device **4000**. The computing device **4000** can implement aspects of the present disclosure, and, in particular, aspects of the wearable device **10** and/or the computing device **2802** of FIG. **28**. The computing device **4000** can communicate with other computing devices.

[0741] The computing device **4000** can include a hardware processor **4002**, a data storage device **4004**, a memory device **4006**, a bus **4008**, a display **4012**, and one or more input/output devices **4014**. A processor **4002** can also be implemented as a combination of computing devices, e.g., a combination of a digital signal processor and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a digital signal processor, or any other such configuration. The processor **4002** can be configured, among other things, to process data, execute instructions to perform one or more functions, such as process one or more physiological signals to obtain one or more measurements, as described herein. The data storage device **4004** can include a magnetic disk, optical disk, or flash drive, etc., and is provided and coupled to the bus **4008** for storing information and instructions. The memory **4006** can include one or more memory devices that store data, including without limitation, random access memory (RAM) and read-only memory (ROM). The computing device **4000** may be coupled via the bus **4008** to a display **4012**, such as a LCD display or touch screen, for displaying information to a user. The computing device **4000** may be coupled via the bus **4008** to one or more input/output devices **4014**. The input device **4014** can include, but is not limited to, a keyboard, mouse, digital pen, microphone, touch screen, gesture recognition system, voice recognition system, imaging device (which may capture eye, hand, head, or body tracking data and/or placement), gamepad, accelerometer, or gyroscope.

Terminology

[0742] Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount. As another example, in certain aspects, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 10 degrees, 5 degrees, 3 degrees, or 1 degree. As another example, in certain aspects, the terms “generally perpendicular” and “substantially perpendicular” refer to a value, amount, or characteristic that departs from exactly perpendicular by less than or equal to 10 degrees, 5 degrees, 3 degrees, or 1 degree.

[0743] Many other variations than those described herein will be apparent from this disclosure. For example, certain acts, events, or functions of any of the algorithms described herein can be performed in a different sequence, can be added, merged, or left out altogether (for example, not all described acts or events are necessary for the practice of the algorithms). Moreover, acts or events can be performed concurrently, for example, through multi-threaded processing, interrupt

processing, or multiple processors or processor cores or on other parallel architectures, rather than sequentially. In addition, different tasks or processes can be performed by different machines and/or computing systems that can function together.

[0744] It is to be understood that not necessarily all such advantages can be achieved in accordance with any particular example of the examples disclosed herein. Thus, the examples disclosed herein can be embodied or carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

[0745] The various illustrative logical blocks, modules, and algorithm steps described in connection with the examples disclosed herein can be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. The described functionality can be implemented in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.

[0746] The various illustrative logical blocks and modules described in connection with the examples disclosed herein can be implemented or performed by a machine, such as a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor can be a microprocessor, but in the alternative, the processor can be a controller, microcontroller, or state machine, combinations of the same, or the like. A processor can include electrical circuitry or digital logic circuitry configured to process computer-executable instructions. In another example, a processor can include an FPGA or other programmable device that performs logic operations without processing computer-executable instructions. A processor can also be implemented as a combination of computing devices, for example, a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. A computing environment can include any type of computer system, including, but not limited to, a computer system based on a microprocessor, a mainframe computer, a digital signal processor, a portable computing device, a device controller, or a computational engine within an appliance, to name a few.

[0747] The steps of a method, process, or algorithm described in connection with the examples disclosed herein can be embodied directly in hardware, in a software module stored in one or more memory devices and executed by one or more processors, or in a combination of the two. A software module can reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of non-transitory computer-readable storage medium, media, or physical computer storage known in the art. An example storage medium can be coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium can be integral to the processor. The storage medium can be volatile or nonvolatile. The processor and the storage medium can reside in an ASIC.

[0748] The apparatuses and methods described herein may be implemented by one or more computer programs executed by one or more processors. The computer programs include processor-executable instructions that are stored on a non-transitory tangible computer readable medium. The computer programs may also include stored data. Non-limiting examples of the non-transitory tangible computer readable medium are nonvolatile memory, magnetic storage, and optical storage.

[0749] The term “substantially” when used in conjunction with the term “real-time” forms a phrase that will be readily understood by a person of ordinary skill in the art. For example, it is readily understood that such language will include speeds in which no or little delay occurs.

[0750] Conditional language used herein, such as, among others, “can,” “might,” “may,” “for example,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain examples include, while other examples do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more examples or that one or more examples necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular example. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Further, the term “each,” as used herein, in addition to having its ordinary meaning, can mean any subset of a set of elements to which the term “each” is applied.

[0751] Disjunctive language such as the phrase “at least one of X, Y, or Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to present that an item, term, etc., may be either X, Y, or Z, or any combination thereof (for example, X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain examples require at least one of X, at least one of Y, or at least one of Z to each be present.

[0752] Unless otherwise explicitly stated, articles such as “a” or “an” should generally be interpreted to include one or more described items. Accordingly, phrases such as “a device configured to” are intended to include one or more recited devices. Such one or more recited devices can also be collectively configured to carry out the stated recitations. For example, “a processor configured to carry out recitations A, B and C” can include a first processor configured to carry out recitation A working in conjunction with a second processor configured to carry out recitations B and C.

[0753] While the above detailed description has shown, described, and pointed out novel features as applied to various examples, it will be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As will be recognized, the inventions described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others.

[0754] Additionally, all publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

Claims

1. (canceled)

2. A wearable health monitoring device configured to be secured to a wearer, comprising: a housing including a front side and a back side, the back side configured to face tissue of the wearer when the device is worn; and a frame positioned by the back side of the housing, said frame defining: a first emitter chamber housing a first plurality of optical emitters; a second emitter chamber housing a second plurality of optical emitters, wherein each emitter of the second plurality of optical emitters is configured to emit optical radiation having a same wavelength as optical radiation emitted from a corresponding emitter of the first plurality of optical emitters; a plurality of detector

chambers housing a plurality of optical detectors configured to detect optical radiation emitted from the first and second plurality of optical emitters, wherein the frame is configured to reduce light from the first and second plurality of optical emitters from reaching the plurality of optical detectors without being attenuated by the tissue of the wearer.

3. The wearable health monitoring device of claim 2, wherein the first and second plurality of optical emitters are configured to emit infrared optical radiation and green optical radiation.
4. The wearable health monitoring device of claim 2, wherein the first plurality of optical emitters comprises: a first optical emitter configured to emit optical radiation having a first wavelength between 620 nm and 660 nm; a second optical emitter configured to emit optical radiation having a second wavelength between 650 nm and 670 nm; and a third optical emitter configured to emit optical radiation having a third wavelength between 900 nm and 910 nm.
5. The wearable health monitoring device of claim 4, wherein the second plurality of optical emitters is configured to emit optical radiation having the first, second, and third wavelengths.
6. The wearable health monitoring device of claim 4, wherein the first plurality of optical emitters comprises: a fourth optical emitter configured to emit optical radiation having a fourth optical wavelength of about 970 nm.
7. The wearable health monitoring device of claim 6, wherein the first plurality of optical emitters comprises: a fifth optical emitter configured to emit optical radiation having a fifth optical wavelength between 525 nm and 650 nm.
8. The wearable health monitoring device of claim 7, wherein the second plurality of optical emitters is configured to emit optical radiation having the first, second, third, fourth, and fifth wavelengths.
9. The wearable health monitoring device of claim 2, wherein the frame comprises a light barrier configured to separate the first emitter chamber from the second emitter chamber, wherein a width of the light barrier is less than 1.40 mm.
10. The wearable health monitoring device of claim 2, wherein the plurality of detector chambers surround the first and second emitter chambers in an annular formation.
11. The wearable health monitoring device of claim 2, wherein the plurality of detector chambers comprise six detector chambers.
12. The wearable health monitoring device of claim 2, wherein the first plurality of optical emitters comprises: a first optical emitter configured to emit optical radiation having a first wavelength; and a second optical emitter configured to emit optical radiation having a second wavelength; wherein the second plurality of optical emitters comprises: a third optical emitter configured to emit optical radiation having the first wavelength; and a fourth optical emitter configured to emit optical radiation having the second wavelength; wherein the first and third optical emitters are separated from a first optical detector of the plurality of optical detectors by a first distance; wherein the second and fourth optical emitters are separated from the first optical detector by a second distance.
13. The wearable health monitoring device of claim 12, wherein the first and second optical emitters are positioned at mirror images from each other across a centerline of the frame that bisects the first optical detector.
14. A wearable health monitoring device configured to be secured to a wearer, comprising: a housing including a front side and a back side, the back side configured to face tissue of the wearer when the device is worn; and a frame positioned by the back side of the housing, said frame defining: a first emitter chamber housing a first plurality of optical emitters, wherein the first plurality of optical emitters is configured to emit at least infrared optical radiation and green optical radiation; a second emitter chamber housing a second plurality of optical emitters, wherein the second plurality of optical emitters is configured to emit at least infrared optical radiation and green optical radiation; a plurality of detector chambers housing a plurality of optical detectors configured to detect optical radiation emitted from the first and second plurality of optical emitters, wherein the frame is configured to reduce light from the first and second plurality of optical

emitters from reaching the plurality of optical detectors without being attenuated by the tissue of the wearer.

15. The wearable health monitoring device of claim 14, wherein the first plurality of optical emitters comprises: a first optical emitter configured to emit optical radiation having a first wavelength between 620 nm and 660 nm; a second optical emitter configured to emit optical radiation having a second wavelength between 650 nm and 670 nm; and a third optical emitter configured to emit optical radiation having a third wavelength between 900 nm and 910 nm.

16. The wearable health monitoring device of claim 15, wherein the second plurality of optical emitters is configured to emit optical radiation having the first, second, and third wavelengths.

17. The wearable health monitoring device of claim 15, wherein the first plurality of optical emitters comprises: a fourth optical emitter configured to emit optical radiation having a fourth optical wavelength of about 970 nm; and a fifth optical emitter configured to emit optical radiation having a fifth optical wavelength between 525 nm and 650 nm.

18. The wearable health monitoring device of claim 2, wherein the first plurality of optical emitters comprises: a first optical emitter configured to emit optical radiation having a first wavelength; and a second optical emitter configured to emit optical radiation having a second wavelength; wherein the second plurality of optical emitters comprises: a third optical emitter configured to emit optical radiation having the first wavelength; and a fourth optical emitter configured to emit optical radiation having the second wavelength; wherein the first and third optical emitters are separated from a first optical detector of the plurality of optical detectors by a first distance; wherein the second and fourth optical emitters are separated from the first optical detector by a second distance.

19. A wearable health monitoring device configured to be secured to a wearer, comprising: a housing including a front side and a back side, the back side configured to face tissue of the wearer when the device is worn; and a frame positioned by the back side of the housing, said frame defining: a first emitter chamber housing a first plurality of optical emitters, wherein the first plurality of optical emitters comprises: a first optical emitter configured to emit optical radiation having a first wavelength; and a second optical emitter configured to emit optical radiation having a second wavelength; a second emitter chamber housing a second plurality of optical emitters, wherein the second plurality of optical emitters comprises: a third optical emitter configured to emit optical radiation having the first wavelength, wherein the first and third optical emitters are each separated from a first optical detector of the plurality of optical detectors by a first distance; and a fourth optical emitter configured to emit optical radiation having the second wavelength, wherein the second and fourth optical emitters are each separated from the first optical detector by a second distance; and a plurality of detector chambers housing a plurality of optical detectors configured to detect optical radiation emitted from the first and second plurality of optical emitters, wherein the frame is configured to reduce light from the first and second plurality of optical emitters from reaching the plurality of optical detectors without being attenuated by the tissue of the wearer.

20. The wearable health monitoring device of claim 19, wherein the first and second optical emitters are positioned at mirror images from each other across a centerline of the frame that bisects the first optical detector.

21. The wearable health monitoring device of claim 19, wherein the first plurality of optical emitters comprises: a first optical emitter configured to emit optical radiation having a first wavelength between 620 nm and 660 nm; a second optical emitter configured to emit optical radiation having a second wavelength between 650 nm and 670 nm; and a third optical emitter configured to emit optical radiation having a third wavelength between 900 nm and 910 nm.
